

Foundations of Independent Component Analysis

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Abstract

We present the mathematical foundations of linear independent component analysis (ICA) models based on standard literature in a self-contained note. It is aimed at readers with a background in measure-theoretic probability theory. We first develop the theory of the characteristic functions of probability measures on $\mathbb{R}^d$, including their analyticity and the way in which they determine and characterise the distributions. We then focus on several identifiability results of ICA models with successively strengthened assumptions on the sources: from merely non-constant, to non-Gaussian, to Gaussian-free independent sources. Under the strictest assumptions, we show that the independent sources are identifiable up to translation, permutation, scales and signs, and this even in the presence of additive Gaussian noise. Furthermore, we present the online equivariant gradient descent ICA algorithm for recovering the independent sources from data, in the standard complete noiseless non-Gaussian ICA setting.

Contents

1	Introduction	2
2	Notation and Conventions	4
3	Characteristic Functions of Probability Measures on $\mathbb{R}^d$	5
3.1	Definition and elementary properties	5
3.2	How characteristic functions determine distributions	6
3.3	Moments, the distinguished logarithm, and cumulants	8
3.4	Analytic characteristic functions	10
3.5	Gaussian, sub-Gaussian and super-Gaussian distributions	12
3.6	Why non-Gaussianity is what makes ICA work	15
4	Identifiability for Non-Constant Independent Sources	17
5	Identifiability for Non-Gaussian Independent Sources	20
6	Identifiability for Gaussian-Free Independent Sources	27
6.1	Gaussian-free random variables	27
6.2	Identifiability in the presence of additive Gaussian noise	31
7	Complete Noiseless ICA: Estimation by Equivariant Gradient Descent	35
7.1	The complete noiseless model and its identifiability	35
7.2	The maximum likelihood objective	38
7.3	The relative gradient: what the “preconditioner” really is	40
7.4	Stationary points, stability, and the choice of nonlinearity	42

7.5	The algorithm	49
7.6	LiNGAM: a causal order removes the permutation	49
7.7	Generalisations: a brief overview	52
A	Proofs of the Classical Results on Characteristic Functions	53
A.1	Uniqueness and inversion	54
A.2	Lévy’s continuity theorem	55
A.3	Entire characteristic functions	56
A.4	Marcinkiewicz’ theorem	57
A.5	Cramér’s decomposition theorem	58
B	A Proof of the Kagan–Linnik–Rao Theorem	59
B.1	Finite differences	60
B.2	Proof of the column dichotomy	62
	References	65

1 Introduction

The linear independent component analysis (ICA) model postulates that an observed p -dimensional random vector X is an affine image of a k -dimensional random vector Z with *mutually independent* components:

$$X = AZ + \mu, \quad A \in \mathbb{R}^{p \times k}, \quad \mu \in \mathbb{R}^p. \quad (1)$$

The components $Z_1, \dots, Z_k$ are called the *sources* and A is called the *mixing matrix*; both are unobserved. The *identifiability question* asks how much of the pair $(A, \mathcal{L}(Z))$ is determined by the law of X alone (Comon, 1994; Hyvärinen and Oja, 2000; Eriksson and Koivunen, 2004).

Some ambiguity is unavoidable and built into the model: we may rescale a column of A and inversely rescale the corresponding source, we may permute the columns of A together with the sources, and we may shift a source into the offset μ . Written out, for any permutation matrix P , any invertible diagonal matrix Λ and any $c \in \mathbb{R}^k$,

$$AZ + \mu = (AP\Lambda)(\Lambda^{-1}P^{-1}(Z - c)) + (Ac + \mu), \quad (2)$$

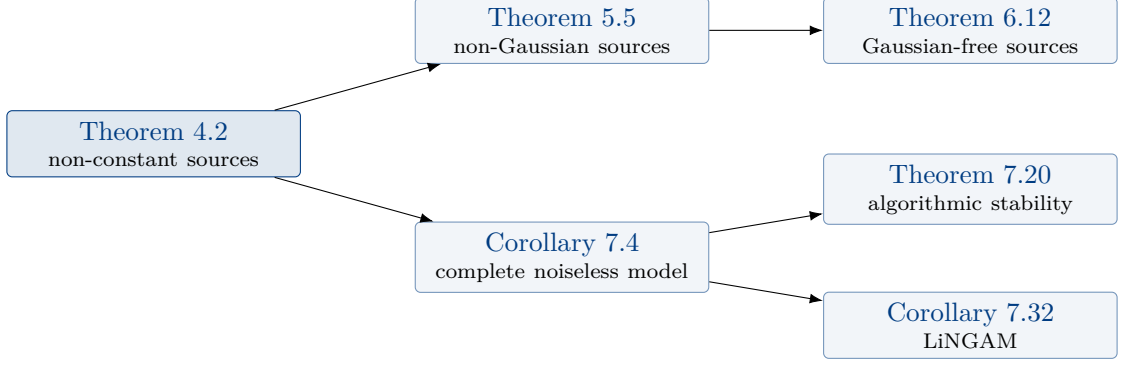
and the right hand side is again a representation of the form Eq. (1) with independent sources. The content of the classical identifiability theory is that, under suitable assumptions, this is *the only* ambiguity – with one important exception: Gaussian sources. That exception is genuine and not an artefact of the proofs. If $Z \sim \mathcal{N}(0, I_k)$ then $QZ \sim \mathcal{N}(0, I_k)$ for every orthogonal Q , so a purely Gaussian source vector can be rotated arbitrarily without changing the law of X .

The results collected here go back to the characterisation theory of the Gaussian law that grew out of the theorems of Cramér (1936), Marcinkiewicz (1939), Darmois (1953) and Skitovich (1954), and that is developed systematically in the monographs of Kagan et al. (1973) and Linnik and Ostrovskii (1977). The ICA community rediscovered and popularised these results, notably through Comon (1994) and Eriksson and Koivunen (2004), and they underpin causal discovery methods such as LiNGAM (Shimizu et al., 2006).

Prerequisites. These notes are written for readers who have taken a measure-theoretic probability course. We assume laws and weak convergence, product measures and Fubini’s theorem, and basic complex analysis up to Morera’s theorem and the identity theorem. Everything else – in particular the entire theory of characteristic functions that the arguments rest on – is developed in Section 3, with precise references for the standard results and full proofs for everything specific to our setting.

Outline and contributions. [Section 2](#) fixes notation. [Section 3](#) develops characteristic functions of probability measures on $\mathbb{R}^d$: their elementary properties, how they determine and characterise distributions (uniqueness, inversion, the Cramér–Wold device, Lévy’s continuity theorem), why local agreement of two characteristic functions is *not* enough, moments and cumulants via the distinguished logarithm, analyticity in a strip and the resulting rigidity, and finally the characterisation of the Gaussian law together with the sub- and super-Gaussian (platykurtic and leptokurtic) distributions that ICA is built on. It closes with the precise sense in which kurtosis acts as a contrast function, and with the observation that the identifiability theory needs strictly less than that: non-normality, not non-zero kurtosis. [Section 4](#) treats sources that are merely assumed to be non-constant and states the fundamental result of [Kagan et al. \(1973\)](#) in the form we need; we take the opportunity to reinstate a proportionality constant that is lost in the statement given there (see [Footnote 1](#)). [Section 5](#) specialises to non-Gaussian sources. The main technical device is [Lemma 5.2](#), which shows how a Gaussian noise vector can be traded against extra columns of the mixing matrix, and back; this is what allows us to reduce the non-Gaussian case to the non-constant case. [Section 6](#) introduces the stronger requirement that the sources be *Gaussian-free*, i.e. that no independent Gaussian noise can be split off them at all. We show in [Theorem 6.7](#) that *every* real-valued random variable decomposes, essentially uniquely, into a Gaussian-free part and independent Gaussian noise, and we then prove the strongest statement of these notes, [Theorem 6.12](#): for full column rank mixing matrices the sources are then identifiable up to permutation, scale and translation even in the presence of additive Gaussian noise with an arbitrary, possibly non-diagonal and degenerate covariance matrix. [Section 7](#) turns to estimation in the case that dominates applications – a square invertible mixing matrix and no noise. We derive the classical identifiability statement as [Corollary 7.4](#), set up the maximum likelihood objective, show that the customary “preconditioner” in the gradient step is not an approximate inverse Hessian but the exact gradient for a natural right-invariant metric, and prove in [Theorem 7.20](#) exactly when a separating solution is a locally asymptotically stable equilibrium of the mean dynamics underlying the online algorithm – which, for correctly specified sources, is precisely when [Corollary 7.4](#) says the model is identifiable. The same section then shows in [Corollary 7.32](#) how a causal ordering removes the last remaining ambiguities, giving the linear non-Gaussian acyclic model LiNGAM, and closes with a short survey of how the model has been generalised. [Section A](#) collects the proofs of the four classical results about characteristic functions that [Section 3](#) quotes: the uniqueness and inversion theorem, Lévy’s continuity theorem, Marcinkiewicz’ theorem and Cramér’s decomposition theorem. [Section B](#) proves [Theorem 4.2](#), the identifiability theorem of [Kagan et al. \(1973\)](#) on which [Section 4](#) onwards rest. The proof is a self-contained finite-difference argument: it needs only the distinguished logarithm, Marcinkiewicz’ theorem, and Fréchet’s functional equation, which we also prove. With these appendices in place, every result in these notes is proved here, with one exception: Bochner’s [Theorem 3.3](#), which we quote and which is used only once, in [Remark 3.8](#), where it could be replaced by a direct computation. Standard background is of course used throughout and is listed at the start of [Section A](#); [Section 7](#) additionally uses standard matrix calculus and, in [Theorem 7.20](#), the unstable-manifold theorem for maps.

How to read this note. The logical skeleton is short. Everything downstream of [Section 4](#) rests on the single result [Theorem 4.2](#), proved in [Section B](#), and branches from there into the two lines of development that the rest of the note pursues:



The upper branch strengthens the hypotheses on the sources and asks how much of the model this buys back; the lower branch fixes the hypotheses at their most convenient and asks how to estimate. A reader interested only in the algorithm may go directly from [Section 3](#) to [Section 7](#), taking [Corollary 7.4](#) on faith.

2 Notation and Conventions

Notation 2.1 (Random objects versus deterministic objects). All random objects are defined on one underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and are denoted by *capital* Latin letters: X for the observed random vector, Z for the source vector, E, V, G, H, N, U for the various (mostly Gaussian) noise and remainder terms. Deterministic matrices are capital Latin or Greek letters ($A, B, P, Q, R, W, \Theta, \Lambda, \Sigma$; in particular W always denotes the deterministic unmixing matrix of [Section 7](#), never a random vector), deterministic vectors and scalars are lower case ($a_j, b_l, \mu, \nu, \xi, \lambda_j, t$). Components of a random vector carry a subscript, $Z = [Z_1, \dots, Z_k]^\top$, and e_j denotes the j -th standard basis vector. Superscripts in brackets, $A^{(1)}, Z^{(2)}$, index the two competing representations of the same observed vector and are never exponents.

Notation 2.2 (Multi-indices and weak convergence). For a multi-index $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d$ we write $|\alpha| := \alpha_1 + \dots + \alpha_d$ for its order, and, for $x \in \mathbb{R}^d$ and a sufficiently differentiable $f: \mathbb{R}^d \rightarrow \mathbb{C}$,

$$x^\alpha := \prod_{j=1}^d x_j^{\alpha_j}, \quad \partial^\alpha f := \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}}, \quad (3)$$

with the convention $x^0 := 1$ and $\partial^0 f := f$. The symbol $\Rightarrow$ denotes weak convergence of laws: for random vectors X_n, X in $\mathbb{R}^d$ we write $X_n \Rightarrow X$ if $\mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)]$ for every bounded continuous $f: \mathbb{R}^d \rightarrow \mathbb{R}$.

Notation 2.3 (Distributional relations). For a random vector X in $\mathbb{R}^p$ we write $\mathcal{L}(X)$ for its law and

$$\varphi_X(t) := \mathbb{E}[\exp(it^\top X)], \quad t \in \mathbb{R}^p, \quad (4)$$

for its characteristic function. We reserve

- $X \sim \mathcal{N}(\mu, \Sigma)$ for “ X is distributed according to the law $\mathcal{N}(\mu, \Sigma)$ ”, and
- $X \stackrel{d}{=} Y$ for “ X and Y have the same law”, i.e. $\mathcal{L}(X) = \mathcal{L}(Y)$, equivalently $\varphi_X = \varphi_Y$.

Stochastic independence is written $X \perp\!\!\!\perp Y$. Mutual independence of a family $\{X_1, \dots, X_k\}$ always means independence of the whole family, not merely pairwise independence. Statements such as “ Z_j is non-constant” are always understood $\mathbb{P}$ -almost surely, i.e. $\mathcal{L}(Z_j)$ is not a Dirac measure δ_c .

Notation 2.4 (Degenerate Gaussian distributions). We call E a (possibly degenerate) *Gaussian* random vector, $E \sim \mathcal{N}(\mu, \Sigma)$ with $\mu \in \mathbb{R}^p$ and a positive semi-definite $\Sigma \in \mathbb{R}^{p \times p}$, if

$$\varphi_E(t) = \exp\left(i t^\top \mu - \frac{1}{2} t^\top \Sigma t\right), \quad t \in \mathbb{R}^p. \quad (5)$$

This includes the degenerate cases: Σ is allowed to be singular, and for $p = 1$, $\sigma^2 = 0$, we have $\mathcal{N}(\mu, 0) = \delta_\mu$, so that constants count as Gaussian. Whenever we want to exclude this case we say *non-degenerate*.

3 Characteristic Functions of Probability Measures on $\mathbb{R}^d$

Every argument in these notes is carried out at the level of characteristic functions. This section collects the facts we need, in the generality in which we need them. We assume familiarity with measure-theoretic probability – laws, weak convergence, product measures, Fubini’s theorem – and we cite the standard results rather than reproving them; everything that is specific to the independent component analysis (ICA) setting is proved in full. Standard references are [Klenke \(2020, Chapter 15\)](#) and [Kallenberg \(2021, Chapters 5–6\)](#) for the general theory, [Lukacs \(1970\)](#) for characteristic functions specifically, and [Kagan et al. \(1973, Chapter 1\)](#) for the characterisation-theoretic material.

3.1 Definition and elementary properties

Definition 3.1 (Characteristic function). Let μ be a probability measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$. Its *characteristic function* (c.f.) is the map

$$\varphi_\mu: \mathbb{R}^d \rightarrow \mathbb{C}, \quad \varphi_\mu(t) := \int_{\mathbb{R}^d} \exp(i t^\top x) \mu(dx). \quad (6)$$

For a random vector X in $\mathbb{R}^d$ with law $\mathcal{L}(X) = \mu$ we write $\varphi_X := \varphi_\mu$, so that $\varphi_X(t) = \mathbb{E}[\exp(i t^\top X)]$. The integral is well defined because $|\exp(i t^\top x)| = 1$, so the integrand is bounded and measurable.

Proposition 3.2 (Elementary properties). *Let X be a random vector in $\mathbb{R}^d$ with characteristic function φ_X . Then:*

- (i) $\varphi_X(0) = 1$ and $|\varphi_X(t)| \leq 1$ for all $t \in \mathbb{R}^d$;
- (ii) φ_X is uniformly continuous on $\mathbb{R}^d$;
- (iii) $\varphi_X(-t) = \overline{\varphi_X(t)}$, and φ_X is real valued if and only if $X \stackrel{d}{=} -X$;
- (iv) φ_X is positive semi-definite: for all $n \in \mathbb{N}$, all $t_1, \dots, t_n \in \mathbb{R}^d$ and all $z_1, \dots, z_n \in \mathbb{C}$,

$$\sum_{j,l=1}^n z_j \overline{z_l} \varphi_X(t_j - t_l) \geq 0; \quad (7)$$

- (v) (affine images) for $A \in \mathbb{R}^{m \times d}$ and $b \in \mathbb{R}^m$,

$$\varphi_{AX+b}(t) = \exp(i t^\top b) \cdot \varphi_X(A^\top t), \quad t \in \mathbb{R}^m; \quad (8)$$

- (vi) (independent sums) if $X \perp\!\!\!\perp Y$ are random vectors in $\mathbb{R}^d$, then

$$\varphi_{X+Y}(t) = \varphi_X(t) \cdot \varphi_Y(t), \quad t \in \mathbb{R}^d; \quad (9)$$

(vii) (one-dimensional projections) for $u \in \mathbb{R}^d$ and $s \in \mathbb{R}$,

$$\varphi_{u^\top X}(s) = \varphi_X(su). \quad (10)$$

Proof. (i) is immediate. For (ii) note that $|\varphi_X(t+h) - \varphi_X(t)| \leq \mathbb{E}[|\exp(i h^\top X) - 1|]$, whose right hand side does not depend on t and tends to 0 as $h \rightarrow 0$ by dominated convergence. (iii): the identity $\exp(i t^\top X) = \exp(-i t^\top X)$ gives $\varphi_X(-t) = \overline{\varphi_X(t)}$, and hence φ_X is real as soon as $X \stackrel{d}{=} -X$, since then $\varphi_X(-t) = \varphi_{-X}(t) = \varphi_X(t)$. Conversely, if φ_X is real then $\varphi_{-X}(t) = \varphi_X(-t) = \overline{\varphi_X(t)} = \varphi_X(t)$ for every t , so $X \stackrel{d}{=} -X$ by [Theorem 3.4](#) (i) below, whose proof does not use this proposition. For (iv) we argue $\sum_{j,l} z_j \bar{z}_l \exp(i(t_j - t_l)^\top X) = |\sum_j z_j \exp(i t_j^\top X)|^2 \geq 0$ after taking expectations. For (v), $\mathbb{E}[\exp(i t^\top (AX + b))] = \exp(i t^\top b) \mathbb{E}[\exp(i (A^\top t)^\top X)]$. (vi) is the factorisation of the expectation of a product of independent bounded random variables, and (vii) is the special case $A = u^\top$, $b = 0$ of (v). $\square$

Properties (8), (9) and (10) are the three identities that carry all the ICA arguments: a linear ICA model is by definition an independent sum of affine images, so its characteristic function factorises into the characteristic functions of the sources evaluated along the columns of the mixing matrix.

Theorem 3.3 (Bochner’s theorem; see [Klenke, 2020](#), Chapter 15; [Lukacs, 1970](#), Section 4.2). *A function $\phi: \mathbb{R}^d \rightarrow \mathbb{C}$ is the characteristic function of a probability measure on $\mathbb{R}^d$ if and only if ϕ is continuous, $\phi(0) = 1$, and ϕ is positive semi-definite in the sense of [Proposition 3.2](#) (iv).*

Beyond the standard background listed at the start of [Section A](#), [Theorem 3.3](#) is the only result from the theory of characteristic functions that any proof in these notes uses without our having proved it: the difficult direction is a genuinely deep theorem, and reproducing it here would take us too far afield. (Two further classical facts are quoted in remarks – Pólya’s criterion in [Remark 3.8](#) and the Lukacs characterisation in [Remark 3.16](#) – but no proof depends on either.) It is what makes statements of the form “ ϕ is a characteristic function” verifiable without exhibiting the underlying law. We shall repeatedly need such statements – for instance in [Theorem 6.7](#), where a supremum over admissible Gaussian factors is taken – although in practice we usually certify them through the more convenient [Theorem 3.7](#) rather than by verifying positive semi-definiteness. The one place where we do verify positive semi-definiteness directly is [Remark 3.8](#).

3.2 How characteristic functions determine distributions

Theorem 3.4 (Uniqueness and inversion). *Let μ, ν be probability measures on $\mathbb{R}^d$.*

(i) *If $\varphi_\mu = \varphi_\nu$ on all of $\mathbb{R}^d$, then $\mu = \nu$. Equivalently, for random vectors X, Y in $\mathbb{R}^d$,*

$$X \stackrel{d}{=} Y \iff \varphi_X = \varphi_Y. \quad (11)$$

(ii) *If in addition $\varphi_\mu \in L^1(\mathbb{R}^d)$, then μ has a bounded continuous Lebesgue density given by the inversion formula*

$$f(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} \exp(-i t^\top x) \varphi_\mu(t) dt. \quad (12)$$

The proof is in [Section A.1](#); see also [Klenke, 2020](#), Chapter 15 and [Kallenberg, 2021](#), Chapter 5.

Theorem 3.5 (Cramér–Wold device; see [Klenke, 2020](#), Chapter 15). *Let X, Y be random vectors in $\mathbb{R}^d$. Then $X \stackrel{d}{=} Y$ if and only if $u^\top X \stackrel{d}{=} u^\top Y$ for every $u \in \mathbb{R}^d$. Likewise, $X_n \Rightarrow X$ weakly if and only if $u^\top X_n \Rightarrow u^\top X$ for every $u \in \mathbb{R}^d$.*

Proof. By [Eq. \(10\)](#) we have $\varphi_{u^\top X}(1) = \varphi_X(u)$, so knowing the laws of all one-dimensional projections is the same as knowing φ_X pointwise, and [Theorem 3.4 \(i\)](#) applies.

For the second statement, suppose first $X_n \Rightarrow X$. By [Theorem 3.7 \(i\)](#) in $\mathbb{R}^d$, $\varphi_{X_n} \rightarrow \varphi_X$ pointwise, so $\varphi_{u^\top X_n}(s) = \varphi_{X_n}(su) \rightarrow \varphi_X(su) = \varphi_{u^\top X}(s)$ for every s ; the limit is continuous at $s = 0$, so [Theorem 3.7 \(ii\)](#) in $d = 1$ gives $u^\top X_n \Rightarrow u^\top X$. Conversely, suppose $u^\top X_n \Rightarrow u^\top X$ for every u . By [Theorem 3.7 \(i\)](#) in $d = 1$ and [Eq. \(10\)](#) at $s = 1$ we get $\varphi_{X_n}(u) \rightarrow \varphi_X(u)$ for every $u \in \mathbb{R}^d$, and φ_X is continuous at the origin, so [Theorem 3.7 \(ii\)](#) yields a random vector X' with $\varphi_{X'} = \varphi_X$ and $X_n \Rightarrow X'$; by [Theorem 3.4 \(i\)](#), $X' \stackrel{d}{=} X$ and hence $X_n \Rightarrow X$. $\square$

Remark 3.6 (Two different theorems of Cramér). [Theorem 3.5](#) is the *Cramér–Wold device*, a statement about reducing $\mathbb{R}^d$ to $\mathbb{R}$. It should not be confused with *Cramér’s decomposition theorem*, [Theorem 3.20](#) below, which says that a Gaussian law has only Gaussian factors. Both are used in these notes, for entirely different purposes.

Theorem 3.7 (Lévy’s continuity theorem). *Let $(X_n)_{n \in \mathbb{N}}$ be random vectors in $\mathbb{R}^d$.*

- (i) *If $X_n \Rightarrow X$ weakly, then $\varphi_{X_n} \rightarrow \varphi_X$ pointwise.*
- (ii) *Conversely, if $\varphi_{X_n}(t) \rightarrow \phi(t)$ for every $t \in \mathbb{R}^d$ and ϕ is continuous at $t = 0$, then ϕ is the characteristic function of a random vector X and $X_n \Rightarrow X$ weakly.*

The proof is in [Section A.2](#); see also [Klenke, 2020](#), Section 15.3.

The continuity requirement in (ii) is not a technicality: for $X_n \sim \mathcal{N}(0, n)$ we have $\varphi_{X_n}(t) = \exp(-nt^2/2) \rightarrow \mathbf{1}_{\{t=0\}}$ pointwise, and the limit – discontinuous at the origin – is not a characteristic function; the mass has escaped to infinity.

Remark 3.8 (Local equality of characteristic functions is *not* enough). [Theorem 3.4](#) requires $\varphi_\mu = \varphi_\nu$ on *all* of $\mathbb{R}^d$. Agreement on a neighbourhood of the origin does not suffice. The classical counterexample goes back to [Pólya \(1949\)](#); in the form given by [Feller \(1971, Chapter XV\)](#) and [Lukacs \(1970, p. 85\)](#) it reads as follows. Let

$$\phi_1(t) := (1 - |t|)_+, \quad (13)$$

the “tent function”. It *is* a characteristic function. Indeed, with $h := \mathbf{1}_{[-1/2, 1/2]}$ one has, for every $t \in \mathbb{R}$,

$$(h * h)(t) = \int_{\mathbb{R}} h(u) h(t - u) du = (1 - |t|)_+ = \phi_1(t), \quad (14)$$

and since h is real-valued and even, $\phi_1(t_j - t_l) = \int_{\mathbb{R}} h(t_j - u) h(t_l - u) du$, whence for all $n \in \mathbb{N}$, $t_1, \dots, t_n \in \mathbb{R}$ and $z_1, \dots, z_n \in \mathbb{C}$

$$\sum_{j,l=1}^n z_j \overline{z_l} \phi_1(t_j - t_l) = \int_{\mathbb{R}} \left| \sum_{j=1}^n z_j h(t_j - u) \right|^2 du \geq 0. \quad (15)$$

So ϕ_1 is continuous, $\phi_1(0) = 1$ and positive semi-definite, and [Theorem 3.3](#) certifies that $\phi_1 = \varphi_{\mu_1}$ for a probability measure μ_1 on $\mathbb{R}$. (Alternatively one may invoke Pólya’s criterion, [Feller \(1971, Section XV.2\)](#): every continuous even $\phi: \mathbb{R} \rightarrow \mathbb{R}$ that is convex on $[0, \infty)$ and satisfies $\phi(0) = 1$ and $\phi(t) \rightarrow 0$ as $t \rightarrow \infty$ is a characteristic function; the tent function is the

standard example, being the maximum of two affine functions on $[0, \infty)$.) Now [Theorem 3.4](#) (ii) may be applied, since $\phi_1 \in L^1(\mathbb{R})$: the law μ_1 is absolutely continuous with density

$$f(x) = \frac{1}{2\pi} \int_{-1}^1 (1 - |t|) e^{-itx} dt = \frac{1}{\pi} \int_0^1 (1 - t) \cos(tx) dt = \frac{1 - \cos x}{\pi x^2}. \quad (16)$$

Let ϕ_2 be the 2-periodic extension of $1 - |t|$ from $[-1, 1]$ to all of $\mathbb{R}$. Expanding the resulting triangular wave into its Fourier series gives

$$\phi_2(t) = \frac{1}{2} + \sum_{n \text{ odd}} \frac{4}{\pi^2 n^2} \cos(n\pi t), \quad (17)$$

which is the characteristic function of the *discrete* law with mass $\frac{1}{2}$ at 0 and mass $2/(\pi^2 n^2)$ at each of $\pm n\pi$ for odd $n \geq 1$; the masses do sum to $\frac{1}{2} + \frac{4}{\pi^2} \sum_{n \text{ odd}} n^{-2} = \frac{1}{2} + \frac{4}{\pi^2} \cdot \frac{\pi^2}{8} = 1$. Then $\phi_1 = \phi_2$ on $[-1, 1]$, while the two laws are as different as they could be: one is absolutely continuous, the other purely atomic.

This is why every local statement in these notes is stated with the explicit qualifier “in a neighbourhood of the origin”, and why such statements are never silently upgraded to statements about laws. The upgrade is legitimate under an analyticity hypothesis; see [Corollary 3.14](#).

3.3 Moments, the distinguished logarithm, and cumulants

Proposition 3.9 (Moments and derivatives). *Let X be a random vector in $\mathbb{R}^d$ and $n \in \mathbb{N}$ with $\mathbb{E}[\|X\|^n] < \infty$. Then $\varphi_X \in C^n(\mathbb{R}^d; \mathbb{C})$ and, for every multi-index α with $|\alpha| \leq n$ and every $t \in \mathbb{R}^d$,*

$$\partial^\alpha \varphi_X(t) = i^{|\alpha|} \mathbb{E}[X^\alpha e^{it^\top X}]; \quad \text{in particular} \quad \partial^\alpha \varphi_X(0) = i^{|\alpha|} \mathbb{E}[X^\alpha]. \quad (18)$$

In the scalar case $d = 1$ this gives the Taylor expansion

$$\varphi_X(t) = \sum_{m=0}^n \frac{\mathbb{E}[X^m]}{m!} (it)^m + o(|t|^n), \quad t \rightarrow 0. \quad (19)$$

Proof. We prove [Eq. \(18\)](#) by induction on $|\alpha|$, the case $\alpha = 0$ being the definition. Suppose it holds for α with $|\alpha| < n$ and let $j \in \{1, \dots, d\}$. For $h \neq 0$,

$$\frac{\partial^\alpha \varphi_X(t + he_j) - \partial^\alpha \varphi_X(t)}{h} = i^{|\alpha|} \mathbb{E} \left[X^\alpha e^{it^\top X} \frac{e^{ihX_j} - 1}{h} \right], \quad (20)$$

and the integrand is dominated in modulus by $|X^\alpha| |X_j| \leq \|X\|^{|\alpha|+1}$, which is integrable because $|\alpha| + 1 \leq n$; here we used $|e^{ihx} - 1| \leq |hx|$. Since $(e^{ihX_j} - 1)/h \rightarrow iX_j$ pointwise as $h \rightarrow 0$, dominated convergence gives $\partial_j \partial^\alpha \varphi_X(t) = i^{|\alpha|+1} \mathbb{E}[X^\alpha X_j e^{it^\top X}]$, which is the claim for the multi-index $\alpha + e_j$. The same domination shows that each such derivative is continuous in t , so $\varphi_X \in C^n$. [Eq. \(19\)](#) is Taylor’s theorem with Peano remainder applied to the C^n function φ_X , together with [Eq. \(18\)](#) at $t = 0$. $\square$

A partial converse holds, and it needs no moment hypothesis at all: for a real-valued random variable Y and $m \in \mathbb{N}$, if φ_Y is $2m$ times differentiable at the single point 0, then $\mathbb{E}[Y^{2m}] < \infty$; the odd moments up to order $2m$ are then finite by Lyapunov’s inequality. This is [Lemma A.3](#), proved in [Section A](#).

All our arguments manipulate *logarithms* of characteristic functions. Since a characteristic function is complex valued and may vanish, this requires a small amount of care, which the following lemma settles once and for all.

Lemma 3.10 (Distinguished logarithm). *Let X be a random vector in $\mathbb{R}^d$. (Uniqueness.) If $U \ni 0$ is a connected open subset of $\mathbb{R}^d$ and $\psi, \tilde{\psi}: U \rightarrow \mathbb{C}$ are continuous with $\psi(0) = \tilde{\psi}(0) = 0$ and $\exp \psi = \exp \tilde{\psi} = \varphi_X$ on U , then $\psi = \tilde{\psi}$; so the germ at the origin is well defined and we may compare distinguished logarithms on any such U . (Existence.) There exist $\delta > 0$ and a continuous function $\psi_X: B_\delta(0) \rightarrow \mathbb{C}$ with $\psi_X(0) = 0$ and*

$$\varphi_X(t) = \exp(\psi_X(t)), \quad t \in B_\delta(0), \quad (21)$$

where $B_\delta(0) := \{t \in \mathbb{R}^d : \|t\| < \delta\}$. We call ψ_X the cumulant generating function of X (near the origin). If $\mathbb{E}[\|X\|^n] < \infty$, then $\psi_X \in C^n$ on $B_\delta(0)$.

Proof. By Proposition 3.2 (i)–(ii) the function φ_X is continuous with $\varphi_X(0) = 1$, so there is a $\delta > 0$ with $|\varphi_X(t) - 1| < 1$ for all $t \in B_\delta(0)$. Hence φ_X maps $B_\delta(0)$ into the open disc of radius 1 around 1, which is contained in the slit plane $\mathbb{C} \setminus (-\infty, 0]$ on which the principal branch Log of the complex logarithm is defined and continuous. Put $\psi_X := \text{Log} \circ \varphi_X$; then ψ_X is continuous, $\psi_X(0) = \text{Log} 1 = 0$ and $\exp \circ \psi_X = \varphi_X$.

For uniqueness, let $U \ni 0$ be any connected open subset of $\mathbb{R}^d$ and let $\psi, \tilde{\psi}: U \rightarrow \mathbb{C}$ be continuous with $\psi(0) = \tilde{\psi}(0) = 0$ and $\exp \psi = \exp \tilde{\psi} = \varphi_X$ on U . Then $\exp(\tilde{\psi} - \psi) \equiv 1$ on U , so $(\tilde{\psi} - \psi)/(2\pi i)$ is a continuous integer-valued function on the connected set U vanishing at the origin, hence identically 0. This covers in particular the case $U = B_\delta(0)$, $\psi = \psi_X$. The regularity statement follows from Proposition 3.9 because Log is holomorphic on the slit plane. $\square$

Definition 3.11 (Cumulants). Let X be a real-valued random variable with $\mathbb{E}[\|X\|^n] < \infty$ and let ψ_X be as in Lemma 3.10. The *cumulants* $\kappa_1(X), \dots, \kappa_n(X)$ of X are the coefficients in the expansion

$$\psi_X(t) = \sum_{m=1}^n \frac{\kappa_m(X)}{m!} (it)^m + o(t^n), \quad t \rightarrow 0, \quad (22)$$

equivalently $\kappa_m(X) = i^{-m} \psi_X^{(m)}(0)$. The first four cumulants are

$$\kappa_1(X) = \mathbb{E}[X], \quad \kappa_2(X) = \text{Var}(X), \quad (23)$$

$$\kappa_3(X) = \mathbb{E}[(X - \mathbb{E}X)^3], \quad \kappa_4(X) = \mathbb{E}[(X - \mathbb{E}X)^4] - 3 \text{Var}(X)^2. \quad (24)$$

Proposition 3.12 (Cumulant calculus). *Let X, Y be real-valued random variables with finite n -th absolute moments and let $a, b \in \mathbb{R}$. Then, for $m \leq n$,*

(i) (additivity) if $X \perp\!\!\!\perp Y$ then $\kappa_m(X + Y) = \kappa_m(X) + \kappa_m(Y)$;

(ii) (homogeneity) $\kappa_m(aX) = a^m \kappa_m(X)$;

(iii) (translation invariance) $\kappa_m(X + b) = \kappa_m(X)$ for $m \geq 2$, and $\kappa_1(X + b) = \kappa_1(X) + b$.

Proof. By Eq. (9) and Lemma 3.10 we have $\psi_{X+Y} = \psi_X + \psi_Y$ on a neighbourhood of the origin, since both sides are continuous, vanish at 0 and exponentiate to φ_{X+Y} ; comparing Taylor coefficients gives (i). For (ii), $\varphi_{aX}(t) = \varphi_X(at)$ by Eq. (8), so $\psi_{aX}(t) = \psi_X(at)$ and the m -th coefficient picks up a^m . For (iii), $\psi_{X+b}(t) = \psi_X(t) + ibt$, which changes only the coefficient of $(it)^1$. $\square$

Additivity is the reason cumulants, rather than moments, are the natural bookkeeping device for ICA: a mixture is an independent sum, so its cumulants are sums of the source cumulants.

3.4 Analytic characteristic functions

Theorem 3.13 (Analyticity in a strip). *Let X be a random vector in $\mathbb{R}^d$ and suppose that there is an $a > 0$ with*

$$\mathbb{E}[\exp(a\|X\|)] < \infty. \quad (25)$$

Then the map

$$z \mapsto \mathbb{E}[\exp(iz^\top X)] \quad (26)$$

is well defined and holomorphic on the tube $S_a := \{z \in \mathbb{C}^d : \|\operatorname{Im} z\| < a\}$, and it restricts to φ_X on $\mathbb{R}^d$. If Eq. (25) holds for every $a > 0$, then φ_X extends to an entire function on $\mathbb{C}^d$.

Proof. We give the argument for $d = 1$, which is the only case used below: every appeal to this theorem in these notes is to a real-valued random variable, and the one place where a multivariate statement is wanted, [Corollary 3.14](#), is reduced to one variable via [Theorem 3.5](#). For general d the same argument applied in each coordinate separately, with $|\cdot|$ replaced by $\|\cdot\|$, yields *separate* holomorphy together with local boundedness on S_a ; joint holomorphy then follows from Osgood's lemma (or, dispensing with boundedness, from Hartogs' theorem). For $z = s + iu$ with $|u| < a$ we have $|\exp(izX)| = \exp(-uX) \leq \exp(a|X|)$, which is integrable by [Eq. \(25\)](#); so $F(z) := \mathbb{E}[\exp(izX)]$ is well defined on the strip S_a , and it is continuous there by dominated convergence. For any closed triangle $\Delta \subseteq S_a$, Fubini's theorem applies because $(z, \omega) \mapsto \exp(izX(\omega))$ is jointly measurable and dominated by the integrable $\exp(a|X|)$ uniformly on the compact set $\partial\Delta$; hence

$$\oint_{\partial\Delta} F(z) dz = \mathbb{E} \left[\oint_{\partial\Delta} \exp(izX) dz \right] = 0, \quad (27)$$

the inner integral vanishing because $z \mapsto \exp(izX)$ is entire for each fixed value of X . By Morera's theorem F is holomorphic on S_a , and $F|_{\mathbb{R}} = \varphi_X$ by construction. $\square$

Corollary 3.14 (Rigidity of analytic characteristic functions). *Let X, Y be random vectors in $\mathbb{R}^d$ both satisfying [Eq. \(25\)](#) for some $a > 0$. If $\varphi_X = \varphi_Y$ on some neighbourhood of the origin in $\mathbb{R}^d$, then $X \stackrel{d}{=} Y$.*

Proof. Replacing a by the smaller of the two constants we may assume that X and Y satisfy [Eq. \(25\)](#) with the same $a > 0$, and let $\delta > 0$ be such that $\varphi_X = \varphi_Y$ on $B_\delta(0)$. Fix $u \in \mathbb{R}^d \setminus \{0\}$. By [Eq. \(10\)](#),

$$\varphi_{u^\top X}(s) = \varphi_X(su) = \varphi_Y(su) = \varphi_{u^\top Y}(s), \quad |s| < \delta/\|u\|. \quad (28)$$

Moreover $|u^\top X| \leq \|u\|\|X\|$, so the real-valued random variables $u^\top X$ and $u^\top Y$ satisfy [Eq. \(25\)](#) with the constant $a/\|u\|$, and by [Theorem 3.13](#) their characteristic functions extend holomorphically to a strip in $\mathbb{C}$. Two functions holomorphic on a connected open subset of $\mathbb{C}$ which agree on a real interval agree on all of it, by the identity theorem in one complex variable; hence $\varphi_{u^\top X} = \varphi_{u^\top Y}$ on all of $\mathbb{R}$ and therefore $u^\top X \stackrel{d}{=} u^\top Y$ by [Theorem 3.4](#) (i). Since u was arbitrary, [Theorem 3.5](#) gives $X \stackrel{d}{=} Y$. $\square$

We deliberately reduced to one complex variable here. In several variables the *naive* form of the identity theorem fails: agreement of two holomorphic functions on a set with accumulation points is not sufficient, as $f(z_1, z_2) = z_1$, vanishing on the uncountable set $\{0\} \times \mathbb{C}$, shows. A d -dimensional argument is nevertheless available – at a real point of a

real neighbourhood on which h vanishes, every complex partial derivative of h agrees with the corresponding real one and hence vanishes, so $h \equiv 0$ on a polydisc and then on all of the connected S_a – but the reduction to one variable is shorter and is all we need.

Corollary 3.14 is exactly what **Remark 3.8** rules out in general: the tent function of that remark is the characteristic function of a law whose density is of order x^{-2} , which has no exponential moments at all.

Proposition 3.15 (Ridge property). *Let $d = 1$ and let X satisfy [Eq. \(25\)](#) for some $a > 0$, and denote again by φ_X the holomorphic extension of [Theorem 3.13](#). Then for all $s \in \mathbb{R}$ and $|u| < a$,*

$$|\varphi_X(s + iu)| \leq \varphi_X(iu) = \mathbb{E}[\exp(-uX)] \in \mathbb{R}_{>0}, \quad (29)$$

that is, on each horizontal line the modulus is maximal on the imaginary axis.

Proof. On the strip we have $\varphi_X(z) = \mathbb{E}[\exp(izX)]$, so

$$|\varphi_X(s + iu)| = |\mathbb{E}[e^{isX} e^{-uX}]| \leq \mathbb{E}[|e^{isX}| e^{-uX}] = \mathbb{E}[e^{-uX}] = \varphi_X(iu), \quad (30)$$

which is real and strictly positive because $e^{-uX} > 0$. $\square$

Remark 3.16 (Ridge property and the moment generating function). Evaluating the holomorphic extension of [Theorem 3.13](#) at $z = -iu$ gives $\varphi_X(-iu) = \mathbb{E}[\exp(uX)] = M_X(u)$, the moment generating function of X ; [Eq. \(29\)](#) then says that $|\varphi_X|$ is maximal on each horizontal line at its purely imaginary point. If φ_X is analytic in *some* neighbourhood of the origin – equivalently, if 0 lies in the interior of $\{u : M_X(u) < \infty\}$ – then its maximal strip of analyticity is exactly the one cut out by that interior ([Lukacs, 1970](#), Theorem 7.1.1); the moment generating function may well be finite at a boundary abscissa while φ_X is analytic only on the open strip. Without the hypothesis at the origin the statement is false: a law with an exponential right tail and a $|x|^{-3}$ left tail has $\{M_X < \infty\} = [0, 1)$ with non-empty interior, yet $\mathbb{E}[X^2] = \infty$, so φ_X is not even twice differentiable at the origin, let alone analytic there.

The following theorem is the single most important analytic input for these notes. It says that the exponential of a polynomial is a characteristic function only in the Gaussian case.

Theorem 3.17 (Marcinkiewicz’ theorem, [Marcinkiewicz, 1939](#)). *Let Y be a real-valued random variable whose characteristic function is of the form*

$$\varphi_Y(t) = \exp(g(t)) \quad (31)$$

in a neighbourhood of the origin, with a (complex) polynomial g . Then $\deg(g) \leq 2$; more precisely, after subtracting from g the constant $g(0) \in 2\pi i \mathbb{Z}$, which changes $\exp(g)$ not at all and the degree only in the trivial case of a constant g ,

$$g(t) = -\frac{1}{2}\sigma^2 t^2 + i\mu t, \quad \sigma^2 \in \mathbb{R}_{\geq 0}, \quad \mu \in \mathbb{R}, \quad (32)$$

and consequently Y has a (possibly degenerate) Gaussian distribution, $Y \sim \mathcal{N}(\mu, \sigma^2)$, where the degenerate case $\sigma^2 = 0$ means $Y \sim \mathcal{N}(\mu, 0) = \delta_\mu$.

The proof is in [Section A.4](#); see also [Kagan et al. \(1973, Lemma 1.4.2\)](#) and [Lukacs \(1970, Section 7.3\)](#).

In the language of [Definition 3.11](#): there is no probability distribution whose cumulant generating function *equals* a polynomial of degree 3 or higher near the origin. The superficially similar statement that the vanishing of all cumulants of order $> n$ already forces the vanishing

of all cumulants of order > 2 is also true, but it is not a restatement of [Theorem 3.17](#): vanishing Taylor coefficients do not by themselves make ψ_Y a polynomial, and an additional analytic argument is required, namely a growth bound on the moments strong enough to make φ_Y entire. For $n = 2$ – the only case we shall need – such a bound is immediate, because then the moments *are* Gaussian moments; we give the argument inside the proof of [Theorem 3.19](#), in the implication (v) $\Rightarrow$ (ii). For general n one argues instead as follows. The number of set partitions of $\{1, \dots, m\}$ all of whose blocks have size at most n is bounded by $C(n)^m (m!)^{1-1/n}$, so the moment–cumulant formula gives $|\mathbb{E}[Y^m]| \leq C'^m (m!)^{1-1/n}$ with a constant $C' \geq 1$ depending on n and on the finitely many non-zero cumulants. Lyapunov’s inequality turns this into a bound of the same shape for the absolute moments, with a larger constant: $\mathbb{E}|Y|^m \leq (\mathbb{E}[Y^{m+1}])^{m/(m+1)} \leq (C')^{m+1} ((m+1)!)^{1-1/n}$ for odd m , and since $((m+1)!)^{1-1/n}/m! = (m+1)^{1-1/n} (m!)^{-1/n}$ one gets $\sum_{m \geq 0} a^m \mathbb{E}|Y|^m/m! < \infty$ for every $a > 0$. By Tonelli’s theorem that sum equals $\mathbb{E}[\exp(a|Y|)]$, so [Eq. \(25\)](#) holds for every $a > 0$ and [Theorem 3.13](#) makes φ_Y entire; then ψ_Y is holomorphic near 0 and equal to its Taylor polynomial there, and [Theorem 3.17](#) applies. We shall not use this generalisation.

Remark 3.18 (How [Theorem 3.17](#) will be used). We shall use [Theorem 3.17](#) in the following two guises.

- (i) If two real-valued random variables Y_1, Y_2 satisfy

$$\varphi_{Y_2}(\lambda t) = \varphi_{Y_1}(t) \exp(g(t)) \quad (33)$$

near the origin with a polynomial g and a scalar $\lambda \in \mathbb{R} \setminus \{0\}$, then Y_2 is Gaussian if and only if Y_1 is Gaussian. Indeed, by [Eq. \(8\)](#) the left hand side is $\varphi_{\lambda Y_2}(t)$, so it suffices to treat $\lambda = 1$ and then to note that λY_2 is Gaussian if and only if Y_2 is, again by [Eq. \(8\)](#) and $\lambda \neq 0$. For $\lambda = 1$: if Y_1 is Gaussian then φ_{Y_1} is itself the exponential of a polynomial of degree ≤ 2 , hence so is φ_{Y_2} near the origin and [Theorem 3.17](#) applies; the converse follows by exchanging the roles and replacing g by $-g$.

- (ii) The sum of an independent non-Gaussian random variable and a Gaussian random variable is again non-Gaussian: if $Y = Y_1 + Y_2$ with $Y_1 \perp\!\!\!\perp Y_2$ and Y_2 Gaussian, then $\varphi_{Y_1}(t) = \varphi_Y(t) \varphi_{Y_2}(t)^{-1}$, and $\varphi_{Y_2}^{-1}$ is the exponential of a polynomial of degree ≤ 2 ; so if Y were Gaussian, Y_1 would be Gaussian as well.

Note that φ_{Y_2} has no zeros for Gaussian Y_2 , so the above quotients are well defined on all of $\mathbb{R}$.

3.5 Gaussian, sub-Gaussian and super-Gaussian distributions

Theorem 3.19 (Characterisation of the Gaussian law). *Let X be a random vector in $\mathbb{R}^d$. The following are equivalent.*

- (i) $X \sim \mathcal{N}(\mu, \Sigma)$ is (possibly degenerate) Gaussian in the sense of [Notation 2.4](#);
- (ii) $\varphi_X(t) = \exp(i t^\top \mu - \frac{1}{2} t^\top \Sigma t)$ for all $t \in \mathbb{R}^d$, for some $\mu \in \mathbb{R}^d$ and some positive semi-definite Σ ;
- (iii) every one-dimensional projection $u^\top X$, $u \in \mathbb{R}^d$, is a (possibly degenerate) Gaussian real-valued random variable.

For $d = 1$ these are further equivalent to each of:

- (iv) ψ_X coincides with a polynomial on a neighbourhood of the origin;
- (v) X has moments of all orders and $\kappa_m(X) = 0$ for all $m \geq 3$.

Proof. (i) $\Leftrightarrow$ (ii) is [Notation 2.4](#) combined with [Theorem 3.4](#) (i).

(ii) $\Rightarrow$ (iii): [Eq. \(10\)](#) gives

$$\varphi_{u^\top X}(s) = \exp\left(is u^\top \mu - \frac{1}{2}s^2 u^\top \Sigma u\right), \quad (34)$$

which is the characteristic function of $\mathcal{N}(u^\top \mu, u^\top \Sigma u)$. (iii) $\Rightarrow$ (ii): put $\mu_u := \mathbb{E}[u^\top X]$ and $\sigma_u^2 := \text{Var}(u^\top X)$, which are finite by (iii); then $\varphi_X(u) = \varphi_{u^\top X}(1) = \exp(i\mu_u - \frac{1}{2}\sigma_u^2)$, and $u \mapsto \mu_u$ is linear while $u \mapsto \sigma_u^2$ is a positive semi-definite quadratic form, so $\mu_u = u^\top \mu$ and $\sigma_u^2 = u^\top \Sigma u$ for $\mu := \mathbb{E}[X]$ and $\Sigma := \text{Cov}(X)$. (ii) $\Rightarrow$ (iv) is clear, and (iv) $\Rightarrow$ (ii) is [Marcinkiewicz's Theorem 3.17](#). (ii) $\Rightarrow$ (v) is immediate from [Eq. \(22\)](#).

For (v) $\Rightarrow$ (ii) some care is needed, because [Eq. \(22\)](#) is only an asymptotic expansion: knowing all Taylor coefficients of ψ_X does *not* by itself make ψ_X a polynomial, so [Theorem 3.17](#) is not directly applicable. Instead, put $\mu := \kappa_1(X)$ and $\sigma^2 := \kappa_2(X)$. The moment-cumulant relations express $\mathbb{E}[X^m]$ as a universal polynomial in $\kappa_1(X), \dots, \kappa_m(X)$; with $\kappa_m(X) = 0$ for all $m \geq 3$ these are exactly the moments of $\mathcal{N}(\mu, \sigma^2)$. Since $\exp(a|x|) \leq \exp(ax) + \exp(-ax)$, monotone convergence gives, for every $a > 0$,

$$\mathbb{E}[\exp(a|X|)] \leq 2 \sum_{k \geq 0} \frac{a^{2k}}{(2k)!} \mathbb{E}[X^{2k}] = 2 \sum_{k \geq 0} \frac{a^{2k}}{(2k)!} \mathbb{E}[N^{2k}] < \infty, \quad (35)$$

where $N \sim \mathcal{N}(\mu, \sigma^2)$. The last series converges because it is dominated by $2 \mathbb{E}[\exp(a|N|)]$, which is finite. So X satisfies [Eq. \(25\)](#) for every $a > 0$, and by [Theorem 3.13](#) its characteristic function is entire, hence equal to its everywhere convergent Taylor series about the origin. By [Proposition 3.9](#) that series is determined by the moments of X , which are those of N , so $\varphi_X = \varphi_N$, which is (ii). $\square$

Theorem 3.20 (Cramér's decomposition theorem, [Cramér, 1936](#)). *If $Y_1 \perp\!\!\!\perp Y_2$ are real-valued random variables such that $Y_1 + Y_2$ is Gaussian, then both Y_1 and Y_2 are Gaussian (possibly degenerate).*

The proof is in [Section A.5](#); see also [Kagan et al., 1973](#), Theorem 1.1.1.

[Theorems 3.17](#) and [3.20](#) are dual to each other and together explain the special role of the Gaussian law: it cannot be built out of non-Gaussian independent pieces, and it cannot be approached by “almost polynomial” cumulant generating functions. The next definition quantifies the deviation from normality that ICA exploits.

Definition 3.21 (Excess kurtosis; sub- and super-Gaussian). Let X be a real-valued random variable with $\mathbb{E}[X^4] < \infty$ and $\sigma^2 := \text{Var}(X) \in (0, \infty)$. Its *excess kurtosis* is

$$\text{kurt}(X) := \frac{\kappa_4(X)}{\sigma^4} = \frac{\mathbb{E}[(X - \mathbb{E}X)^4]}{\sigma^4} - 3. \quad (36)$$

We call X

- *super-Gaussian* (or *leptokurtic*) if $\text{kurt}(X) > 0$,
- *sub-Gaussian* (or *platykurtic*) if $\text{kurt}(X) < 0$,
- *mesokurtic* if $\text{kurt}(X) = 0$.

This is the convention of the ICA literature ([Comon, 1994](#); [Hyvärinen et al., 2001](#)); see [Remark 3.25](#) for the clash with the concentration-theoretic use of the same words, and [Remark 3.24](#) for what the sign of kurt does and does not say about tails. For the laws customarily used as ICA source models the picture is the familiar one: the super-Gaussian

ones are peaked at the mode and heavy tailed – the Laplace and Student laws used to model the “sparse” sources of natural images and audio – while the sub-Gaussian ones are flat-topped, the uniform law being the prototype. [Remark 3.24](#) shows that this correspondence is a property of those particular families rather than a general implication.

Proposition 3.22 (Kurtosis through the characteristic function). *Let X be a real-valued random variable with $\mathbb{E}[X^4] < \infty$. Then*

$$\kappa_4(X) = \psi_X^{(4)}(0) = \frac{d^4}{dt^4} \text{Log } \varphi_X(t) \Big|_{t=0}, \quad (37)$$

and for independent real-valued random variables $X_1, \dots, X_k$ with finite fourth moments and weights $w \in \mathbb{R}^k$,

$$\kappa_4\left(\sum_{j=1}^k w_j X_j\right) = \sum_{j=1}^k w_j^4 \kappa_4(X_j). \quad (38)$$

Proof. The first identity is [Definition 3.11](#) with $m = 4$, using $i^{-4} = 1$ and [Lemma 3.10](#). The second follows from [Proposition 3.12](#) (i)–(ii). $\square$

Remark 3.23 (kurt = 0 does not mean Gaussian). By [Theorem 3.19](#) (v) a non-degenerate Gaussian random variable is mesokurtic, but the converse fails badly. Consider the symmetric three-point law

$$\mathbb{P}[X = a] = \mathbb{P}[X = -a] = p, \quad \mathbb{P}[X = 0] = 1 - 2p, \quad (39)$$

with $a > 0$ and $p \in (0, \frac{1}{2})$. Then $\varphi_X(t) = 1 - 2p + 2p \cos(at)$, $\sigma^2 = 2pa^2$, $\mathbb{E}[X^4] = 2pa^4$, and hence

$$\kappa_4(X) = 2pa^4(1 - 6p), \quad \text{kurt}(X) = \frac{1 - 6p}{2p}. \quad (40)$$

For $p = \frac{1}{6}$ this vanishes, yet X is a three-point law and certainly not Gaussian. Consequently, non-Gaussianity is a strictly weaker requirement than $\text{kurt} \neq 0$ – which is precisely why the identifiability theorems of [Sections 5](#) and [6](#) are formulated in terms of non-normality rather than in terms of kurtosis.

Remark 3.24 (Kurtosis and tail weight). Excess kurtosis is routinely glossed as “tail weight”. The gloss is convenient, but it is not a theorem, and it is worth being precise about what kurt does and does not measure.

What it measures. Let $Z := (X - \mathbb{E}X)/\sigma$ be the standardisation of X , so that $\mathbb{E}[Z] = 0$ and $\mathbb{E}[Z^2] = 1$. Since $\text{Var}(Z^2) = \mathbb{E}[Z^4] - (\mathbb{E}[Z^2])^2 = \mathbb{E}[Z^4] - 1$, we have the identity

$$\text{kurt}(X) = \mathbb{E}[Z^4] - 3 = \text{Var}(Z^2) - 2 = \mathbb{E}[(Z^2 - 1)^2] - 2. \quad (41)$$

The quantity $Z^2 - 1$ vanishes exactly at $Z = \pm 1$, that is at $X = \mathbb{E}X \pm \sigma$, so excess kurtosis is the dispersion of X about *those two points*: probability mass moved away from $\mathbb{E}X \pm \sigma$ increases it, whether it moves outwards into the tails or inwards towards the centre. This is Moors’ reading of kurtosis as a measure of dispersion around $\mu \pm \sigma$ ([Moors, 1986](#)); [Balanda and MacGillivray \(1988\)](#) phrase it as the location- and scale-free movement of probability mass from the “shoulders” of a distribution into its centre *and* its tails, and argue that the notion is irreducibly vague, admitting several inequivalent formalisations. [Eq. \(41\)](#) also yields the sharp bound $\text{kurt} \geq -2$, with equality if and only if $Z^2 = 1$ almost surely, i.e. exactly when X takes the two values $\mathbb{E}X \pm \sigma$ with probability $\frac{1}{2}$ each – any affine image of the Rademacher

law, whose entry in [Table 1](#) records this value. For the same reason kurtosis is not a measure of peakedness either ([Kaplansky, 1945](#); [Darlington, 1970](#)).

What it does not measure. Read “tail weight” as the asymptotic decay of the density, or of $\mathbb{P}[|X| > x]$, relative to a Gaussian. Under that reading the sign of kurt and tail weight are logically independent, in both directions.

- (a) *Light tails, arbitrarily large kurtosis.* The three-point law [Eq. \(39\)](#) is supported on $\{-a, 0, a\}$, so its tails are as light as tails can be; yet $\text{kurt}(X) = (1 - 6p)/(2p) \rightarrow \infty$ as $p \downarrow 0$, by [Eq. \(40\)](#).
- (b) *Heavy tails, negative kurtosis.* Let X have the mixture law

$$\mathcal{L}(X) = (1 - \varepsilon) \text{Unif}[-1, 1] + \varepsilon \text{Laplace}(1), \quad \varepsilon = 10^{-3}. \quad (42)$$

Then $\sigma^2 = 67/200$ and $\text{kurt}(X) = -4515/4489 \approx -1.006$, so X is sub-Gaussian in the sense of [Definition 3.21](#). Its density equals $\frac{1}{2}\varepsilon e^{-|x|}$ for $|x| > 1$, and therefore exceeds the density of the Gaussian law of the same variance for $|x| \geq x_0$ with $x_0 \approx 2.56$, by a factor of about $3.1 \cdot 10^5$ at $|x| = 4$.

- (c) *Heavy tails, no kurtosis at all.* The Cauchy law, and the Student t_ν laws with $\nu \leq 4$, have $\mathbb{E}[X^4] = \infty$, so kurt is not even defined; cf. [Table 1](#).

Under a different reading of “tail” the verdict changes. If tail weight is taken to mean the propensity to produce observations far from the mean *measured in units of σ* , rather than asymptotic decay, then kurtosis is a tail functional, and [Westfall \(2014\)](#) argues that this is its only unambiguous interpretation. Example (a) shows what separates the two readings: the three-point law has compact support, but its atoms sit at $\pm a = \pm\sigma/\sqrt{2p}$, that is arbitrarily many standard deviations from the mean.

Convention. In these notes “sub-Gaussian” and “super-Gaussian” always mean [Definition 3.21](#), the sign of kurt, and never a statement about tails. Where tails are meant – in [Table 1](#), in [Remark 3.25](#) and in [Corollary 6.9](#) – they are named explicitly and always refer to asymptotic decay.

Remark 3.25 (Two incompatible meanings of “sub-Gaussian”). In concentration of measure and high-dimensional statistics, a real-valued random variable X is called *sub-Gaussian* when its tails are dominated by those of a Gaussian law, equivalently – up to the value of the constant σ – when

$$\mathbb{E}[\exp(\lambda(X - \mathbb{E}X))] \leq \exp\left(\frac{1}{2}\lambda^2\sigma^2\right), \quad \lambda \in \mathbb{R}, \quad (43)$$

for some $\sigma > 0$. By [Theorem 3.13](#) this forces φ_X to be entire, and the displayed bound gives $|\varphi_X(z)| \leq \mathbb{E}[e^{|\text{Im } z| |X|}] \leq 2 \exp(|\text{Im } z| |\mathbb{E}X| + \frac{1}{2}\sigma^2 |\text{Im } z|^2)$, so φ_X is of order at most 2. This is *not* the notion of [Definition 3.21](#), and neither notion implies the other. Indeed, the three-point law [Eq. \(39\)](#) with $p < \frac{1}{6}$ is bounded, hence sub-Gaussian in the concentration sense, while [Eq. \(40\)](#) gives $\text{kurt}(X) = (1 - 6p)/(2p) > 0$, so it is super-Gaussian in the sense of [Definition 3.21](#). Conversely, the mixture of [Remark 3.24](#) (b) has exponential tails and so is *not* sub-Gaussian in the concentration sense, while its excess kurtosis is negative, so it is sub-Gaussian in the sense of [Definition 3.21](#). *Throughout these notes, “sub-Gaussian” and “super-Gaussian” always refer to Definition 3.21.*

3.6 Why non-Gaussianity is what makes ICA work

The classical heuristic behind ICA is the central limit theorem: a normalised mixture $w^\top Z = \sum_j w_j Z_j$ of many independent sources looks “more Gaussian” than any individual source, so

Law of X	$\varphi_X(t)$	$\text{kurt}(X)$	type	analyticity of φ_X
$\mathcal{N}(0, \sigma^2)$	$e^{-\sigma^2 t^2/2}$	0	Gaussian	entire, order 2
Uniform on $[-a, a]$	$\frac{\sin(at)}{at}$	$-\frac{6}{5}$	sub-Gaussian	entire, order 1
Rademacher on $\{\pm 1\}$	$\cos t$	-2	sub-Gaussian	entire, order 1
Three-point Eq. (39)	$1 - 2p + 2p \cos(at)$	$\frac{1 - 6p}{2p}$	sub- iff $p > \frac{1}{6}$	entire, order 1
Laplace, scale b	$\frac{1}{1 + b^2 t^2}$	3	super-Gaussian	strip $ \text{Im } z < b^{-1}$
Student t_ν , $\nu > 4$	$\frac{(\sqrt{\nu} t)^{\nu/2} K_{\nu/2}(\sqrt{\nu} t)}{\Gamma(\nu/2) 2^{\nu/2-1}}$	$\frac{6}{\nu - 4}$	super-Gaussian	not analytic at 0
Cauchy, scale γ	$e^{-\gamma t }$	undefined	no moments	not differentiable at 0

Table 1: Characteristic functions, excess kurtosis and domain of analyticity for the standard examples. $K_{\nu/2}$ denotes the modified Bessel function of the second kind. Power-law tails destroy analyticity at the origin; exponential tails give a strip; bounded or Gaussian tails give an entire function.

one may hope to recover the sources by searching for the directions in which the mixture is *least* Gaussian. [Eq. \(38\)](#) turns this heuristic into a precise statement.

Proposition 3.26 (Kurtosis as a contrast function). *Let $Z_1, \dots, Z_k$ be independent real-valued random variables with $\mathbb{E}[Z_j] = 0$, $\text{Var}(Z_j) = 1$ and $\mathbb{E}[Z_j^4] < \infty$, and let $w \in \mathbb{R}^k$ with $\|w\|_2 = 1$. Then $\text{Var}(w^\top Z) = 1$ and*

$$|\text{kurt}(w^\top Z)| = \left| \sum_{j=1}^k w_j^4 \text{kurt}(Z_j) \right| \leq \max_{1 \leq j \leq k} |\text{kurt}(Z_j)|. \quad (44)$$

If $\max_j |\text{kurt}(Z_j)| > 0$, then equality holds if and only if $w = \pm e_j$ for an index j attaining the maximum; so the maximisers of $w \mapsto |\text{kurt}(w^\top Z)|$ on the unit sphere are exactly the signed coordinate directions of maximal $|\text{kurt}|$. If instead $\text{kurt}(Z_j) = 0$ for every j – in particular if all sources are Gaussian – then both sides vanish for every w and the criterion carries no information at all.

Proof. Since $\text{Var}(Z_j) = 1$ and $\|w\|_2 = 1$ we get $\kappa_2(w^\top Z) = 1$ by [Proposition 3.12](#), so $\text{kurt}(w^\top Z) = \kappa_4(w^\top Z)$, and [Eq. \(38\)](#) gives the stated identity. Then

$$\left| \sum_j w_j^4 \text{kurt}(Z_j) \right| \leq \sum_j w_j^4 |\text{kurt}(Z_j)| \leq \max_j |\text{kurt}(Z_j)| \cdot \sum_j w_j^4 \leq \max_j |\text{kurt}(Z_j)|, \quad (45)$$

where the last step uses $\sum_j w_j^4 \leq (\sum_j w_j^2)^2 = 1$. Write $M := \max_j |\text{kurt}(Z_j)|$ and assume $M > 0$. Then equality in the last inequality, which reads $M \sum_j w_j^4 \leq M$, holds if and only if $\sum_j w_j^4 = 1$, i.e. if and only if exactly one coordinate of w is non-zero and thus $w = \pm e_j$; equality in the middle inequality then forces $|\text{kurt}(Z_j)| = M$. If $M = 0$ all three quantities vanish for every w , which is the last assertion. $\square$

[Proposition 3.26](#) is the theoretical justification of the kurtosis-based contrast functions of [Comon \(1994\)](#) and [Hyvärinen et al. \(2001\)](#), and it already displays the two phenomena that the rest of these notes make exact:

- (i) non-Gaussian sources are recoverable, and the recovery is only ever determined up to sign, scale and the ordering of the coordinates – the ambiguities of [Eq. \(1\)](#);

- (ii) Gaussian sources are not recoverable at all, since the contrast degenerates. Compare [Remark 3.27](#).

What the following sections add is that neither fourth moments nor any moments at all are actually needed: the correct hypothesis is non-normality, in the sense of [Theorem 3.19](#), and the correct tool is [Theorem 3.17](#) rather than a contrast function.

Remark 3.27 (Gaussian sources are genuinely unidentifiable). Let $Z \sim \mathcal{N}(0, I_k)$ and let $A \in \mathbb{R}^{p \times k}$. For every orthogonal $Q \in O(k)$ we have $QZ \sim \mathcal{N}(0, I_k)$ by [Eq. \(8\)](#), so QZ again has mutually independent components, while

$$AZ \stackrel{d}{=} (AQ^\top)(QZ). \quad (46)$$

So A and $AQ^\top$ are indistinguishable from the law of $X = AZ$, and the ambiguity is a whole orthogonal group rather than only the discrete permutations and the diagonal rescalings of [Section 1](#). This is why every identifiability statement below either excludes Gaussian sources, or explicitly quantifies the remaining Gaussian ambiguity.

4 Identifiability for Non-Constant Independent Sources

We begin with the weakest set of assumptions on the sources: they are mutually independent and non-constant, and nothing else. Before any uniqueness statement can be made we have to normalise the representation, because two trivial mechanisms produce genuinely different representations of the same random vector: zero columns of the mixing matrix, and columns that are proportional to each other. [Remark 4.1](#) removes both.

Remark 4.1 (Normalising a representation). Consider a representation of a p -dimensional random vector $X \in \mathbb{R}^p$:

$$X = AZ + \mu = \sum_{j=1}^k a_j Z_j + \mu, \quad (47)$$

where $Z = [Z_1, \dots, Z_k]^\top \in \mathbb{R}^k$ is a random vector with mutually independent components $\{Z_1, \dots, Z_k\}$, where $\mu \in \mathbb{R}^p$ is a (non-random) column vector, and where $A = [a_1, \dots, a_k] \in \mathbb{R}^{p \times k}$ is a (non-random) matrix with column vectors $a_j \in \mathbb{R}^p$.

Step 1: removing zero columns and constant sources. By deleting the zero columns $a_j = 0$ and by absorbing every almost surely constant summand $a_j Z_j$ into μ , we arrive at a representation of X in which A has no zero column and in which every component of Z is almost surely non-constant.

Step 2: removing proportional columns. The columns of A may still be proportional to each other, say $a_j = \lambda a_l$ with a proportionality constant $\lambda \in \mathbb{R} \setminus \{0\}$ and $j \neq l$. This is a true ambiguity of the representation, since we can always merge the two corresponding sources,

$$a_j Z_j + a_l Z_l = a_l (\lambda Z_j + Z_l) =: a_l \tilde{Z}_l. \quad (48)$$

For a first identifiability result we therefore have to aggregate these variables. Proportionality is an equivalence relation on $\mathbb{R}^p \setminus \{0\}$, so we can choose columns $\tilde{a}_1, \dots, \tilde{a}_{k'}$ of A forming a system of representatives of $(\{a_1, \dots, a_k\} \setminus \{0\})/\sim$; that is, a set of non-zero columns of A that are pairwise non-proportional and such that every non-zero a_j is proportional to a (then necessarily unique) $\tilde{a}_l$, written $a_j \propto \tilde{a}_l$. Denoting the corresponding proportionality constants

by λ_j , i.e. $a_j = \lambda_j \tilde{a}_l$, we obtain

$$X = AZ + \mu \quad (49)$$

$$= \sum_{j=1}^k a_j Z_j + \mu \quad (50)$$

$$= \sum_{l=1}^{k'} \tilde{a}_l \underbrace{\left(\sum_{a_j \propto \tilde{a}_l} \lambda_j Z_j \right)}_{=: \tilde{Z}_l} + \mu \quad (51)$$

$$= \sum_{l=1}^{\tilde{k}} \tilde{a}_l \tilde{Z}_l + \underbrace{\sum_{l=\tilde{k}+1}^{k'} \tilde{a}_l \tilde{Z}_l}_{=: \tilde{\mu}} + \mu \quad (52)$$

$$= \tilde{A} \tilde{Z} + \tilde{\mu}, \quad (53)$$

where – after possibly re-indexing – $\tilde{k} \leq k'$ denotes the number of indices l for which $\tilde{Z}_l$ is *not* almost surely constant, so that the sum inside $\tilde{\mu}$ collects exactly the almost surely constant $\tilde{Z}_l$ and $\tilde{\mu}$ is (a.s. equal to) a deterministic vector, and where $\tilde{A} := [\tilde{a}_1, \dots, \tilde{a}_{\tilde{k}}]$ and $\tilde{Z} := [\tilde{Z}_1, \dots, \tilde{Z}_{\tilde{k}}]^\top$.

If Step 1 has already been carried out, so that every Z_j is non-constant, then in fact $\tilde{k} = k'$ and the second sum is empty: an independent sum of non-constant random variables is never a.s. constant, because $|\varphi_{Y_1} \varphi_{Y_2}| \equiv 1$ forces $|\varphi_{Y_1}| \equiv 1$ and hence Y_1 degenerate. We nevertheless state Step 2 in the general form, so that it can be applied on its own.

We have thus found a representation of X in which the columns of $\tilde{A}$ are non-zero and pairwise non-proportional and in which every component of $\tilde{Z}$ is almost surely non-constant. Moreover, if the components of Z are mutually independent then so are those of $\tilde{Z}$, since each Z_j enters exactly one $\tilde{Z}_l$.

Now that the existence of such normalised representations is established, we restrict attention to them and investigate their uniqueness – up to the transformations that are unavoidable.

Theorem 4.2 (Identifiability for independent non-constant sources; [Kagan et al., 1973](#), Chapter 10, Lemma 10.2.3 and Theorem 10.3.1). *Let $X \in \mathbb{R}^p$ be a p -dimensional random vector with two representations*

$$A^{(1)} Z^{(1)} + \mu^{(1)} = X = A^{(2)} Z^{(2)} + \mu^{(2)}, \quad (54)$$

with the following properties for $i = 1, 2$:

- (i) $A^{(i)} \in \mathbb{R}^{p \times k^{(i)}}$ is a (non-random) matrix whose columns are non-zero and pairwise non-proportional;
- (ii) $\mu^{(i)} \in \mathbb{R}^p$ is a (non-random) column vector;
- (iii) $Z^{(i)} \in \mathbb{R}^{k^{(i)}}$ is a random vector such that
 - (a) its $k^{(i)}$ components $\{Z_1^{(i)}, \dots, Z_{k^{(i)}}^{(i)}\}$ are mutually independent, and
 - (b) each component $Z_j^{(i)}$ is almost surely non-constant, i.e. $\mathcal{L}(Z_j^{(i)})$ is not a Dirac measure, $j = 1, \dots, k^{(i)}$.

Then

$$\mu^{(2)} - \mu^{(1)} \in \text{im } A^{(1)} = \text{im } A^{(2)}, \quad \text{rank}(A^{(1)}) = \text{rank}(A^{(2)}). \quad (55)$$

In particular there exist $c^{(1)} \in \mathbb{R}^{k^{(1)}}$ and $c^{(2)} \in \mathbb{R}^{k^{(2)}}$ with $\mu^{(2)} - \mu^{(1)} = A^{(1)}c^{(1)} = A^{(2)}c^{(2)}$.

Furthermore, the following statements hold.

- (1) If the l -th column of $A^{(2)}$ is not proportional to any column of $A^{(1)}$, then $Z_l^{(2)}$ is Gaussian.
- (2) Assume that the l -th column of $A^{(2)}$ is proportional to the j -th column of $A^{(1)}$ with proportionality constant¹ $\lambda \in \mathbb{R} \setminus \{0\}$, i.e. $a_l^{(2)} = \lambda \cdot a_j^{(1)}$. Then there exists a (complex) polynomial g such that the characteristic functions of $Z_l^{(2)}$ and $Z_j^{(1)}$ satisfy, in a neighbourhood of the origin,

$$\varphi_{Z_l^{(2)}}(\lambda t) = \varphi_{Z_j^{(1)}}(t) \cdot \exp(g(t)). \quad (56)$$

In particular, by Marcinkiewicz' Theorem 3.17 together with Remark 3.18, $Z_l^{(2)}$ is Gaussian if and only if $Z_j^{(1)}$ is Gaussian.

The proof is in Section B. It is a self-contained finite-difference argument that uses nothing from the sections in between, so the reader may turn to it at any point; we have deferred it only because it is considerably longer than the statement and would interrupt the development here. The first part of the theorem is elementary, and we record it right away as Proposition 4.4 below.

Remark 4.3 (Reading Theorem 4.2). Theorem 4.2 is best read as a dichotomy at the level of columns. Every column of $A^{(2)}$ either

- (i) is proportional to a column of $A^{(1)}$, in which case, by Eq. (56), the two associated sources agree up to that proportionality factor and a factor $\exp(g)$ with g a polynomial; or
- (ii) is not, in which case its source is forced to be Gaussian, and by Remark 3.27 we should not have expected to recover it in the first place.

In case (i), Theorem 4.2 by itself bounds neither $\deg(g)$ nor the direction in which the perturbation acts; under the additional hypotheses of Theorem 5.5 the factor becomes an independent Gaussian perturbation of one of the two sources. All the work in the following sections consists of ruling out the second alternative by strengthening the assumptions on the sources, and of bookkeeping the Gaussian perturbation in the first.

Theorem 4.2 is the engine of everything that follows. Its first part – the statement about images, ranks and offsets – needs none of the machinery of Section B, so we prove it here; only the column dichotomy (1)–(2) is postponed.

Proposition 4.4 (The first part of Theorem 4.2). *Under the hypotheses of Theorem 4.2, the affine hull of the support of $\mathcal{L}(X)$ equals $\mu^{(i)} + \text{im } A^{(i)}$ for $i = 1, 2$. Consequently*

$$\mu^{(2)} - \mu^{(1)} \in \text{im } A^{(1)} = \text{im } A^{(2)}, \quad \text{rank}(A^{(1)}) = \text{rank}(A^{(2)}). \quad (57)$$

Proof. Fix i and abbreviate $A := A^{(i)}$, $Z := Z^{(i)}$, $\mu := \mu^{(i)}$, $k := k^{(i)}$. Since the components of Z are mutually independent, $\mathcal{L}(Z)$ is the product of the laws of its components, and the

¹ Note that this proportionality constant was not reintroduced in (Kagan et al., 1973, Theorem 10.3.1) after it had been removed “without loss of generality” in (Kagan et al., 1973, Lemmata 10.2.4 and 10.2.5). It matters here, since we do not normalise the columns of $A^{(i)}$.

support of a finite product of Borel probability measures on second countable spaces – in particular on $\mathbb{R}$ – is the product of the supports: $\text{supp } \mathcal{L}(Z) = \prod_{j=1}^k \text{supp } \mathcal{L}(Z_j)$. Each factor contains at least two points, because Z_j is almost surely non-constant. Hence, for each j , choosing two distinct points of $\text{supp } \mathcal{L}(Z_j)$ and fixing an arbitrary point of $\text{supp } \mathcal{L}(Z_l)$ in every other coordinate $l \neq j$ produces two elements of $\text{supp } \mathcal{L}(Z)$ whose difference is a non-zero multiple of e_j . The direction space of the affine hull of $\text{supp } \mathcal{L}(Z)$ therefore contains every e_j , so

$$\text{aff}(\text{supp } \mathcal{L}(Z)) = \mathbb{R}^k. \quad (58)$$

Now let $T(z) := Az + \mu$. For a continuous map T one has $\text{supp } \mathcal{L}(T(Z)) = \overline{T(\text{supp } \mathcal{L}(Z))}$: the inclusion “ $\supseteq$ ” holds because $T^{-1}(U)$ is an open neighbourhood of any preimage point and therefore has positive mass, and “ $\subseteq$ ” because the closed set $\overline{T(\text{supp } \mathcal{L}(Z))}$ has $\mathcal{L}(T(Z))$ -measure 1. Affine hulls are unchanged by taking closures, since affine subspaces are closed, and they commute with affine maps. Hence

$$\text{aff}(\text{supp } \mathcal{L}(X)) = T(\text{aff}(\text{supp } \mathcal{L}(Z))) = A\mathbb{R}^k + \mu = \mu + \text{im } A. \quad (59)$$

Applying this to $i = 1$ and $i = 2$ and equating the two descriptions of the same affine subspace gives $\mu^{(1)} + \text{im } A^{(1)} = \mu^{(2)} + \text{im } A^{(2)}$, whence $\text{im } A^{(1)} = \text{im } A^{(2)}$ (the two direction spaces coincide) and $\mu^{(2)} - \mu^{(1)} \in \text{im } A^{(1)}$. Equality of ranks is equality of the dimensions of these images. $\square$

5 Identifiability for Non-Gaussian Independent Sources

[Theorem 4.2](#) exhibits a clear distinction between Gaussian and non-Gaussian components in a representation $X = AZ + \mu$: Gaussian components tend to be non-identifiable. It therefore makes sense to separate them off from the beginning and to work with a representation in which the Gaussian part appears as one additive noise vector.

Remark 5.1 (Separated representations). Start from a normalised representation as in [Remark 4.1](#) and write it as

$$X = AZ + \mu = \sum_{j=1}^k a_j Z_j + \mu \quad (60)$$

$$= \sum_{l=1}^{\hat{k}} a_l Z_l + \underbrace{\sum_{l=\hat{k}+1}^k a_l Z_l}_{=: \tilde{E} \sim \mathcal{N}(\tilde{\mu}, \tilde{\Sigma})} + \mu \quad (61)$$

$$= \tilde{A}\tilde{Z} + \tilde{E}, \quad (62)$$

where – after possibly re-indexing – $\hat{k}$ is the number of non-Gaussian components and the second sum collects all Gaussian components Z_l into a single p -variate (possibly degenerate) Gaussian random vector $\tilde{E} \sim \mathcal{N}(\tilde{\mu}, \tilde{\Sigma})$, where $\tilde{Z} := [Z_1, \dots, Z_{\hat{k}}]^\top$ is a vector of mutually independent, non-constant, non-Gaussian components with $\tilde{Z} \perp\!\!\!\perp \tilde{E}$, and where the mixing matrix $\tilde{A} := [a_1, \dots, a_{\hat{k}}]$ has non-zero, pairwise non-proportional columns.

Before stating the general identifiability result – which will be based on [Theorem 4.2](#) – we investigate how the above outsourcing of the Gaussian components can be *reverted*. This is the technical heart of this section: we need to be able to move Gaussian mass back and forth between the noise vector and additional columns of the mixing matrix.

Lemma 5.2 (Trading Gaussian noise against columns of the mixing matrix). *Let $X \in \mathbb{R}^p$ be a p -variate random vector with a representation*

$$X = AZ + E, \quad (63)$$

where

- $A \in \mathbb{R}^{p \times k}$ is a (non-random) matrix with non-zero, pairwise non-proportional columns,
- $Z \in \mathbb{R}^k$ is a random vector with mutually independent, non-constant, non-Gaussian components, and
- $E \in \mathbb{R}^p$ is a (possibly degenerate) Gaussian random vector, $E \sim \mathcal{N}(\mu, \Sigma)$, with mean vector $\mu \in \mathbb{R}^p$ and positive semi-definite covariance matrix $\Sigma \in \mathbb{R}^{p \times p}$ of rank $r \leq p$, such that $Z \perp\!\!\!\perp E$.

Then there exist a matrix $B \in \mathbb{R}^{p \times r}$ of rank r and a random vector $V \sim \mathcal{N}(0, I_r)$ such that

$$\Sigma = BB^\top, \quad E = BV + \mu \text{ a.s.}, \quad Z \perp\!\!\!\perp V, \quad (64)$$

which yields the representation

$$X = AZ + BV + \mu. \quad (65)$$

If $r = 0$ then B is the empty $p \times 0$ matrix, V the empty vector and $E = \mu$ a.s. If $r \geq 2$ then B can moreover be chosen such that no column of B is proportional to any column of A .

Furthermore, for either choice of B and for every $r \geq 0$, we obtain a normalised representation in the sense of [Remark 4.1](#):

$$X = \begin{bmatrix} A & \tilde{B} \end{bmatrix} \begin{bmatrix} \tilde{Z} \\ \tilde{V} \end{bmatrix} + \mu, \quad (66)$$

where

- $\tilde{B} \in \mathbb{R}^{p \times \tilde{r}}$ with $\tilde{r} \leq r$ consists of those columns of B that are not proportional to any column of A , and $\tilde{V} \sim \mathcal{N}(0, I_{\tilde{r}})$ collects the corresponding components of V ;
- $\tilde{Z} = Z + \hat{V}$ with $\hat{V} \sim \mathcal{N}(0, \Gamma)$ for a diagonal matrix Γ (possibly with zeros on the diagonal), where $\{Z, \hat{V}, \tilde{V}\}$ is mutually independent;
- every component of $\tilde{Z}$ is non-Gaussian, the random vector $[\tilde{Z}^\top, \tilde{V}^\top]^\top$ has mutually independent non-constant components, and the matrix $[A, \tilde{B}]$ has non-zero, pairwise non-proportional columns.

Proof. The first statement follows directly from the spectral decomposition of the symmetric positive semi-definite matrix Σ (equivalently, from its singular value decomposition):

$$\Sigma = Q \begin{bmatrix} \Delta_r & 0 \\ 0 & 0 \end{bmatrix} Q^\top, \quad (67)$$

with an orthogonal matrix $Q \in O(p) \subseteq \mathbb{R}^{p \times p}$ and a diagonal matrix $\Delta_r \in \mathbb{R}^{r \times r}$ with strictly positive diagonal entries. We may then put

$$B := Q \begin{bmatrix} \Delta_r^{1/2} \\ 0 \end{bmatrix}, \quad V := \begin{bmatrix} \Delta_r^{-1/2} & 0 \end{bmatrix} Q^\top (E - \mu), \quad (68)$$

which yields exactly the desired properties:

$$\text{rank}(B) = r, \quad \Sigma = BB^\top, \quad E = BV + \mu \text{ a.s.}, \quad V \sim \mathcal{N}(0, I_r), \quad Z \perp V. \quad (69)$$

Note that if $G \in O(r)$ is orthogonal, then replacing B by BG and V by $G^\top V$ preserves all of these properties. For $r \geq 2$ this freedom can be used to make every column of BG non-proportional to every column of A ; the technical details are the content of [Lemma 5.4](#).

In any case, let $\tilde{B}$ be the matrix consisting of those columns of $B = [b_1, \dots, b_r]$ that are not proportional to any column of A , and let $\tilde{V}$ be the random vector of the corresponding components of V . After re-indexing we may assume these to be the first $\tilde{r}$ columns of B , so that $\tilde{B} = [b_1, \dots, b_{\tilde{r}}]$. For $l > \tilde{r}$ we then have $b_l = \lambda_l a_{j_l}$ for a unique index j_l and some $\lambda_l \in \mathbb{R} \setminus \{0\}$. With this we obtain

$$X = AZ + E \quad (70)$$

$$= AZ + BV + \mu \quad (71)$$

$$= \sum_{j=1}^k a_j Z_j + \sum_{l=1}^r b_l V_l + \mu \quad (72)$$

$$= \sum_{j=1}^k a_j Z_j + \sum_{l=1}^{\tilde{r}} b_l V_l + \sum_{l=\tilde{r}+1}^r b_l V_l + \mu \quad (73)$$

$$= \sum_{j=1}^k a_j Z_j + \sum_{l=1}^{\tilde{r}} b_l V_l + \sum_{l=\tilde{r}+1}^r a_{j_l} \lambda_l V_l + \mu \quad (74)$$

$$= \sum_{j=1}^k a_j \underbrace{\left(Z_j + \sum_{b_l \propto a_j} \lambda_l V_l \right)}_{=: \tilde{Z}_j} + \sum_{l=1}^{\tilde{r}} b_l V_l + \mu \quad (75)$$

$$= \sum_{j=1}^k a_j \tilde{Z}_j + \sum_{l=1}^{\tilde{r}} b_l V_l + \mu \quad (76)$$

$$= \begin{bmatrix} A & \tilde{B} \end{bmatrix} \begin{bmatrix} \tilde{Z} \\ \tilde{V} \end{bmatrix} + \mu. \quad (77)$$

Each $\tilde{Z}_j$ is non-Gaussian, being the sum of the non-Gaussian Z_j and independent Gaussian noise, by Marcinkiewicz' [Theorem 3.17](#), cf. [Remark 3.18](#) (ii). The components of $[\tilde{Z}^\top, \tilde{V}^\top]^\top$ are mutually independent because the components of $[Z^\top, V^\top]^\top$ are mutually independent and each component V_l enters exactly one component of $[\tilde{Z}^\top, \tilde{V}^\top]^\top$. The columns of $\tilde{B}$ are pairwise non-proportional because $\text{rank}(B) = r$ makes the columns of B linearly independent, and they are non-proportional to the columns of A by the very choice of $\tilde{B}$. So the matrix $[A, \tilde{B}]$ has non-zero, pairwise non-proportional columns. $\square$

The following lemma provides the rotation used in the proof above. It is the only place where we need a genuinely geometric argument.

Lemma 5.3 (Avoiding finitely many hyperplanes). *Let $k \geq 1$ and let $\mathcal{A} \subseteq \mathbb{R}^k \setminus \{0\}$ be a finite set. Then there exists $q \in \mathbb{R}^k$ with $q^\top a \neq 0$ for every $a \in \mathcal{A}$. In fact the set of such q is open and its complement is a Lebesgue null set.*

Proof. For $a \neq 0$ the orthogonal complement $a^\perp = \{v \in \mathbb{R}^k : v^\top a = 0\}$ is a linear subspace of dimension $k - 1$, hence closed and a Lebesgue null set. A finite union of null sets is a null set,

so $\mathbb{R}^k \setminus \bigcup_{a \in \mathcal{A}} a^\perp$ has full Lebesgue measure and is in particular non-empty; it is open as the complement of a finite union of closed sets. $\square$

Lemma 5.4 (Rotating a frame away from finitely many directions). *Let $p \geq k \geq 2$, let $\mathcal{A} \subseteq \mathbb{R}^p \setminus \{0\}$ be a finite set of non-trivial (column) vectors in $\mathbb{R}^p$, and let $B \in \mathbb{R}^{p \times k}$ be a matrix with $\text{rank}(B) = k$. Then there exists an orthogonal matrix $Q \in O(k) \subseteq \mathbb{R}^{k \times k}$ such that for every $a \in \mathcal{A}$ and every column vector c of $C := BQ \in \mathbb{R}^{p \times k}$ the set $\{a, c\}$ is linearly independent, i.e. a is not proportional to c (and vice versa).*

Proof. Put $B^\dagger := (B^\top B)^{-1} B^\top \in \mathbb{R}^{k \times p}$, which is a left inverse of B by the assumption $\text{rank}(B) = k$.

We first discard from $\mathcal{A}$ those a with $B^\top a = 0$. Such an a is automatically linearly independent of every column c of BQ , whatever $Q \in O(k)$: indeed $c \in \text{im } B$, so $a = \lambda c$ with $\lambda \neq 0$ would give $a \in \text{im } B$, whereas $B^\top a = 0$ says $a \perp \text{im } B$, and together these force $a = 0$, which is excluded. Note also that $B^\dagger a = 0$ if and only if $B^\top a = 0$, since $B^\top B$ is invertible. Replacing $\mathcal{A}$ by $\{a \in \mathcal{A} : B^\top a \neq 0\}$ – if this is empty any $Q \in O(k)$ will do – we may therefore assume $B^\dagger a \neq 0$ for every $a \in \mathcal{A}$, and consider the sets

$$\begin{aligned}\tilde{\mathcal{A}} &:= \{B^\dagger a \mid a \in \mathcal{A}\} \subseteq \mathbb{R}^k \setminus \{0\}, \\ \mathcal{A}' &:= \tilde{\mathcal{A}} \cup \{t_{\tilde{a}} \mid \tilde{a} \in \tilde{\mathcal{A}}\} \subseteq \mathbb{R}^k,\end{aligned}$$

where for $\tilde{a} \in \tilde{\mathcal{A}}$ the column vector $t_{\tilde{a}} \in \mathbb{R}^k$ is any fixed non-trivial vector with $t_{\tilde{a}}^\top \tilde{a} = 0$; such a vector exists because $k \geq 2$. By Lemma 5.3 there exists a vector $q_1 \in \mathbb{R}^k$ that is not orthogonal to any vector in $\mathcal{A}'$, and without loss of generality we may scale it to unit norm, $\|q_1\|_2 = 1$.

It follows that q_1 is not proportional to any $\tilde{a} \in \tilde{\mathcal{A}}$: otherwise $q_1 = \lambda \tilde{a}$ with $\lambda \in \mathbb{R}$, and hence $t_{\tilde{a}}^\top q_1 = \lambda t_{\tilde{a}}^\top \tilde{a} = 0$, contradicting the choice of q_1 .

Since q_1 is also not orthogonal to any element of $\tilde{\mathcal{A}}$, the set $\tilde{\mathcal{A}}$ is disjoint from the orthogonal complement

$$q_1^\perp := \{v \in \mathbb{R}^k \mid v^\top q_1 = 0\},$$

which is a $(k-1)$ -dimensional subspace of $\mathbb{R}^k$ and thus admits an orthonormal basis $q_2, \dots, q_k$. We can now build the orthogonal matrix

$$Q := [q_1, \dots, q_k] \in O(k) \subseteq \mathbb{R}^{k \times k}.$$

Assume, by way of contradiction, that there exist $a \in \mathcal{A}$ and a column vector c of $C := BQ$, say $c = Ce_j = Bq_j$, such that $\{a, c\}$ is linearly dependent. Since $a, c \neq 0$ there is then a $\lambda \in \mathbb{R} \setminus \{0\}$ with $a = \lambda c$, and multiplying by $B^\dagger$ gives

$$\tilde{a} := B^\dagger a = \lambda \cdot B^\dagger c = \lambda \cdot B^\dagger BQe_j = \lambda \cdot q_j.$$

For $j = 1$ this contradicts $q_1 \neq \lambda^{-1} \tilde{a}$ for all $\tilde{a} \in \tilde{\mathcal{A}}$. For $j \geq 2$ it contradicts $\tilde{a} \in \tilde{\mathcal{A}}$ together with $\tilde{\mathcal{A}} \cap \text{span}\{q_2, \dots, q_k\} = \emptyset$. So $\{a, c\}$ is linearly independent for every $a \in \mathcal{A}$ and every column vector c of BQ , which proves the claim. $\square$

Theorem 5.5 (Identifiability for independent non-Gaussian sources). *Let $X \in \mathbb{R}^p$ be a random vector and assume that we have two representations*

$$A^{(1)}Z^{(1)} + E^{(1)} = X = A^{(2)}Z^{(2)} + E^{(2)}, \quad (78)$$

with the following properties for $i = 1, 2$:

- (i) $A^{(i)} \in \mathbb{R}^{p \times k^{(i)}}$ is a (non-random) matrix whose columns are non-zero and pairwise non-proportional;
- (ii) $E^{(i)} \in \mathbb{R}^p$ is a p -variate Gaussian random vector, $E^{(i)} \sim \mathcal{N}(\mu^{(i)}, \Sigma^{(i)})$, possibly degenerate, with mean vector $\mu^{(i)} \in \mathbb{R}^p$ and positive semi-definite covariance matrix $\Sigma^{(i)}$;
- (iii) $Z^{(i)} \in \mathbb{R}^{k^{(i)}}$ is a random vector such that
 - (a) its $k^{(i)}$ components $\{Z_1^{(i)}, \dots, Z_{k^{(i)}}^{(i)}\}$ are mutually independent, and
 - (b) each component $Z_j^{(i)}$ is a (non-constant) non-Gaussian random variable, for $j = 1, \dots, k^{(i)}$;
- (iv) $E^{(i)}$ is independent of $Z^{(i)}$: $E^{(i)} \perp\!\!\!\perp Z^{(i)}$.

Then $k^{(1)} = k^{(2)} =: k$, and there exist a permutation matrix $P = P(\rho) \in \mathbb{R}^{k \times k}$, given by $Pe_j = e_{\rho(j)}$ for a permutation ρ , and an invertible diagonal matrix $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_k) \in \mathbb{R}^{k \times k}$ such that

$$A^{(2)} = A^{(1)} P \Lambda, \quad \text{rank}(A^{(2)}) = \text{rank}(A^{(1)}), \quad (79)$$

and such that for every $j = 1, \dots, k$ there exists a (complex) polynomial g_j with

$$\varphi_{Z_j^{(2)}}(\lambda_j t) = \varphi_{Z_{\rho(j)}^{(1)}}(t) \cdot \exp(g_j(t)) \quad (80)$$

in a neighbourhood of the origin.

Furthermore, and in particular, $A^{(2)}$ has a left inverse ($\text{rank}(A^{(2)}) = k$) if and only if $A^{(1)}$ has a left inverse ($\text{rank}(A^{(1)}) = k$). If this is the case, then for each $j = 1, \dots, k$ separately we have $\deg(g_j) \leq 2$ and there exist a constant $\nu_j \in \mathbb{R}$ and a (possibly degenerate) random variable $G_j \sim \mathcal{N}(0, \sigma_j^2)$ such that at least one of the following two cases holds, according to the sign of the degree-two coefficient of g_j :

$$(\lambda_j Z_j^{(2)} + \nu_j) \stackrel{d}{=} Z_{\rho(j)}^{(1)} + G_j, \quad G_j \perp\!\!\!\perp Z_{\rho(j)}^{(1)}, \quad \text{if } g_j'' \leq 0, \quad (81)$$

$$Z_{\rho(j)}^{(1)} \stackrel{d}{=} (\lambda_j Z_j^{(2)} + \nu_j) + G_j, \quad G_j \perp\!\!\!\perp Z_j^{(2)}, \quad \text{if } g_j'' \geq 0. \quad (82)$$

If $g_j'' \neq 0$ exactly one of the two cases applies; if $g_j'' = 0$ then $G_j = 0$ and both hold and coincide. The polynomial g_j is determined near the origin only up to an additive constant in $2\pi i \mathbb{Z}$, because $\varphi_{Z_{\rho(j)}^{(1)}}$ does not vanish there; in particular $\deg(g_j)$ and g_j'' are well defined.

Proof. Apply [Lemma 5.2](#) to both sides to obtain the representations

$$\begin{bmatrix} A^{(1)} & \tilde{B}^{(1)} \end{bmatrix} \begin{bmatrix} \tilde{Z}^{(1)} \\ \tilde{V}^{(1)} \end{bmatrix} + \mu^{(1)} = X = \begin{bmatrix} A^{(2)} & \tilde{B}^{(2)} \end{bmatrix} \begin{bmatrix} \tilde{Z}^{(2)} \\ \tilde{V}^{(2)} \end{bmatrix} + \mu^{(2)}, \quad (83)$$

which satisfy the requirements of [Theorem 4.2](#).

Since $\tilde{Z}_j^{(2)}$ is non-Gaussian, [Theorem 4.2](#) implies that the j -th column $a_j^{(2)}$ of $A^{(2)}$ is proportional to some column of $[A^{(1)}, \tilde{B}^{(1)}]$. All components of $\tilde{V}^{(1)}$ are Gaussian whereas those of $\tilde{Z}^{(1)}$ are non-Gaussian, so [Theorem 4.2](#) forces $a_j^{(2)}$ to be proportional to a column of $A^{(1)}$, say to $a_{\rho(j)}^{(1)}$. The index $\rho(j)$ is unique because the columns of $A^{(1)}$ are pairwise non-proportional, and $j \mapsto \rho(j)$ is injective: $\rho(j) = \rho(j')$ would give $a_j^{(2)} \propto a_{\rho(j)}^{(1)} \propto a_{j'}^{(2)}$, contradicting pairwise non-proportionality of the columns of $A^{(2)}$; hence $k^{(2)} \leq k^{(1)}$. Since the same argument applies to all columns of $A^{(2)}$, and also with the roles of $A^{(1)}$ and $A^{(2)}$

interchanged, we obtain a bijection between the columns of $A^{(1)}$ and those of $A^{(2)}$, together with their proportionality constants. This already proves the first part of the claim, namely $k^{(1)} = k^{(2)} =: k$ and the representation

$$A^{(2)} = A^{(1)} P \Lambda$$

for some permutation matrix $P = P(\rho)$ with permutation ρ and some invertible diagonal matrix $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_k)$. Indeed, multiplying by the unit vector e_j from the right gives

$$a_j^{(2)} = A^{(2)} e_j = A^{(1)} P \Lambda e_j = A^{(1)} P \lambda_j e_j = A^{(1)} \lambda_j e_{\rho(j)} = \lambda_j a_{\rho(j)}^{(1)}.$$

Again by [Theorem 4.2](#) there exists a polynomial $\tilde{g}_j$ with

$$\varphi_{\tilde{Z}_j^{(2)}}(\lambda_j t) = \varphi_{\lambda_j \tilde{Z}_j^{(2)}}(t) = \varphi_{\tilde{Z}_{\rho(j)}^{(1)}}(t) \cdot \exp(\tilde{g}_j(t)) \quad (84)$$

in a neighbourhood of the origin. [Lemma 5.2](#) gives us the representations

$$\tilde{Z}_j^{(2)} = Z_j^{(2)} + \hat{V}_j^{(2)}, \quad Z_j^{(2)} \perp\!\!\!\perp \hat{V}_j^{(2)}, \quad (85)$$

$$\tilde{Z}_{\rho(j)}^{(1)} = Z_{\rho(j)}^{(1)} + \hat{V}_{\rho(j)}^{(1)}, \quad Z_{\rho(j)}^{(1)} \perp\!\!\!\perp \hat{V}_{\rho(j)}^{(1)}, \quad (86)$$

with Gaussian $\hat{V}_j^{(2)}$ and $\hat{V}_{\rho(j)}^{(1)}$. Taking characteristic functions yields

$$\varphi_{\lambda_j \tilde{Z}_j^{(2)}}(t) = \varphi_{\lambda_j Z_j^{(2)}}(t) \cdot \varphi_{\lambda_j \hat{V}_j^{(2)}}(t), \quad (87)$$

$$\varphi_{\tilde{Z}_{\rho(j)}^{(1)}}(t) = \varphi_{Z_{\rho(j)}^{(1)}}(t) \cdot \varphi_{\hat{V}_{\rho(j)}^{(1)}}(t). \quad (88)$$

Combining this with the previous display we get

$$\varphi_{Z_j^{(2)}}(\lambda_j t) = \varphi_{\lambda_j Z_j^{(2)}}(t) = \varphi_{\lambda_j \tilde{Z}_j^{(2)}}(t) \cdot \varphi_{\lambda_j \hat{V}_j^{(2)}}(t)^{-1} \quad (89)$$

$$= \varphi_{\tilde{Z}_{\rho(j)}^{(1)}}(t) \cdot \exp(\tilde{g}_j(t)) \cdot \varphi_{\lambda_j \hat{V}_j^{(2)}}(t)^{-1} \quad (90)$$

$$= \varphi_{Z_{\rho(j)}^{(1)}}(t) \cdot \underbrace{\varphi_{\hat{V}_{\rho(j)}^{(1)}}(t) \cdot \exp(\tilde{g}_j(t)) \cdot \varphi_{\lambda_j \hat{V}_j^{(2)}}(t)^{-1}}_{=: \exp(g_j(t))} \quad (91)$$

$$= \varphi_{Z_{\rho(j)}^{(1)}}(t) \cdot \exp(g_j(t)), \quad (92)$$

where we used that the logarithm of the characteristic function of a (possibly degenerate) Gaussian distribution is a polynomial of degree ≤ 2 and that such characteristic functions have no zeros. Hence g_j is a well-defined polynomial. This proves the main claim.

Now assume in addition that $\text{rank}(A^{(1)}) = k \leq p$. Then $A^{(1)}$ has the left inverse

$$A^{(1)\dagger} := (A^{(1)\top} A^{(1)})^{-1} A^{(1)\top}. \quad (93)$$

Multiplying the original representation [Eq. \(78\)](#) by $e_{\rho(j)}^\top A^{(1)\dagger}$ from the left gives

$$Z_{\rho(j)}^{(1)} + e_{\rho(j)}^\top A^{(1)\dagger} E^{(1)} = e_{\rho(j)}^\top A^{(1)\dagger} (A^{(1)} Z^{(1)} + E^{(1)}) \quad (94)$$

$$= e_{\rho(j)}^\top A^{(1)\dagger} (A^{(2)} Z^{(2)} + E^{(2)}) \quad (95)$$

$$= e_{\rho(j)}^\top A^{(1)\dagger} (A^{(1)} P \Lambda Z^{(2)} + E^{(2)}) \quad (96)$$

$$= e_{\rho(j)}^\top P \Lambda Z^{(2)} + e_{\rho(j)}^\top A^{(1)\dagger} E^{(2)} \quad (97)$$

$$= e_j^\top \Lambda Z^{(2)} + e_{\rho(j)}^\top A^{(1)\dagger} E^{(2)} \quad (98)$$

$$= \lambda_j Z_j^{(2)} + e_{\rho(j)}^\top A^{(1)\dagger} E^{(2)}, \quad (99)$$

with Gaussian

$$N_j^{(i)} := e_{\rho(j)}^\top A^{(1)\dagger} E^{(i)} \sim \mathcal{N}(\nu_j^{(i)}, (\tau_j^{(i)})^2), \quad (100)$$

which are independent of the corresponding $Z^{(i)}$. Taking characteristic functions in

$$Z_{\rho(j)}^{(1)} + N_j^{(1)} = \lambda_j Z_j^{(2)} + N_j^{(2)} \quad (101)$$

gives

$$\varphi_{Z_{\rho(j)}^{(1)}}(t) \cdot \varphi_{N_j^{(1)}}(t) = \varphi_{\lambda_j Z_j^{(2)}}(t) \cdot \varphi_{N_j^{(2)}}(t). \quad (102)$$

Now put

$$\sigma_j := \sqrt{(\tau_j^{(2)})^2 - (\tau_j^{(1)})^2}, \quad \nu_j := \nu_j^{(2)} - \nu_j^{(1)}, \quad G_j \sim \mathcal{N}(0, \sigma_j^2). \quad (103)$$

Treating the two cases $\tau_j^{(1)} \geq \tau_j^{(2)}$ and $\tau_j^{(1)} \leq \tau_j^{(2)}$ separately we obtain

$$\varphi_{\lambda_j Z_j^{(2)}}(t) = \varphi_{Z_{\rho(j)}^{(1)}}(t) \cdot \varphi_{N_j^{(1)}}(t) \cdot \varphi_{N_j^{(2)}}(t)^{-1} \quad (104)$$

$$= \varphi_{Z_{\rho(j)}^{(1)}}(t) \cdot \varphi_{G_j - \nu_j}(t), \quad \tau_j^{(1)} \geq \tau_j^{(2)}, \quad (105)$$

$$\varphi_{Z_{\rho(j)}^{(1)}}(t) = \varphi_{\lambda_j Z_j^{(2)}}(t) \cdot \varphi_{N_j^{(2)}}(t) \cdot \varphi_{N_j^{(1)}}(t)^{-1} \quad (106)$$

$$= \varphi_{\lambda_j Z_j^{(2)}}(t) \cdot \varphi_{G_j + \nu_j}(t), \quad \tau_j^{(1)} \leq \tau_j^{(2)}. \quad (107)$$

This shows that we are in one of the two cases

$$\lambda_j Z_j^{(2)} \stackrel{d}{=} Z_{\rho(j)}^{(1)} + G_j - \nu_j, \quad G_j \perp\!\!\!\perp Z_{\rho(j)}^{(1)}, \quad \tau_j^{(1)} \geq \tau_j^{(2)}, \quad (108)$$

$$Z_{\rho(j)}^{(1)} \stackrel{d}{=} \lambda_j Z_j^{(2)} + G_j + \nu_j, \quad G_j \perp\!\!\!\perp Z_j^{(2)}, \quad \tau_j^{(1)} \leq \tau_j^{(2)}, \quad (109)$$

which is the claim, once the case distinction is expressed through g_j . For that, compare the two displays above with [Eq. \(56\)](#). Near the origin $\varphi_{Z_{\rho(j)}^{(1)}}$ does not vanish, so dividing gives

$$\exp(g_j(t)) = \frac{\varphi_{\lambda_j Z_j^{(2)}}(t)}{\varphi_{Z_{\rho(j)}^{(1)}}(t)} = \begin{cases} \varphi_{G_j - \nu_j}(t), & \tau_j^{(1)} \geq \tau_j^{(2)}, \\ \varphi_{G_j + \nu_j}(t)^{-1}, & \tau_j^{(1)} \leq \tau_j^{(2)}, \end{cases} \quad (110)$$

near the origin. By [Notation 2.4](#) the right hand side equals $\exp(\mp \frac{1}{2} \sigma_j^2 t^2 - i \nu_j t)$ in the two cases, so, applying the uniqueness part of [Lemma 3.10](#) to $\varphi_{G_j \mp \nu_j}$ and its exponent $\pm g_j$,

$$g_j(t) \equiv \mp \frac{1}{2} \sigma_j^2 t^2 - i \nu_j t \pmod{2\pi i \mathbb{Z}}, \quad g_j'' = \mp \sigma_j^2. \quad (111)$$

In particular $\deg(g_j) \leq 2$, the coefficient g_j'' is real, $\sigma_j^2 = |g_j''|$, and $g_j'' \leq 0$ holds exactly in the first case and $g_j'' \geq 0$ exactly in the second. $\square$

Remark 5.6 (The four remaining ambiguities). [Theorem 5.5](#) states that, under the assumption of independent non-Gaussian sources and full column rank of the mixing matrix, the distributions of the sources can be recovered up to translation, scale, permutation and componentwise additive Gaussian noise. These are the four ambiguities the theorem leaves. Each of the first three ambiguities can be removed by a normalisation convention:

- (1) the translation ambiguity by centring all random variables, e.g. by subtracting their means (assuming these exist);
- (2) the scale ambiguity by rescaling, e.g. by dividing by the standard deviations (assuming these exist and are finite). Beware that this pins the scale down to a sign only when the fourth ambiguity is absent: in the setting of [Theorem 5.5](#) the first case gives $\lambda_j^2 \text{Var}(Z_j^{(2)}) = \text{Var}(Z_{\rho(j)}^{(1)}) + \sigma_j^2$, so standardising both source vectors leaves $\lambda_j = \pm\sqrt{1 + \sigma_j^2}$. In the noiseless model $\sigma_j = 0$ and only the sign survives, as [Corollary 7.4](#) records with $\Lambda = \text{diag}(\pm 1, \dots, \pm 1)$;
- (3) the permutation ambiguity by enforcing a recognisable ordering of the sources, as is done for causal models such as the linear non-Gaussian acyclic model (LiNGAM) of [Shimizu et al. \(2006\)](#); there the ordering is not merely a convention but is itself identified, see [Corollary 7.32](#).

The fourth ambiguity – componentwise additive Gaussian noise – is of a different nature, since it changes the law of the sources rather than just their parametrisation. Removing it is the subject of [Section 6](#).

6 Identifiability for Gaussian-Free Independent Sources

[Theorem 5.5](#) left us with one irreducible ambiguity: the sources are determined only up to an additive Gaussian perturbation. That ambiguity is caused by sources that still contain some Gaussian noise which could equally well be attributed to the noise vector. In this section we remove it by strengthening the hypothesis on the sources from “non-Gaussian” to “no Gaussian noise can be split off at all”.

6.1 Gaussian-free random variables

Definition 6.1 (Gaussian-free). A real-valued random variable Z is called *Gaussian-free* if for every decomposition

$$Z \stackrel{d}{=} U_1 + U_2, \quad U_1 \perp\!\!\!\perp U_2, \quad (112)$$

neither U_1 nor U_2 is a non-degenerate Gaussian random variable. Equivalently: there is no $\sigma > 0$ and no real-valued random variable Y with $Y \perp\!\!\!\perp G$, $G \sim \mathcal{N}(0, \sigma^2)$ and $Z \stackrel{d}{=} Y + G$; that is, $\mathcal{L}(Z)$ has no non-degenerate Gaussian convolution factor.

Remark 6.2 (Gaussian-freeness is affine invariant). Whether Z is Gaussian-free depends only on $\mathcal{L}(Z)$, and it is invariant under invertible affine maps: if Z is Gaussian-free, $\lambda \neq 0$ and $c \in \mathbb{R}$, then $\lambda Z + c$ is Gaussian-free as well. Indeed, a decomposition $\lambda Z + c \stackrel{d}{=} U_1 + U_2$ with $U_1 \perp\!\!\!\perp U_2$ and $U_2 \sim \mathcal{N}(m, \tau^2)$, $\tau > 0$, would give $Z \stackrel{d}{=} \lambda^{-1}(U_1 - c) + \lambda^{-1}U_2$ with $\lambda^{-1}U_2 \sim \mathcal{N}(m/\lambda, \tau^2/\lambda^2)$ again non-degenerate, contradicting Gaussian-freeness of Z . We use this silently whenever a rescaled source is called Gaussian-free, for instance in the proof of [Theorem 6.12](#).

Remark 6.3 (Gaussian-free versus non-Gaussian). A *non-constant* Gaussian-free random variable is automatically non-Gaussian: a non-degenerate Gaussian Z fails to be Gaussian-free, because the trivial decomposition $Z \stackrel{d}{=} Z + 0$ already exhibits a non-degenerate Gaussian factor. The converse fails, by [Example 6.10](#) (d), and constants are Gaussian-free but degenerate – which is why non-constancy has to be assumed separately. Consequently, in [Theorem 6.12](#) the Gaussian-free hypothesis re-derives the non-Gaussianity demanded by [Theorem 5.5](#) for those components that are assumed Gaussian-free, and for them only the non-constancy has

to be carried over. In part (1) of that theorem, where only $Z^{(1)}$ is assumed Gaussian-free, the non-Gaussianity of $Z^{(2)}$ remains a genuine hypothesis and cannot be dropped: with $p = 2$, $A^{(1)} = [e_1]$, $Z^{(1)}$ Rademacher and $E^{(1)} = e_2 N$ for $N \sim \mathcal{N}(0, 1)$, the same X also equals $A^{(2)} Z^{(2)}$ with $A^{(2)} = [e_1, e_2]$, $Z^{(2)} = [Z^{(1)}, N]^\top$ and $E^{(2)} = 0$, and the conclusion fails because $Z_2^{(2)}$ is Gaussian.

The interpretation is that from a Gaussian-free random variable one cannot shave off any further Gaussian noise: it carries, in this sense, the cleanest possible signal.

Remark 6.4 (Terminology). The classical name for the property in [Definition 6.1](#) is that $\mathcal{L}(Z)$ has *no Gaussian component*, or equivalently *no normal component*, where “component” means *convolution factor*. This is the vocabulary of the arithmetic of probability distributions of [Linnik and Ostrovskii \(1977\)](#), whose chapters are devoted to distributions “with a Gaussian component”, and it is used in exactly this sense in the ICA literature ([Eriksson and Koivunen, 2006](#)). We prefer the adjective *Gaussian-free* for two reasons: it does not collide with the “components” $Z_1, \dots, Z_k$ of a random vector, which we would otherwise have to call components having no components, and it does not collide with the “Gaussian components” of a Gaussian mixture model, which are mixture summands rather than convolution factors.

A related but genuinely different classical notion is the Khinchin–Linnik class I_0 of distributions having no *indecomposable* factors. Here a law is a *factor* of μ if μ is its convolution with some law – so every law is a factor of itself – and μ is *indecomposable* if it is non-degenerate and in every factorisation $\mu = \mu_1 * \mu_2$ one of μ_1, μ_2 is degenerate. The two notions are logically independent: $\mathcal{N}(0, 1)$ lies in I_0 by Cramér’s [Theorem 3.20](#), since all its factors are Gaussian and every non-degenerate Gaussian is decomposable, but it is not Gaussian-free; whereas a non-degenerate Bernoulli law is Gaussian-free by [Corollary 6.9](#) below yet is itself indecomposable – a convolution of two laws with $|A|$ and $|B|$ support points has at least $|A| + |B| - 1$ of them, so two non-degenerate factors would force at least three – and hence is not in I_0 .

To make [Definition 6.1](#) quantitative we measure how much Gaussian noise *can* be split off.

Definition 6.5 (Splittable Gaussian scales). For a real-valued random variable Z and $s \in \mathbb{R}_{\geq 0}$ set

$$g_s(t) := \varphi_Z(t) \exp\left(\frac{1}{2}s^2 t^2\right), \quad t \in \mathbb{R}, \quad (113)$$

and put

$$S(Z) := \{s \in \mathbb{R}_{\geq 0} : g_s \text{ is a characteristic function}\}, \quad (114)$$

as well as

$$\sigma_{\max}(Z) := \sup S(Z) \in [0, \infty]. \quad (115)$$

We call $\sigma_{\max}(Z)$ the *maximal Gaussian scale* of Z .

The name is not standard. It is justified by the following reading: $s \in S(Z)$ if and only if a $\mathcal{N}(0, s^2)$ factor can be split off Z , so $\sigma_{\max}(Z)$ is the supremum of the standard deviations of the Gaussians that Z contains as convolution factors – and by [Lemma 6.6](#) (iii) below that supremum is attained, so it really is the largest such standard deviation.

Lemma 6.6 (Structure of $S(Z)$). *Let Z be a real-valued random variable. Then:*

(i) *for every $s \in S(Z)$ we have the tail bound*

$$|\varphi_Z(t)| \leq \exp\left(-\frac{1}{2}s^2 t^2\right), \quad t \in \mathbb{R}; \quad (116)$$

- (ii) $\sigma_{\max}(Z) < \infty$, always;
- (iii) $S(Z) = [0, \sigma_{\max}(Z)]$; in particular the supremum is attained;
- (iv) Z is Gaussian-free if and only if $\sigma_{\max}(Z) = 0$.

Proof. (i) If $s \in S(Z)$ then g_s is a characteristic function, so $|g_s(t)| \leq 1$ by Proposition 3.2 (i), which is Eq. (116).

(ii) Suppose $S(Z)$ were unbounded and pick $s_n \in S(Z)$ with $s_n \rightarrow \infty$. Fix $t \neq 0$. By Eq. (116), $|\varphi_Z(t)| \leq \exp(-\frac{1}{2}s_n^2 t^2) \rightarrow 0$, so $\varphi_Z(t) = 0$. Thus φ_Z vanishes on $\mathbb{R} \setminus \{0\}$ while $\varphi_Z(0) = 1$, contradicting the continuity of φ_Z (Proposition 3.2 (ii)).

(iii) First, $0 \in S(Z)$, and $S(Z)$ is downward closed: if $s \in S(Z)$ and $0 \leq s' \leq s$, then

$$g_{s'}(t) = g_s(t) \cdot \exp\left(-\frac{1}{2}(s^2 - s'^2)t^2\right) \quad (117)$$

is a product of the characteristic function g_s with that of a $\mathcal{N}(0, s^2 - s'^2)$ variable, hence a characteristic function by Eq. (9). So $S(Z)$ is an interval containing 0, and it is bounded by (ii). It remains to see that $\sigma := \sigma_{\max}(Z)$ itself lies in $S(Z)$. Choose $s_n \in S(Z)$ with $s_n \uparrow \sigma$. For every $t \in \mathbb{R}$ we have the pointwise convergence $g_{s_n}(t) \rightarrow g_\sigma(t)$, and g_σ is continuous at $t = 0$ because φ_Z and $\exp$ are. By Lévy's continuity Theorem 3.7 (ii), g_σ is a characteristic function, i.e. $\sigma \in S(Z)$.

(iv) If $\sigma_{\max}(Z) = \sigma > 0$ then, writing $\varphi_Z(t) = g_\sigma(t) \exp(-\frac{1}{2}\sigma^2 t^2)$ with the characteristic function g_σ from (iii), we exhibit a non-degenerate Gaussian convolution factor of $\mathcal{L}(Z)$, so Z is not Gaussian-free. Conversely, if Z is not Gaussian-free, say $Z \stackrel{d}{=} U + V$ with $U \perp\!\!\!\perp V$ and $V \sim \mathcal{N}(\mu, \tau^2)$, $\tau > 0$, then

$$\varphi_Z(t) \exp\left(\frac{1}{2}\tau^2 t^2\right) = \varphi_U(t) \exp(i\mu t) = \varphi_{U+\mu}(t) \quad (118)$$

is a characteristic function, so $\tau \in S(Z)$ and $\sigma_{\max}(Z) \geq \tau > 0$. $\square$

Lemma 6.6 puts us in a position to prove the following, which turns the informal wish of the introduction to this section into a theorem: *every* real-valued random variable splits into a Gaussian-free signal and independent Gaussian noise, and it does so in essentially one way.

Theorem 6.7 (Gaussian splitting). *Let Z be a real-valued random variable and put $\sigma := \sigma_{\max}(Z) \in [0, \infty)$. Then there exist a Gaussian-free random variable Y and a random variable $G \sim \mathcal{N}(0, \sigma^2)$ with $Y \perp\!\!\!\perp G$ such that*

$$Z \stackrel{d}{=} Y + G. \quad (119)$$

The decomposition is maximal and unique up to a translation: if $Z \stackrel{d}{=} Y' + G'$ with $Y' \perp\!\!\!\perp G'$, Y' Gaussian-free and $G' \sim \mathcal{N}(m, \tau^2)$, then $\tau = \sigma$ and $Y' \stackrel{d}{=} Y - m$.

Proof. By Lemma 6.6 (iii) we have $\sigma \in S(Z)$, so that $t \mapsto g_\sigma(t) = \varphi_Z(t) \exp(\frac{1}{2}\sigma^2 t^2)$ is the characteristic function of some random variable Y , and

$$\varphi_Z(t) = g_\sigma(t) \cdot \exp\left(-\frac{1}{2}\sigma^2 t^2\right) = \varphi_Y(t) \cdot \varphi_G(t) \quad (120)$$

for $G \sim \mathcal{N}(0, \sigma^2)$ chosen independent of Y ; this is Eq. (119) by Eq. (9) and Theorem 3.4 (i). We allow the degenerate case $\sigma = 0$, where $G = 0$ and $Y \stackrel{d}{=} Z$.

To see that Y is Gaussian-free, suppose $Y \stackrel{d}{=} U + V$ with $U \perp\!\!\!\perp V$ and $V \sim \mathcal{N}(\mu, \tau^2)$, $\tau > 0$. Then, as in the proof of [Lemma 6.6](#) (iv), $\varphi_Y(t) \exp(\frac{1}{2}\tau^2 t^2)$ is a characteristic function, hence so is

$$\varphi_Z(t) \exp\left(\frac{1}{2}(\sigma^2 + \tau^2)t^2\right) = \varphi_Y(t) \exp\left(\frac{1}{2}\tau^2 t^2\right), \quad (121)$$

so that $\bar{s} := \sqrt{\sigma^2 + \tau^2} \in S(Z)$. But then

$$\sigma_{\max}(Z) = \sigma < \sqrt{\sigma^2 + \tau^2} = \bar{s} \in S(Z), \quad (122)$$

contradicting the definition of $\sigma_{\max}$.

For the uniqueness statement, let $Z \stackrel{d}{=} Y' + G'$ be as stated. Then $\varphi_Z(t) \exp(\frac{1}{2}\tau^2 t^2) = \varphi_{Y'}(t) \exp(imt)$ is a characteristic function, so $\tau \in S(Z)$ and hence $\tau \leq \sigma$ by [Lemma 6.6](#) (iii). If we had $\tau < \sigma$, then

$$\varphi_{Y'}(t) \exp\left(\frac{1}{2}(\sigma^2 - \tau^2)t^2\right) = \varphi_Z(t) \exp\left(\frac{1}{2}\sigma^2 t^2\right) \exp(-imt) = \varphi_{Y-m}(t) \quad (123)$$

would be a characteristic function, exhibiting a $\mathcal{N}(0, \sigma^2 - \tau^2)$ factor of $\mathcal{L}(Y')$ and contradicting that Y' is Gaussian-free. So $\tau = \sigma$, and then $\varphi_{Y'}(t) = \varphi_Z(t) \exp(\frac{1}{2}\sigma^2 t^2) \exp(-imt) = \varphi_{Y-m}(t)$. $\square$

Remark 6.8 (A dichotomy, not a trichotomy). For every real-valued random variable Z exactly one of the following holds.

- (1) $\sigma_{\max}(Z) = 0$. Equivalently, Z is Gaussian-free. This is the degenerate case of [Theorem 6.7](#), with $G = 0$ and $Y \stackrel{d}{=} Z$.
- (2) $0 < \sigma_{\max}(Z) < \infty$. Here [Theorem 6.7](#) gives a proper decomposition $Z \stackrel{d}{=} Y + G$ with a non-degenerate Gaussian G , a Gaussian-free Y , and $Y \perp\!\!\!\perp G$.

One might expect a third case, in which arbitrarily large Gaussian factors can be shaved off; [Lemma 6.6](#) (ii) shows that this cannot happen. The reason is [Eq. \(116\)](#): splitting off a $\mathcal{N}(0, s^2)$ factor forces the characteristic function to decay at least as fast as $\exp(-\frac{1}{2}s^2 t^2)$, and no characteristic function can decay faster than every Gaussian without vanishing identically off the origin.

The tail bound [Eq. \(116\)](#) is also the most convenient practical criterion, alongside a support argument.

Corollary 6.9 (Two sufficient criteria). *Let Z be a real-valued random variable. Each of the following implies that Z is Gaussian-free.*

- (i) $\text{supp } \mathcal{L}(Z) \neq \mathbb{R}$; this holds in particular if Z is bounded, or bounded from one side.
- (ii) $\exp(\frac{1}{2}s^2 t^2)|\varphi_Z(t)|$ is unbounded in t for every $s > 0$; in particular, this holds whenever $\liminf_{|t| \rightarrow \infty} (-\log|\varphi_Z(t)|)/t^2 = 0$.

Proof. (i) The support of a convolution of two laws is the closure of the sum of their supports. A non-degenerate Gaussian law has support $\mathbb{R}$, so any law with a non-degenerate Gaussian convolution factor has support $\mathbb{R}$. (ii) is the contrapositive of [Eq. \(116\)](#), together with [Lemma 6.6](#) (iv). $\square$

Example 6.10 (Which laws are Gaussian-free?). (a) By [Corollary 6.9](#) (i), every law whose support is not all of $\mathbb{R}$ is Gaussian-free: the uniform, Rademacher and three-point laws of [Table 1](#), every Bernoulli law, and also the exponential, Gamma, Poisson and χ^2 laws, which are supported on a half line.

- (b) By [Corollary 6.9](#) (ii), every law whose characteristic function decays more slowly than any Gaussian is Gaussian-free. This covers the Laplace law, with $\varphi_Z(t) = (1 + b^2 t^2)^{-1}$ decaying polynomially; the Cauchy law, with $\varphi_Z(t) = \exp(-\gamma|t|)$; and the Student t_ν laws, whose characteristic functions decay exponentially. So the standard super-Gaussian sources of ICA are all Gaussian-free.
- (c) A non-degenerate Gaussian law is of course not Gaussian-free, with $\sigma_{\max}(\mathcal{N}(\mu, \sigma^2)) = \sigma$.
- (d) The instructive example is $Z := R + G$ with R Rademacher, $G \sim \mathcal{N}(0, 1)$ and $R \perp\!\!\!\perp G$, i.e. the Gaussian mixture $\frac{1}{2}\mathcal{N}(-1, 1) + \frac{1}{2}\mathcal{N}(1, 1)$. Here $\varphi_Z(t) = \cos(t) \exp(-\frac{1}{2}t^2)$, so $\varphi_Z(t) \exp(\frac{1}{2}s^2 t^2) = \cos(t) \exp(\frac{1}{2}(s^2 - 1)t^2)$ is unbounded for $s > 1$ and hence $\sigma_{\max}(Z) = 1$, with Gaussian-free part R – determined, as [Theorem 6.7](#) says, only up to a translation. By [Remark 3.18](#) (ii) the variable Z is *non-Gaussian*, so it is an admissible source for [Theorem 5.5](#), but it is *not* Gaussian-free and therefore not admissible for [Theorem 6.12](#) below. This shows that the hypothesis of this section is strictly stronger than that of [Section 5](#).

Remark 6.11 (The infinitely divisible case). For readers familiar with the Lévy–Khintchine representation ([Kallenberg, 2021](#), Chapter 7), there is a clean characterisation. Let Z be infinitely divisible with triplet (a, ν, γ) , i.e.

$$\varphi_Z(t) = \exp\left(i\gamma t - \frac{1}{2}at^2 + \int_{\mathbb{R}} \left(e^{itx} - 1 - itx\mathbf{1}_{\{|x| \leq 1\}}\right) \nu(dx)\right), \quad (124)$$

with $a \geq 0$ and a Lévy measure ν satisfying $\int \min(1, x^2) \nu(dx) < \infty$. Then Z is Gaussian-free if and only if $a = 0$, and in fact $\sigma_{\max}(Z)^2 = a$.

Indeed, if $a > 0$ then splitting the factor $\exp(-\frac{1}{2}at^2)$ off φ_Z leaves the characteristic function of the infinitely divisible law with triplet $(0, \nu, \gamma)$, so $\sqrt{a} \in S(Z)$. Conversely, assume $a = 0$ and let $s \in S(Z)$. Taking moduli,

$$-\log|\varphi_Z(t)| = \int_{\mathbb{R}} (1 - \cos(tx)) \nu(dx), \quad (125)$$

and this is $o(t^2)$ as $|t| \rightarrow \infty$: on $\{|x| \leq 1\}$ we have $(1 - \cos(tx))/t^2 \leq x^2/2$, which is ν -integrable there and tends to 0 pointwise, so dominated convergence applies; on $\{|x| > 1\}$ the integral is bounded by $2\nu(|x| > 1) < \infty$. On the other hand [Eq. \(116\)](#) gives $-\log|\varphi_Z(t)| \geq \frac{1}{2}s^2 t^2$. Comparing the two forces $s = 0$, so $\sigma_{\max}(Z) = 0$.

The same comparison gives the identity $\sigma_{\max}(Z)^2 = a$ for general $a \geq 0$, not just the two implications. Indeed, without assuming $a = 0$,

$$-\log|\varphi_Z(t)| = \frac{1}{2}at^2 + \int_{\mathbb{R}} (1 - \cos(tx)) \nu(dx) = \frac{1}{2}at^2 + o(t^2), \quad |t| \rightarrow \infty, \quad (126)$$

by the estimate just given, while [Eq. \(116\)](#) bounds the left hand side below by $\frac{1}{2}s^2 t^2$ for every $s \in S(Z)$. Dividing by t^2 and letting $|t| \rightarrow \infty$ yields $s^2 \leq a$, hence $\sigma_{\max}(Z)^2 \leq a$; combined with $\sqrt{a} \in S(Z)$ this is equality. Note that this argument never assumes that the cofactor is itself infinitely divisible – which matters, since factors of infinitely divisible laws need not be infinitely divisible.

6.2 Identifiability in the presence of additive Gaussian noise

In the following we explicitly assume that the sources of our ICA model have been separated according to [Theorem 6.7](#), so that they are mutually independent and Gaussian-free. Note that Gaussian random variables are considered noise and are usually not properly identifiable

in ICA models anyway, cf. [Remark 3.27](#). Assuming that the proper signals in an ICA model are Gaussian-free is therefore a reasonable assumption, even though it is more restrictive than the usual assumption of mere non-normality – and by [Example 6.10](#) (b) it is satisfied by all the standard super-Gaussian source models. The reward for this stronger assumption is a stronger identifiability result: it identifies the sources inside a mixture even in the presence of additive *and dependent* Gaussian noise.

Theorem 6.12 (Identifiability for independent Gaussian-free sources with additive Gaussian noise). *Let $X \in \mathbb{R}^p$ be a random vector and assume that we have two representations of X satisfying all the assumptions of [Theorem 5.5](#):*

$$A^{(1)}Z^{(1)} + E^{(1)} = X = A^{(2)}Z^{(2)} + E^{(2)}. \quad (127)$$

Assume furthermore that $A^{(1)}$ (equivalently, by [Theorem 5.5](#), $A^{(2)}$) has full column rank, $\text{rank}(A^{(1)}) = k \leq p$.

- (1) *If all components of $Z^{(1)}$ are Gaussian-free, then there exist a permutation matrix $P \in \mathbb{R}^{k \times k}$, an invertible diagonal matrix $\Lambda \in \mathbb{R}^{k \times k}$ and a (possibly degenerate) Gaussian random vector $H \sim \mathcal{N}(\xi, \Delta)$ with mean vector $\xi \in \mathbb{R}^k$ and diagonal, possibly degenerate, covariance matrix $\Delta \in \mathbb{R}^{k \times k}$, independent of $Z^{(1)}$ and of $E^{(2)}$, such that*

$$A^{(2)} = A^{(1)}P\Lambda, \quad (128)$$

$$Z^{(2)} \stackrel{d}{=} \Lambda^{-1}P^{-1}(Z^{(1)} + H), \quad H \perp\!\!\!\perp (Z^{(1)}, E^{(2)}), \quad (129)$$

$$E^{(1)} \stackrel{d}{=} A^{(1)}H + E^{(2)}, \quad (130)$$

$$\mu^{(1)} = A^{(1)}\xi + \mu^{(2)}, \quad (131)$$

$$\Sigma^{(1)} = A^{(1)}\Delta A^{(1)\top} + \Sigma^{(2)}. \quad (132)$$

- (2) *If, in addition, all components of $Z^{(2)}$ are Gaussian-free as well, then $H = \xi$ and $\Delta = 0$, and we further get*

$$Z^{(2)} \stackrel{d}{=} \Lambda^{-1}P^{-1}(Z^{(1)} + \xi), \quad E^{(2)} \stackrel{d}{=} E^{(1)} - A^{(1)}\xi, \quad \Sigma^{(2)} = \Sigma^{(1)}. \quad (133)$$

Proof. Part (1). By [Theorem 5.5](#) we already have $A^{(2)} = A^{(1)}P\Lambda$, and for each $j = 1, \dots, k$ separately one of the two cases

$$(\lambda_j Z_j^{(2)} + \nu_j) \stackrel{d}{=} Z_{\rho(j)}^{(1)} + G_j, \quad G_j \perp\!\!\!\perp Z_{\rho(j)}^{(1)}, \quad (134)$$

$$Z_{\rho(j)}^{(1)} \stackrel{d}{=} (\lambda_j Z_j^{(2)} + \nu_j) + G_j, \quad G_j \perp\!\!\!\perp Z_j^{(2)}, \quad (135)$$

with some $\nu_j \in \mathbb{R}$ and $G_j \sim \mathcal{N}(0, \sigma_j^2)$. Since $Z_{\rho(j)}^{(1)}$ is Gaussian-free, the second case exhibits a Gaussian convolution factor of $\mathcal{L}(Z_{\rho(j)}^{(1)})$ and can therefore only occur with $\sigma_j^2 = 0$, i.e. $G_j = 0$; but then it is subsumed by the first case. We may therefore assume the first case for all $j = 1, \dots, k$, which we rearrange into

$$e_j^\top Z^{(2)} = Z_j^{(2)} \stackrel{d}{=} \lambda_j^{-1} Z_{\rho(j)}^{(1)} - \lambda_j^{-1} \nu_j + \lambda_j^{-1} G_j, \quad G_j \perp\!\!\!\perp Z_{\rho(j)}^{(1)}, \quad (136)$$

$$= e_j^\top \Lambda^{-1} P^{-1} Z^{(1)} - e_j^\top \Lambda^{-1} \nu + e_j^\top \Lambda^{-1} G, \quad (137)$$

$$= e_j^\top \Lambda^{-1} P^{-1} (Z^{(1)} - P\nu + PG), \quad (138)$$

$$= e_j^\top \Lambda^{-1} P^{-1} (Z^{(1)} + H), \quad (139)$$

where we used the abbreviations

$$\nu := [\nu_1, \dots, \nu_k]^\top, \quad \xi := -P\nu, \quad (140)$$

$$G := [G_1, \dots, G_k]^\top \sim \mathcal{N}(0, \Gamma), \quad \Gamma := \text{diag}(\sigma_1^2, \dots, \sigma_k^2), \quad (141)$$

$$H := \xi + PG \sim \mathcal{N}(\xi, \Delta), \quad \Delta := P\Gamma P^\top. \quad (142)$$

Note that the G_j were constructed componentwise, so we may and do take G to have independent components; this implies that $\Delta = P\Gamma P^\top$ is again a diagonal matrix, and hence that H has independent components as well. Together with the fact that the components of $Z^{(1)}$ and of $Z^{(2)}$ are each mutually independent, we may gather all components into the single distributional equation

$$Z^{(2)} \stackrel{d}{=} \Lambda^{-1} P^{-1} (Z^{(1)} + H), \quad H \perp\!\!\!\perp Z^{(1)}. \quad (143)$$

By the continuous mapping theorem we may multiply [Eq. \(143\)](#) by $A^{(2)} = A^{(1)}P\Lambda$ and obtain

$$A^{(2)}Z^{(2)} \stackrel{d}{=} A^{(1)}Z^{(1)} + A^{(1)}H, \quad H \perp\!\!\!\perp Z^{(1)}, \quad (144)$$

which in terms of characteristic functions reads

$$\varphi_{A^{(2)}Z^{(2)}}(t) = \varphi_{A^{(1)}Z^{(1)}}(t) \cdot \varphi_{A^{(1)}H}(t). \quad (145)$$

Plugging this into the characteristic functions of the original model equation,

$$\varphi_{A^{(2)}Z^{(2)}}(t) \cdot \varphi_{E^{(2)}}(t) = \varphi_{A^{(1)}Z^{(1)}}(t) \cdot \varphi_{E^{(1)}}(t), \quad (146)$$

we obtain an identity that we would like to divide by $\varphi_{A^{(1)}Z^{(1)}}$. Here we may not simply cancel the factor $\varphi_{A^{(1)}Z^{(1)}}$, since a characteristic function may well have zeros: for $p = k = 1$, $A^{(1)} = 1$ and $Z^{(1)} \sim \text{Unif}[-1, 1]$ – an admissible Gaussian-free, non-Gaussian source – one has $\varphi_{Z^{(1)}}(t) = \sin(t)/t$, which vanishes at $t = \pi$. We argue locally instead. By continuity and $\varphi_{A^{(1)}Z^{(1)}}(0) = 1$ there is a $\delta > 0$ with $\varphi_{A^{(1)}Z^{(1)}} \neq 0$ on the ball $B_\delta(0) \subseteq \mathbb{R}^p$, and cancelling there gives

$$\varphi_{A^{(1)}H}(t) \cdot \varphi_{E^{(2)}}(t) = \varphi_{E^{(1)}}(t), \quad t \in B_\delta(0). \quad (147)$$

Both sides of [Eq. \(147\)](#) are characteristic functions of (possibly degenerate) multivariate Gaussian distributions, hence of the form $\exp(q(t))$ with polynomials q of degree ≤ 2 and $q(0) = 0$; here we may in addition take H independent of $E^{(2)}$: all assertions here are statements about laws, so we may enlarge the underlying probability space (replacing Ω by $\Omega \times [0, 1]$) and realise an H with law $\mathcal{N}(\xi, \Delta)$ independent of the pair $(Z^{(1)}, E^{(2)})$. Writing the two sides as $\exp(q_1)$ and $\exp(q_2)$, we get $\exp(q_1 - q_2) \equiv 1$ on $B_\delta(0)$, so the continuous function $(q_1 - q_2)/(2\pi i)$ takes values in $\mathbb{Z}$ on the connected set $B_\delta(0)$ and vanishes at the origin; hence $q_1 = q_2$ on $B_\delta(0)$. Two polynomials that agree on a non-empty open subset of $\mathbb{R}^p$ agree everywhere, so $q_1 = q_2$ identically. Hence [Eq. \(147\)](#) in fact holds for every $t \in \mathbb{R}^p$, and [Theorem 3.4](#) gives

$$A^{(1)}H + E^{(2)} \stackrel{d}{=} E^{(1)}, \quad H \perp\!\!\!\perp (Z^{(1)}, E^{(2)}), \quad (148)$$

and comparing mean vectors and covariance matrices yields

$$A^{(1)}\xi + \mu^{(2)} = \mu^{(1)}, \quad (149)$$

$$A^{(1)}\Delta A^{(1)\top} + \Sigma^{(2)} = \Sigma^{(1)}. \quad (150)$$

This proves all claims of part (1).

Part (2). If the components $Z_j^{(2)}$ are Gaussian-free as well, then so are the affinely transformed variables $\lambda_j Z_j^{(2)} + \nu_j$ by [Remark 6.2](#), and the only possibility left is

$$(\lambda_j Z_j^{(2)} + \nu_j) \stackrel{d}{=} Z_{\rho(j)}^{(1)} + G_j, \quad G_j \perp\!\!\!\perp Z_{\rho(j)}^{(1)}, \quad (151)$$

with $\sigma_j^2 = 0$ and thus $G_j = 0$. This shows $H = \xi$ and $\Delta = 0$, and everything else follows from part (1). $\square$

Remark 6.13 (What [Theorem 6.12](#) buys us). Compared with [Theorem 5.5](#), the stronger assumption on the sources removes the fourth ambiguity of [Remark 5.6](#) – but only part (2) removes it entirely. Part (1) makes it one-sided: any residual Gaussian must sit in $Z^{(2)}$ and is exactly compensated in $E^{(2)}$, which is what the vector H records. It is not vacuous. Take $p = k = 1$, $A^{(1)} = A^{(2)} = [1]$, $Z^{(1)}$ Rademacher (Gaussian-free) with $E^{(1)} \sim \mathcal{N}(0, 1)$, against $Z^{(2)} = Z^{(1)} + E^{(1)}$ (non-constant and, by [Remark 3.18](#) (ii), non-Gaussian) with $E^{(2)} = 0$: both representations satisfy every hypothesis of part (1), yet $\mathcal{L}(Z^{(2)})$ is absolutely continuous while $\mathcal{L}(Z^{(1)})$ has two atoms, so no affine map carries one to the other. Three aspects deserve emphasis.

- (i) The Gaussian noise vectors $E^{(i)}$ are *not* assumed to have independent components, nor a non-degenerate covariance matrix. They may be arbitrarily dependent across the p observed coordinates, and they may be partially deterministic. Nevertheless, under the hypothesis of part (2), the sources are pinned down up to permutation, scale and translation.
- (ii) Part (2) is a genuine two-sided statement: as soon as *both* candidate source vectors are Gaussian-free, not only the sources but also the noise laws agree, $\Sigma^{(1)} = \Sigma^{(2)}$, so the whole model $(A, \mathcal{L}(Z), \mathcal{L}(E))$ is identified up to the unavoidable permutation, scaling and shift.
- (iii) The Gaussian-free assumption is not a restriction on which *observations* can be modelled, only on how the model is parametrised. Indeed, by [Theorem 6.7](#) every source splits as $Z_j \stackrel{d}{=} Y_j + G_j$ with Y_j Gaussian-free and $G_j \sim \mathcal{N}(0, \sigma_j^2)$ independent of Y_j , and moving the Gaussian part into the noise is the trivial direction of [Lemma 5.2](#): writing a_j for the j -th column of A ,

$$a_j Z_j \stackrel{d}{=} a_j Y_j + a_j G_j, \quad a_j G_j \sim \mathcal{N}(0, \sigma_j^2 a_j a_j^\top) \perp\!\!\!\perp Y_j, \quad (152)$$

so that $a_j G_j$ may simply be added to E – which is why we allow E to have an arbitrary, possibly degenerate covariance matrix. Since the Z_j are mutually independent, the pairs (Y_j, G_j) may be taken jointly independent, so all k Gaussian parts can be moved at once. ([Lemma 5.2](#) proves the harder converse direction, moving Gaussian noise out of E and into extra columns of A .) One caveat: if Z_j was itself Gaussian, then its Gaussian-free part Y_j is a *constant*, and the corresponding column must then be deleted and absorbed into the offset μ , exactly as in Step 1 of [Remark 4.1](#), before [Theorem 6.12](#) can be applied to the reduced model. What the assumption buys is that this splitting and pruning has been carried out, so that no Gaussian mass is left sitting ambiguously between Z and E .

In practice, part (1) is the statement one applies when a candidate solution is compared against the ground truth, and part (2) is the statement one applies when both are outputs of a procedure that is guaranteed to return Gaussian-free sources.

7 Complete Noiseless ICA: Estimation by Equivariant Gradient Descent

The previous sections answered the question of *what* is identifiable. This section answers the complementary question of *how* one estimates it, in the special case that is by far the most common in applications: a square invertible mixing matrix and no noise. We first specialise the identifiability theory to that case (Section 7.1), then set up the maximum likelihood objective (Section 7.2), then explain what the customary “preconditioner” in the gradient step really is (Section 7.3), and finally analyse when the resulting algorithm converges to a separating solution (Section 7.4).

Example 7.1 (The cocktail party problem). Two loudspeakers in a room play different pieces of music, modelled as real-valued signals $Z_1(t)$ and $Z_2(t)$ indexed by time t . Two microphones at different positions each record a mixture of the two,

$$X_1(t) = a_{11}Z_1(t) + a_{12}Z_2(t), \quad X_2(t) = a_{21}Z_1(t) + a_{22}Z_2(t). \quad (153)$$

The task is to recover the two pieces of music $Z_j(t)$ from the two recorded mixtures $X_i(t)$. What makes this more than a matrix inversion is that *both* the mixing matrix $A = (a_{ij})$ and the sources Z_j are unknown; all that is assumed is that the two sources are statistically independent. Corollary 7.4 below says that this is enough – provided at most one source is Gaussian.

7.1 The complete noiseless model and its identifiability

Assumption 7.2 (Complete noiseless ICA model). We observe $X(1), \dots, X(T)$ in $\mathbb{R}^p$, all with the same law, and assume:

- (i) *Independent sources*. There are random vectors $Z(t) = [Z_1(t), \dots, Z_k(t)]^\top$ in $\mathbb{R}^k$ whose components $\{Z_1(t), \dots, Z_k(t)\}$ are mutually independent for each fixed t , and almost surely non-constant. Nothing is assumed about the dependence *across* t .
- (ii) *Linear noiseless mixing*. $X(t) = AZ(t)$ for all t , with a matrix $A \in \mathbb{R}^{p \times k}$ that does not depend on t .
- (iii) *Completeness*. $p = k$ and A is invertible. We write $W := A^{-1}$ and denote the rows of W by $w_1^\top, \dots, w_k^\top$.
- (iv) *Non-Gaussianity*. At most one of the components $Z_1(t), \dots, Z_k(t)$ is Gaussian.

For the estimation part we shall in addition normalise:

- (v) *Centring and scale*. $\mathbb{E}[Z_j(t)] = 0$ for all j and t , and the scale of each Z_j is fixed either by $\text{Var}(Z_j) = 1$ or by the fixed-point normalisation Eq. (182); see Remark 7.3.

Since the law of $X(t)$ does not depend on t , we drop the argument and write $X = AZ$ for a generic sample whenever only the common law matters.

Remark 7.3 (The two scale conventions). Rescaling the sources is without loss of generality: it is the reparametrisation $X = (AD)(D^{-1}Z)$ with D diagonal and invertible, which preserves Assumption 7.2 (i)–(iv). Centring them, however, requires centring the data as well, since Assumption 7.2 (ii) carries no offset; that is step (1) of Remark 7.9. Two scale conventions appear below, and they are *not* the same. The first is $\text{Var}(Z_j) = 1$, used wherever only second-order quantities matter (Remark 7.9); it requires the extra assumption $\mathbb{E}[Z_j^2] < \infty$, which Assumption 7.2 does not otherwise impose. The second is the fixed-point scale of Lemma 7.18, which is the one Theorem 7.20 needs and which is generically *not* the unit-variance scale.

The identifiability statement usually quoted in the data-science literature is the following. It is exactly the specialisation of [Theorem 4.2](#) to a square invertible mixing matrix, with the “at most one Gaussian” clause handled by an extra argument at the end of the proof.

Corollary 7.4 (Identifiability of the complete noiseless model; [Comon, 1994](#), Theorem 11). *Let X be a random vector in $\mathbb{R}^k$ with two representations*

$$A^{(1)}Z^{(1)} + \mu^{(1)} = X = A^{(2)}Z^{(2)} + \mu^{(2)}, \quad (154)$$

where for $i = 1, 2$:

(i) $A^{(i)} \in \mathbb{R}^{k \times k}$ is invertible and $\mu^{(i)} \in \mathbb{R}^k$;

(ii) the components of $Z^{(i)}$ are mutually independent and almost surely non-constant;

and, for $i = 1$ only,

(iii) at most one component of $Z^{(1)}$ is Gaussian.

Then there exist a permutation matrix $P \in \mathbb{R}^{k \times k}$, an invertible diagonal matrix $\Lambda \in \mathbb{R}^{k \times k}$ and a vector $c \in \mathbb{R}^k$ such that

$$A^{(2)} = A^{(1)}P\Lambda, \quad Z^{(1)} = P\Lambda Z^{(2)} + c \quad \text{almost surely.} \quad (155)$$

If moreover both source vectors are centred with unit variances, then $\Lambda = \text{diag}(\pm 1, \dots, \pm 1)$ and $c = 0$: the sources are determined up to permutation and sign, and the columns of the mixing matrix up to permutation and sign.

Proof. An invertible matrix has non-zero, pairwise non-proportional columns, so both representations satisfy the hypotheses of [Theorem 4.2](#).

Step 1: matching the non-Gaussian columns. Let $N_i := \{j : Z_j^{(i)} \text{ is non-Gaussian}\}$. Let $l \in N_2$. By [Theorem 4.2](#) (1) the column $a_l^{(2)}$ must be proportional to some column $a_{\rho(l)}^{(1)}$ of $A^{(1)}$, and by [Theorem 4.2](#) (2) the source $Z_{\rho(l)}^{(1)}$ is then non-Gaussian, i.e. $\rho(l) \in N_1$. The index $\rho(l)$ is unique because the columns of $A^{(1)}$ are pairwise non-proportional, and the map $\rho: N_2 \rightarrow N_1$ is injective: if $\rho(l) = \rho(l')$ then $a_l^{(2)} \propto a_{\rho(l)}^{(1)} \propto a_{\rho(l')}^{(1)} \propto a_{l'}^{(2)}$, contradicting pairwise non-proportionality of the columns of $A^{(2)}$. Hence $|N_2| \leq |N_1|$, and by symmetry $|N_1| = |N_2| =: n$, with $\rho: N_2 \rightarrow N_1$ a bijection. In particular the two representations have Gaussian sources in equal number, $k - n$ each. This is why hypothesis (iii) needs to be imposed on one of the two representations only: assuming it for $i = 1$ gives $n \geq k - 1$, and hence at most one Gaussian component of $Z^{(2)}$ as well. In either case $n \in \{k - 1, k\}$.

Step 2: reduction to a single matrix. Put $C := (A^{(1)})^{-1}A^{(2)}$, which is invertible, so that $A^{(2)} = A^{(1)}C$; it therefore suffices to prove $C = P\Lambda$. Multiplying [Eq. \(154\)](#) by $(A^{(1)})^{-1}$ gives, almost surely,

$$Z^{(1)} = CZ^{(2)} + d, \quad d := (A^{(1)})^{-1}(\mu^{(2)} - \mu^{(1)}), \quad (156)$$

so a factorisation $C = P\Lambda$ gives the second assertion as well, with $c := d$. For $l \in N_2$ we have $a_l^{(2)} = \lambda_l a_{\rho(l)}^{(1)}$ with $\lambda_l \neq 0$, hence $Ce_l = \lambda_l e_{\rho(l)}$: every column of C indexed by N_2 is a non-zero multiple of a standard basis vector.

Step 3: the case $n = k$. Here $N_2 = \{1, \dots, k\}$ and ρ is a permutation, so every column of C is of that form and $C = P(\rho)\Lambda$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_k)$ invertible.

Step 4: the case $n = k - 1$. Let $l_0 \notin N_2$ and $k_0 \notin N_1$ be the two exceptional indices, so that $Z_{l_0}^{(2)}$ and $Z_{k_0}^{(1)}$ are Gaussian, and non-degenerate because they are non-constant. By Step

2 every column of C except the l_0 -th is a non-zero multiple of a standard basis vector, the corresponding indices $\rho(l)$ exhausting $\{1, \dots, k\} \setminus \{k_0\}$. Reading off rows, [Eq. \(156\)](#) becomes

$$Z_{k_0}^{(1)} = \beta Z_{l_0}^{(2)} + d_{k_0}, \quad \beta := C_{k_0 l_0}, \quad (157)$$

$$Z_j^{(1)} = \lambda_{\rho^{-1}(j)} Z_{\rho^{-1}(j)}^{(2)} + \beta_j Z_{l_0}^{(2)} + d_j, \quad \beta_j := C_{j l_0}, \quad j \neq k_0. \quad (158)$$

Invertibility of C forces $\beta \neq 0$. Suppose, for contradiction, that $\beta_j \neq 0$ for some $j \neq k_0$. Abbreviate $U := Z_{\rho^{-1}(j)}^{(2)}$, $\lambda := \lambda_{\rho^{-1}(j)} \neq 0$ and $G := Z_{l_0}^{(2)} \sim \mathcal{N}(m, \tau^2)$ with $\tau > 0$; note $U \perp\!\!\!\perp G$. The components $Z_j^{(1)}$ and $Z_{k_0}^{(1)}$ are independent, so for all $s, t \in \mathbb{R}$,

$$\varphi_U(s\lambda) \varphi_G(s\beta_j + t\beta) = \varphi_U(s\lambda) \varphi_G(s\beta_j) \varphi_G(t\beta), \quad (159)$$

where the constants d_j, d_{k_0} have cancelled. Since φ_U is continuous with $\varphi_U(0) = 1$, it is non-zero in a neighbourhood of the origin, so we may cancel it there and take logarithms of the nowhere vanishing Gaussian characteristic functions. The two exponents are the quadratics of [Notation 2.4](#); their difference is continuous, takes values in $2\pi i \mathbb{Z}$ on the connected set where the identity holds, and vanishes at $(0, 0)$, hence is identically zero. This gives, for all small s, t ,

$$im(s\beta_j + t\beta) - \frac{1}{2}\tau^2(s\beta_j + t\beta)^2 = im(s\beta_j + t\beta) - \frac{1}{2}\tau^2((s\beta_j)^2 + (t\beta)^2), \quad (160)$$

i.e. $\tau^2\beta_j\beta st = 0$ for all small s, t . As $\tau > 0$ and $\beta \neq 0$ this forces $\beta_j = 0$, a contradiction.

Hence $\beta_j = 0$ for all $j \neq k_0$, so the l_0 -th column of C equals βe_{k_0} and $C = P\Lambda$ for a permutation matrix P and an invertible diagonal Λ . With [Eq. \(156\)](#) this proves the first display, and $A^{(2)} = A^{(1)}C = A^{(1)}P\Lambda$.

The normalised form. If both source vectors are centred, taking expectations in $Z^{(1)} = P\Lambda Z^{(2)} + c$ gives $c = 0$. If in addition all variances equal 1, then each component of $Z^{(1)}$ equals λ times a component of $Z^{(2)}$, where λ is the corresponding diagonal entry of Λ ; taking variances gives $1 = \lambda^2$, so $\lambda = \pm 1$. $\square$

Remark 7.5 (Two Gaussian sources already destroy identifiability). Hypothesis (iii) of [Corollary 7.4](#) cannot be weakened. Let $k = 2$ and $Z \sim \mathcal{N}(0, I_2)$, $A^{(1)} = I_2$, so $X = Z$. For any orthogonal $Q \in O(2)$ put $A^{(2)} := Q^\top$ and $Z^{(2)} := QZ$. Then $Z^{(2)} \sim \mathcal{N}(0, I_2)$ again has independent components and $A^{(2)}Z^{(2)} = Q^\top QZ = X$, so the whole rotation group is compatible with the observed law; cf. [Remark 3.27](#). The same construction applies inside any two-dimensional Gaussian block, which is why at most one Gaussian source is allowed. This degeneracy reappears analytically in [Proposition 7.22](#) below: two Gaussian sources force the stability quantities to satisfy $\zeta_j \zeta_l = 1$, so the linearised dynamics acquires a neutral direction. With the true score the implication reverses as well, by [Corollary 7.23](#) ($\zeta_j \geq 1$ with equality only in the Gaussian case). When in addition the two source variances agree, the neutral direction is the antisymmetric one, i.e. the infinitesimal version of the rotation Q above; for unequal variances it is $(\sigma_j^2, -\sigma_l^2)$ instead.

Remark 7.6 (Relation to the general theory). [Corollary 7.4](#) is the noiseless, square, full-rank corner of the theory of [Sections 5 and 6](#). Comparing: [Theorem 5.5](#) allows a rectangular A and additive Gaussian noise but must then leave the sources undetermined up to an extra additive Gaussian term; [Theorem 6.12](#) removes that term at the price of the Gaussian-free hypothesis. In the complete noiseless model there is no noise to hide anything in, so the weakest hypothesis – at most one Gaussian source – already suffices, and the conclusion holds almost surely rather than merely in distribution.

7.2 The maximum likelihood objective

From here on we work under [Assumption 7.2](#) and additionally assume that each source Z_j has a strictly positive Lebesgue density p_{Z_j} . The model is parametrised by the unmixing matrix $W = A^{-1}$ together with k model source densities $p_1, \dots, p_k$, which are a modelling choice and need not equal the true densities $p_{Z_1}, \dots, p_{Z_k}$:

$$\theta = (W, p_1, \dots, p_k), \quad p_Z(z) = p_1(z_1) \cdots p_k(z_k). \quad (161)$$

Proposition 7.7 (Model density and log-likelihood). *Under [Assumption 7.2](#), the density of $X = W^{-1}Z$ induced by the parameter θ is*

$$q_\theta(x) = |\det W| \prod_{j=1}^k p_j(w_j^\top x), \quad x \in \mathbb{R}^k. \quad (162)$$

Consequently the population log-likelihood, with X distributed according to the true law q , is

$$\mathcal{L}(\theta) := \mathbb{E}_{X \sim q}[\log q_\theta(X)] = \log |\det W| + \sum_{j=1}^k \mathbb{E}_{X \sim q}[\log p_j(w_j^\top X)], \quad (163)$$

and its empirical counterpart is

$$\mathcal{L}_T(\theta) := \frac{1}{T} \sum_{t=1}^T \log q_\theta(X(t)) = \log |\det W| + \frac{1}{T} \sum_{t=1}^T \sum_{j=1}^k \log p_j(w_j^\top X(t)). \quad (164)$$

Moreover, if q has a density and finite differential entropy,

$$\mathcal{L}(\theta) = -\text{KL}(q \parallel q_\theta) - H(q), \quad H(q) := -\mathbb{E}_{X \sim q}[\log q(X)]. \quad (165)$$

Proof. [Eq. \(162\)](#) is the change-of-variables formula for the diffeomorphism $z = Wx$ with Jacobian determinant $\det W$: for a Borel set S , $\mathbb{P}[X \in S] = \mathbb{P}[Z \in WS] = \int_{WS} p_Z(z) dz = \int_S p_Z(Wx) |\det W| dx$. [Eqs. \(163\)](#) and [\(164\)](#) follow by taking logarithms and using $\log \prod_j = \sum_j \log$. For [Eq. \(165\)](#), $\text{KL}(q \parallel q_\theta) = \mathbb{E}_q[\log q] - \mathbb{E}_q[\log q_\theta] = -H(q) - \mathcal{L}(\theta)$. $\square$

Since $H(q)$ does not depend on θ , [Eq. \(165\)](#) says that maximising $\mathcal{L}$ is the same as minimising the Kullback–Leibler divergence from the true law to the model.

Remark 7.8 (The empirical objective does not need independence over time). [Eq. \(164\)](#) is the exact log-likelihood only if the samples $X(1), \dots, X(T)$ are independent. [Assumption 7.2](#) (i) deliberately does not assume this: in the cocktail-party application consecutive samples of an audio signal are strongly dependent. Nothing below breaks, because we never use [Eq. \(164\)](#) as a likelihood. We use it as an *M-estimation criterion* whose population version [Eq. \(163\)](#) depends only on the common law of $X(t)$, and all our statements – the stationarity conditions of [Proposition 7.17](#) and the stability analysis of [Theorem 7.20](#) – are statements about that population criterion. Provided $(Z(t))_t$ is stationary and ergodic – enough for a law of large numbers, so that $\mathcal{L}_T \rightarrow \mathcal{L}$ almost surely – dependence across t affects only the rate of that convergence, not the location of the optima. Some such hypothesis is needed: if $Z(t) \equiv Z(1)$ for every t , then all five conditions of [Assumption 7.2](#) hold, yet $\mathcal{L}_T = \log q_\theta(x(1))$ for every T , and this is unbounded above (send a row of W towards a direction orthogonal to $x(1)$ while $\log |\det W| \rightarrow \infty$).

Remark 7.9 (Centring and whitening). Two preprocessing steps are customary.

- (1) *Centring*: replace $X(t)$ by $X(t) - \mathbb{E}[X]$, so that $\mathbb{E}[X] = 0$; since A is invertible this is equivalent to $\mathbb{E}[Z] = 0$, and being a deterministic shift of each Z_j separately it preserves [Assumption 7.2](#) (i). In practice one subtracts the sample mean $\bar{X}$, which is an approximation to this whose validity again needs the law of large numbers of [Remark 7.8](#): subtracting $\bar{X}$ mixes the time points and can destroy the independence of the components at a fixed t .
- (2) *Whitening*: assume $\mathbb{E}\|Z\|^2 < \infty$ (so that $\Sigma_X := \text{Cov}(X)$ exists and is invertible) and replace X by $X' := \Sigma_X^{-1/2} X$, whose mixing matrix is $A' := \Sigma_X^{-1/2} A$. With centred unit-variance sources this gives

$$I_k = \text{Cov}(X') = A' \text{Cov}(Z) (A')^\top = A' (A')^\top, \quad (166)$$

with the *symmetric* square root $\Sigma_X^{-1/2}$, which exists and is invertible because the Z_j are non-constant with finite variance; that is, the mixing matrix of the whitened data is orthogonal. For algorithms that constrain W to $O(k)$ this reduces the number of free parameters from k^2 to the $k(k-1)/2$ dimensions of $O(k)$, and it is what most ICA algorithms exploit. The relative-gradient update [Eq. \(172\)](#) does not preserve orthogonality, so it gains no such reduction.

The algorithm developed below needs neither step in order to be well defined. By [Proposition 7.15](#) its trajectory depends on A only through the initial global system matrix $R_0 = W_0 A$, so whitening cannot change the shape of the dynamics – but it does change R_0 , replacing it by an orthogonal matrix when $W_0 = \Sigma_X^{-1/2}$, and a better conditioned starting point is a genuine practical benefit. Centring is advisable in any case, because [Theorem 7.20](#) assumes $\mathbb{E}[Z] = 0$.

Proposition 7.10 (Gradient of the log-likelihood). *Fix source densities $p_1, \dots, p_k$ that are differentiable and strictly positive, and write*

$$\eta_j := (\log p_j)' = \frac{p_j'}{p_j} \quad (167)$$

for the model score of the j -th source ([Remark 7.11](#)). Put $C := WX$, and write $\eta(C)$ for the vector with components $\eta_j(C_j)$. Then, under integrability conditions permitting differentiation under the expectation,

$$\frac{\partial \mathcal{L}}{\partial W_{ij}} = \mathbb{E}[\eta_i(C_i) X_j] + (W^{-1})_{ji}, \quad \text{that is} \quad \nabla_W \mathcal{L} = W^{-\top} + \mathbb{E}[\eta(C) X^\top]. \quad (168)$$

Proof. Differentiating [Eq. \(163\)](#) term by term, $\partial_{W_{ij}} \log p_l(w_l^\top X) = \delta_{li} \eta_l(w_l^\top X) X_j$ by the chain rule, and summing over l leaves the single term $\eta_i(C_i) X_j$. For the determinant, Jacobi's formula gives $\partial_{W_{ij}} \log |\det W| = (W^{-1})_{ji}$, i.e. $\nabla_W \log |\det W| = W^{-\top}$. $\square$

Remark 7.11 (Two scores: the model score and the true score). From here to the end of the section, two different objects compete for the name “score”, and almost every subtlety in what follows is a statement about their relationship. We keep them notationally apart.

- $\eta_j := (\log p_j)'$ of [Eq. \(167\)](#), the score of the *model* density p_j . This is a modelling choice, fixed by the user before the algorithm is run, and it is the function the algorithm actually evaluates – in [Algorithm 7.28](#) it is applied to the current estimate C_j of the j -th source. We also call it the *nonlinearity*. Nothing forces it to be the score of the *true* source density. The three used in practice are in fact scores of perfectly good densities – just not, in general, of the right one: – $\tanh$ is the score of the hyperbolic secant density $\frac{1}{\pi} \text{sech}$, with kurt = +2; $\tanh$ –id is the score of $\propto \cosh(u) e^{-u^2/2}$, i.e. of

the mixture $\frac{1}{2}(\mathcal{N}(-1, 1) + \mathcal{N}(1, 1))$, with kurt $= -\frac{1}{2}$; and $-u^3$ is the score of $\propto e^{-u^4/4}$, with kurt ≈ -0.81 . Choosing η_j thus amounts to positing a source law, and the three choices posit a leptokurtic, a mildly platykurtic and a platykurtic one respectively. Not every nonlinearity is a score, though: $+u^3$ would require the non-integrable $e^{u^4/4}$.

- $\eta_j^* := (\log p_{Z_j})'$, the score of the *true* density p_{Z_j} of the source Z_j . This is a property of the data generating process, and it is unknown.

The maximum likelihood derivation of [Section 7.2](#) would use η_j^* if it could; the algorithm uses η_j because it must. [Theorem 7.20](#) is a statement about an arbitrary η_j , and this is exactly what makes it useful: it says how far η_j may deviate from η_j^* before the separating solution stops attracting. [Corollary 7.23](#) is the special case $\eta_j = \eta_j^*$. Note that η_j is evaluated along the trajectory at $C_j = (RZ)_j$, a mixture of the sources, whereas η_j^* belongs to Z_j alone; the two are brought into contact only through their values on the sources, i.e. at the fixed point $R = I_k$, where $C_j = Z_j$. A third score makes a single appearance, after [Proposition 7.17](#): $(\log p_{C_j})'$, the true score of the current *estimate*, which coincides with η_j^* at $R = I_k$.

7.3 The relative gradient: what the “preconditioner” really is

A plain gradient ascent step $W \leftarrow W + \alpha \nabla_W \mathcal{L}$ requires the matrix inverse $W^{-\top}$ at every step, which is both expensive and numerically delicate. The standard remedy is to multiply the gradient by $W^\top W$ on the right, giving the update

$$W \leftarrow W + \alpha (\nabla_W \mathcal{L}) W^\top W. \quad (169)$$

This is often presented as a Newton step with an approximate inverse Hessian. It is not, and [Remark 7.14](#) explains what it is instead: an *exact* gradient, taken with respect to a different – and for this problem far more natural – Riemannian metric on the group of invertible matrices.

Definition 7.12 (Relative gradient). Let F be a differentiable real-valued function on the group $GL(k)$ of invertible $k \times k$ matrices. The *relative gradient* of F at W is the matrix $\nabla^{\text{rel}} F(W)$ determined by the first-order expansion under *multiplicative* perturbations,

$$F((I_k + \varepsilon)W) = F(W) + \langle \nabla^{\text{rel}} F(W), \varepsilon \rangle + o(\|\varepsilon\|), \quad \varepsilon \in \mathbb{R}^{k \times k}, \quad (170)$$

where $\langle M, N \rangle := \text{tr}(M^\top N)$ is the Frobenius inner product.

Proposition 7.13 (Relative gradient of the ICA likelihood). *For any differentiable F on $GL(k)$ we have $\nabla^{\text{rel}} F(W) = (\nabla F(W))W^\top$, so that a relative gradient ascent step $W \leftarrow (I_k + \alpha \nabla^{\text{rel}} F(W))W$ is exactly [Eq. \(169\)](#). For the ICA log-likelihood [Eq. \(163\)](#),*

$$\nabla^{\text{rel}} \mathcal{L}(W) = I_k + \mathbb{E}[\eta(C)C^\top], \quad C = WX, \quad (171)$$

and the update rule reads

$$W \leftarrow \left(I_k + \alpha (I_k + \mathbb{E}[\eta(C)C^\top]) \right) W. \quad (172)$$

Proof. By the ordinary chain rule, $F((I_k + \varepsilon)W) = F(W + \varepsilon W) = F(W) + \langle \nabla F(W), \varepsilon W \rangle + o(\|\varepsilon\|)$, and $\langle \nabla F(W), \varepsilon W \rangle = \text{tr}(\nabla F(W)^\top \varepsilon W) = \text{tr}(W \nabla F(W)^\top \varepsilon) = \langle \nabla F(W)W^\top, \varepsilon \rangle$, which is [Eq. \(170\)](#) with $\nabla^{\text{rel}} F(W) = \nabla F(W)W^\top$. Substituting [Eq. \(168\)](#) and using $W^{-\top}W^\top = I_k$ and $X^\top W^\top = C^\top$,

$$\nabla^{\text{rel}} \mathcal{L}(W) = \left(W^{-\top} + \mathbb{E}[\eta(C)X^\top] \right) W^\top = I_k + \mathbb{E}[\eta(C)X^\top W^\top] = I_k + \mathbb{E}[\eta(C)C^\top]. \quad (173)$$

□

Remark 7.14 (It is a metric, not a Hessian). Differentiating [Eq. \(168\)](#) once more gives the true Hessian of $\mathcal{L}$,

$$\frac{\partial^2 \mathcal{L}}{\partial W_{lm} \partial W_{ij}} = \delta_{il} \mathbb{E}[\eta'_i(C_i) X_j X_m] - (W^{-1})_{jl} (W^{-1})_{mi}, \quad (174)$$

using $\partial(W^{-1})_{ji}/\partial W_{lm} = -(W^{-1})_{jl}(W^{-1})_{mi}$. The operator implicitly used in [Eq. \(169\)](#) is, in coordinates,

$$K_{(lm),(ij)} = \delta_{li} (W^\top W)_{jm} = \delta_{li} \sum_{r=1}^k W_{rj} W_{rm}, \quad (175)$$

since $[(\nabla \mathcal{L}) W^\top W]_{lm} = \sum_{ij} K_{(lm),(ij)} (\nabla \mathcal{L})_{ij}$. Two observations settle the matter. First, K does not depend on the data at all, whereas [Eq. \(174\)](#) does, through $\mathbb{E}[\eta'_i(C_i) X_j X_m]$; so K cannot be the inverse of [Eq. \(174\)](#), not even approximately, uniformly in the source distribution. Second, K has an exact interpretation that has nothing to do with curvature: it is the inverse metric tensor of the right-invariant Riemannian metric on $GL(k)$ in which distances are measured multiplicatively, and [Eq. \(172\)](#) is the corresponding steepest-ascent step. This is Amari's *natural gradient* (Amari et al., 1996; Amari, 1998) and Cardoso and Laheld's *relative gradient* (Cardoso and Laheld, 1996); MacKay's derivation of exactly the update [Eq. \(172\)](#) calls it the *covariant* algorithm (MacKay, 1996). The essential point for what follows is that [Eq. \(172\)](#) involves *no approximation whatsoever*: it is the exact gradient of the exact objective for the right-invariant metric just described. (Not for a different *chart*: a gradient in a chart is the gradient for the flat metric that chart induces, and no flat left- or right-invariant metric exists on $GL(k)$ for $k \geq 2$.) The only approximation in the whole algorithm is the choice of the model scores η_j – that is, the extent to which they fail to be the true scores η_j^* – and that is the subject of [Section 7.4](#).

The reason to prefer this particular gradient is the following invariance, which is why the algorithm is used at all.

Proposition 7.15 (Equivariance). *Write $R := WA$ for the global system matrix, so that $C = WX = RZ$. Then the update [Eq. \(172\)](#) induces on R the update*

$$R \leftarrow \left(I_k + \alpha (I_k + \mathbb{E}[\eta(RZ)(RZ)^\top]) \right) R, \quad (176)$$

which does not involve A . Consequently, if $(W_n)_{n \geq 0}$ and $(W'_n)_{n \geq 0}$ are the iterates produced from data $X = AZ$ and $X' = A'Z$ with the same sources and with initialisations satisfying $W_0 A = W'_0 A'$, then $W_n A = W'_n A'$ and $W_n X = W'_n X'$ for all n .

Proof. Multiplying [Eq. \(172\)](#) on the right by A and using $C = WX = WAZ = RZ$ turns the update for W into [Eq. \(176\)](#), in which A no longer appears. The second statement follows by induction: the hypothesis $W_n A = W'_n A'$ makes both iterates apply the same map [Eq. \(176\)](#) to the same matrix R_n , so $W_{n+1} A = W'_{n+1} A'$; and $W_n X = R_n Z = W'_n X'$. $\square$

The trajectory of the recovered signal C therefore depends on the mixing matrix only through $R_0 = W_0 A$: two mixings A and MA give identical C -trajectories provided the initialisations are co-transformed, however badly conditioned M is. This is *not* the statement that the conditioning of A is irrelevant; see [Remark 7.16](#).

Remark 7.16 (Equivariant, relative, natural, covariant). Four names circulate for the same update, and they name four different things about it. *Equivariant* (Cardoso and Laheld,

1996) names the property proved in [Proposition 7.15](#): writing $\Phi_n(x, W_0)$ for the n -th iterate produced from data x and initialisation W_0 , that proposition says

$$\Phi_n(Mx, W_0 M^{-1}) = \Phi_n(x, W_0) M^{-1}, \quad M \in GL(k), \quad (177)$$

which is equivariance of the map $(x, W_0) \mapsto W_n$ under the action $x \mapsto Mx$, $W_0 \mapsto W_0 M^{-1}$ of $GL(k)$. Note that the initialisation is co-transformed: for a *fixed* W_0 , such as the default $W_0 = I_k$ of [Algorithm 7.28](#), the iterates are not equivariant, which is the point already made in [Remark 7.9](#). *Relative gradient* ([Cardoso and Laheld, 1996](#)) names the construction: the gradient with respect to multiplicative perturbations, [Definition 7.12](#). *Natural gradient* ([Amari, 1998](#)) names the same object as steepest ascent for a Riemannian metric on $GL(k)$. *Covariant* ([MacKay, 1996](#)) borrows the physicists’ sense of the word – form-invariant under a change of parametrisation.

We prefer *equivariant*. It names a property of the algorithm that we have actually proved, rather than a property of the metric used to derive it; it is the established term in the source-separation literature; and “covariant” is doubly unfortunate here, since it invites confusion with covariance matrices, and since in the tensor-calculus sense of the word the object K of [Remark 7.14](#) raises an index and so produces a *contravariant* quantity. When the emphasis is on the derivation rather than on the algorithm, “relative gradient” and “natural gradient” are both accurate.

7.4 Stationary points, stability, and the choice of nonlinearity

We now address the question that the derivation so far has left open: the true scores η_j^* are unknown, so the algorithm is run with a fixed guess η_j in their place ([Remark 7.11](#)). Why should it still work?

Proposition 7.17 (Stationary points are nonlinear decorrelations). *W is a stationary point of the relative gradient, i.e. $\nabla^{\text{rel}} \mathcal{L}(W) = 0$, if and only if*

$$\mathbb{E}[\eta_j(C_j) C_l] = -\delta_{jl}, \quad j, l = 1, \dots, k. \quad (178)$$

Proof. Immediate from [Eq. \(171\)](#): the (j, l) entry of $I_k + \mathbb{E}[\eta(C) C^\top]$ is $\delta_{jl} + \mathbb{E}[\eta_j(C_j) C_l]$. $\square$

The k diagonal equations of [Eq. \(178\)](#) fix the scale of each recovered component; the $k(k-1)$ off-diagonal equations say that each transformed component $\eta_j(C_j)$ is orthogonal to every untransformed C_l , $l \neq j$, and hence – the data being centred – uncorrelated with it. This is a *nonlinear decorrelation* condition. If the model score η_j happens to coincide with the true score $(\log p_{C_j})'$ of the current estimate C_j , and $c p_{C_j}(c) \rightarrow 0$ as $|c| \rightarrow \infty$, then the j -th diagonal equation holds automatically: integration by parts gives $\mathbb{E}[\eta_j(C_j) C_j] = \int c p'_{C_j}(c) dc = -1$.

By [Proposition 7.15](#) the dynamics lives on the global system matrix R , and by [Corollary 7.4](#) the separating solutions are exactly the matrices $R = P\Lambda$. Relabelling and rescaling the sources, it suffices to analyse the fixed point $R = I_k$. When the η_j differ across components this relabelling matters: the conditions below are to be read for the pairing of η_j with the source that P sends to position j , and a different pairing has different ζ ’s and its own verdict. That is exactly what makes a wrong assignment of model scores to components unstable. We first record that the scale can always be arranged.

Lemma 7.18 (The fixed-point scale exists and is unique). *Let Y be a real-valued random variable that is not almost surely 0 and let the model score η be one of*

$$\eta(u) = -\tanh(u), \quad \eta(u) = \tanh(u) - u, \quad \eta(u) = -u^3, \quad (179)$$

with $\mathbb{E}[|Y|] < \infty$ in the first case, $\mathbb{E}[Y^2] < \infty$ in the second and $\mathbb{E}[Y^4] < \infty$ in the third. Then there is exactly one $c > 0$ with

$$\mathbb{E}[\eta(cY) cY] = -1. \quad (180)$$

Proof. In all three cases write $h(c) := -\mathbb{E}[\eta(cY) cY] = \mathbb{E}[\varphi(cY)]$ with $\varphi(u) := u \tanh(u)$, $\varphi(u) := u(u - \tanh(u))$ and $\varphi(u) := u^4$ respectively. Each φ is non-negative, even, vanishes only at 0 and is strictly increasing in $|u|$; for the second this is because $u - \tanh u$ has the sign of u and $|u - \tanh u|$ increases in $|u|$. Hence $c \mapsto \varphi(cY(\omega))$ is non-decreasing in $c > 0$ for every ω , and strictly increasing whenever $Y(\omega) \neq 0$. So h is continuous by dominated convergence on compacts, strictly increasing, $h(0^+) = 0$, and $h(c) \rightarrow \infty$ as $c \rightarrow \infty$ by monotone convergence. The intermediate value theorem gives a $c > 0$ with $h(c) = 1$, and strict monotonicity makes it unique. $\square$

Remark 7.19 (Why the normalisation makes ζ_j well defined). The quantities β_j , γ_j and ζ_j of Eq. (183) below are *not* invariant under rescaling the source: replacing Z_j by cZ_j changes $\beta_j = \mathbb{E}[\eta'_j(Z_j)]$ and $\sigma_j^2 = \mathbb{E}[Z_j^2]$ separately, and in general changes their product. This is not a defect but a consequence of the fact that the fixed point under analysis is $R = I_k$, which presupposes a choice of scale for the sources. Lemma 7.18 says that for the three nonlinearities in use there is exactly one such choice compatible with stationarity *up to sign*, namely the one making Eq. (182) hold; the sign is invisible to $\beta_j, \gamma_j, \zeta_j$, because η'_j is even for all three, and it is exactly the sign ambiguity of Corollary 7.4. Once that normalisation is imposed, ζ_j is a well-defined functional of $\mathcal{L}(Z_j)$ and of the model score η_j alone, and the stability conditions Eq. (186) are statements about the source law rather than about an arbitrary scaling. This is what makes the entries of Table 2 meaningful.

Theorem 7.20 (Local stability of the separating solution). *Let $k \geq 2$, let Z have mutually independent components with $\mathbb{E}[Z_j] = 0$ and $\sigma_j^2 := \mathbb{E}[Z_j^2] \in (0, \infty)$, let the model scores $\eta_1, \dots, \eta_k$ of Remark 7.11 be arbitrary twice continuously differentiable functions, and assume*

$$\sup_{u \in \mathbb{R}} |\eta''_j(u)| < \infty \quad \text{for all } j, \quad \mathbb{E}[\|Z\|^3] < \infty \quad (181)$$

(see Remark 7.21). Assume finally the normalisation

$$\mathbb{E}[\eta_j(Z_j) Z_j] = -1, \quad j = 1, \dots, k, \quad (182)$$

holds. Then $R = I_k$ is a stationary point of Eq. (176). Define

$$\beta_j := \mathbb{E}[\eta'_j(Z_j)], \quad \gamma_j := \mathbb{E}[\eta'_j(Z_j) Z_j^2], \quad \zeta_j := -\beta_j \sigma_j^2. \quad (183)$$

Writing $R = I_k + \varepsilon$, the update Eq. (176) linearised at $\varepsilon = 0$ decouples into the k scalar recursions

$$\varepsilon_{jj} \mapsto (1 + \alpha(\gamma_j - 1)) \varepsilon_{jj} \quad (184)$$

and the $\binom{k}{2}$ two-dimensional recursions

$$\begin{bmatrix} \varepsilon_{jl} \\ \varepsilon_{lj} \end{bmatrix} \mapsto (I_2 + \alpha M_{jl}) \begin{bmatrix} \varepsilon_{jl} \\ \varepsilon_{lj} \end{bmatrix}, \quad M_{jl} := \begin{bmatrix} \beta_j \sigma_l^2 & -1 \\ -1 & \beta_l \sigma_j^2 \end{bmatrix}, \quad j \neq l. \quad (185)$$

Consequently the conditions

$$\gamma_j < 1 \quad \text{for all } j, \quad \zeta_j > 0 \quad \text{for all } j, \quad \zeta_j \zeta_l > 1 \quad \text{for all } j \neq l \quad (186)$$

are sufficient for $R = I_k$ to be locally asymptotically stable for all sufficiently small step sizes $\alpha > 0$; if any one of them is reversed strictly, then $R = I_k$ is unstable for every $\alpha > 0$.

Proof. Stationarity at $R = I_k$: the (j, l) entry of $I_k + \mathbb{E}[\eta(Z)Z^\top]$ is $\delta_{jl} + \mathbb{E}[\eta_j(Z_j)Z_l]$, which vanishes for $j = l$ by Eq. (182) and for $j \neq l$ because independence and $\mathbb{E}[Z_l] = 0$ give $\mathbb{E}[\eta_j(Z_j)Z_l] = \mathbb{E}[\eta_j(Z_j)]\mathbb{E}[Z_l] = 0$.

Now let $R = I_k + \varepsilon$ and $C = RZ$, so $C_j = Z_j + \sum_m \varepsilon_{jm} Z_m$. Write $G(R) := I_k + \mathbb{E}[\eta(C)C^\top]$. Since Eq. (176) reads $R \mapsto (I_k + \alpha G(R))R$ and $G(I_k) = 0$, the induced map on ε is $\varepsilon \mapsto \varepsilon + \alpha G(I_k + \varepsilon)\varepsilon = \varepsilon + \alpha G_1(\varepsilon) + O(\|\varepsilon\|^2)$, where G_1 is the derivative of G at I_k . Expanding to first order,

$$\eta_j(C_j) = \eta_j(Z_j) + \eta'_j(Z_j) \sum_m \varepsilon_{jm} Z_m + O(\|\varepsilon\|^2), \quad (187)$$

so that

$$\mathbb{E}[\eta_j(C_j)C_l] = \underbrace{\mathbb{E}[\eta_j(Z_j)Z_l]}_{=-\delta_{jl}} + \sum_m \varepsilon_{lm} \mathbb{E}[\eta_j(Z_j)Z_m] + \sum_m \varepsilon_{jm} \mathbb{E}[\eta'_j(Z_j)Z_m Z_l] + O(\|\varepsilon\|^2). \quad (188)$$

In the second sum $\mathbb{E}[\eta_j(Z_j)Z_m]$ equals -1 for $m = j$ and 0 otherwise, contributing $-\varepsilon_{lj}$. In the third sum, independence and $\mathbb{E}[Z_m] = 0$ leave only $m = l$ when $l \neq j$, contributing $\varepsilon_{jl} \mathbb{E}[\eta'_j(Z_j)]\mathbb{E}[Z_l^2] = \beta_j \sigma_l^2 \varepsilon_{jl}$; and only $m = j$ when $l = j$, contributing $\gamma_j \varepsilon_{jj}$. Hence

$$G_{jl}(I_k + \varepsilon) = \begin{cases} (\gamma_j - 1) \varepsilon_{jj}, & l = j, \\ \beta_j \sigma_l^2 \varepsilon_{jl} - \varepsilon_{lj}, & l \neq j, \end{cases} \quad (189)$$

up to $O(\|\varepsilon\|^2)$, which is exactly Eqs. (184) and (185).

For the stability criterion, $I_k + \alpha M$ is a contraction for all sufficiently small $\alpha > 0$ as soon as every eigenvalue of M has strictly negative real part, and it fails to be one, for every $\alpha > 0$, as soon as some eigenvalue has strictly positive real part; eigenvalues on the imaginary axis are undecided. For the scalar blocks the relevant condition is $\gamma_j - 1 < 0$. Each M_{jl} is real symmetric, so its eigenvalues are real and both are negative if and only if $\text{tr } M_{jl} < 0$ and $\det M_{jl} > 0$, that is

$$\beta_j \sigma_l^2 + \beta_l \sigma_j^2 < 0, \quad \beta_j \beta_l \sigma_j^2 \sigma_l^2 > 1. \quad (190)$$

In terms of $\zeta_j = -\beta_j \sigma_j^2$ the second condition reads $\zeta_j \zeta_l > 1$, and the first reads $\zeta_j \sigma_l^2 / \sigma_j^2 + \zeta_l \sigma_j^2 / \sigma_l^2 > 0$. If $\zeta_j \zeta_l > 1$ then ζ_j and ζ_l have the same sign; if both were negative the first condition would fail, so both are positive, and conversely $\zeta_j, \zeta_l > 0$ implies the first condition. This is Eq. (186).

It remains to check that each strict reversal produces an eigenvalue with strictly positive real part. For $\gamma_j > 1$ the diagonal multiplier $1 + \alpha(\gamma_j - 1)$ exceeds 1 for every $\alpha > 0$. For $\zeta_j \zeta_l < 1$ we get $\det M_{jl} = \zeta_j \zeta_l - 1 < 0$, so M_{jl} has one positive and one negative eigenvalue. For $\zeta_j < 0$, pick any $l \neq j$ (possible since $k \geq 2$): if $\zeta_j \zeta_l < 1$ we are in the previous case; otherwise $\zeta_j \zeta_l \geq 1$ forces $\zeta_l < 0$, and then $\text{tr } M_{jl} = -\zeta_j \sigma_l^2 / \sigma_j^2 - \zeta_l \sigma_j^2 / \sigma_l^2 > 0$, so at least one eigenvalue of M_{jl} is positive. Finally, if $I_k + \alpha M$ has an eigenvalue $1 + \alpha \lambda$ with $\text{Re } \lambda > 0$, then $|1 + \alpha \lambda|^2 = 1 + 2\alpha \text{Re } \lambda + \alpha^2 |\lambda|^2 > 1$ for every $\alpha > 0$, so the linearised map is expanding in the corresponding direction. Since the remainder in Eqs. (184) and (185) is $O(\|\varepsilon\|^2)$ and the linear part is symmetric, the unstable-manifold theorem for maps then makes $R = I_k$ unstable for the nonlinear iteration as well. $\square$

Remark 7.21 (Reading Theorem 7.20). Four comments on the statement.

- (i) *What the regularity condition is for.* The perturbation $\Delta_j := \sum_m \varepsilon_{jm} Z_m$ that appears in the proof is *not* bounded almost surely – only $|\Delta_j| \leq \|\varepsilon\| \|Z\|$ – so the mean value theorem has to be applied at an intermediate point that can be far from Z_j . A bound on η''_j that is uniform on all of $\mathbb{R}$ removes the difficulty, and $\mathbb{E}\|Z\|^3 < \infty$ then makes the Taylor remainder $O(\|\varepsilon\|^2)$.

- (ii) *What it covers.* Eq. (181) holds for $\eta_j = -\tanh$ and $\eta_j = \tanh - \text{id}$, but not for the cubic $\eta_j(u) = -u^3$, whose second derivative $\eta_j''(u) = -6u$ is unbounded. For the cubic one uses instead the polynomial variant of the same estimate, valid whenever $\mathbb{E}\|Z\|^4 < \infty$: there $\mathbb{E}[\eta_j(C_j)C_l]$ is an explicit polynomial in ε of degree at most four whose coefficients are moments of Z of order at most four.
- (iii) *The borderline.* The only cases left undecided by the linearisation are $\gamma_j = 1$ and $\zeta_j \zeta_l = 1$ with $\zeta_j, \zeta_l > 0$. The apparent third borderline case $\zeta_j = 0$ is decided: it gives $\zeta_j \zeta_l = 0 < 1$, hence instability.
- (iv) *The case $k = 1$.* There is then nothing to separate, and only the condition $\gamma_1 < 1$ survives.

Theorem 7.20 is the stability analysis of Amari et al. (1997), recast in the notation of these notes; see also Cardoso (1998b, Section VI).

The quantity ζ_j therefore decides everything. The next three results compute it in the cases of interest. The first explains, in one line, why Gaussian sources are the exact borderline – for *every* choice of model score, correctly specified or not.

Proposition 7.22 (Gaussian sources lie exactly on the stability boundary). *In the setting of Theorem 7.20, suppose $Z_j \sim \mathcal{N}(0, \sigma_j^2)$ with $\sigma_j > 0$, and suppose in addition that $\eta_j(z)p(z) \rightarrow 0$ as $|z| \rightarrow \infty$, where p denotes the $\mathcal{N}(0, \sigma_j^2)$ density (see Remark 7.24). Then $\zeta_j = 1$, whatever the model score η_j – correctly specified or not. Consequently, if two sources are Gaussian, then $\zeta_j \zeta_l = 1$ and Eq. (186) fails.*

Proof. Let p be the $\mathcal{N}(0, \sigma_j^2)$ density, so that $z p(z) = -\sigma_j^2 p'(z)$. Integration by parts, whose boundary term $[\eta_j p]_{-\infty}^{\infty}$ vanishes by hypothesis, gives Stein’s identity

$$\mathbb{E}[\eta_j(Z_j)Z_j] = \int \eta_j(z) z p(z) dz = -\sigma_j^2 \int \eta_j(z) p'(z) dz = \sigma_j^2 \int \eta_j'(z) p(z) dz = \sigma_j^2 \beta_j. \quad (191)$$

The normalisation Eq. (182) says that the left hand side is -1 , so $\zeta_j = -\beta_j \sigma_j^2 = 1$. $\square$

Corollary 7.23 (The true score: stability matches identifiability exactly). *In the setting of Theorem 7.20, but requiring of each η_j only one continuous derivative and assuming neither Eq. (181) nor Eq. (182), suppose the model is correctly specified, that is*

$$\eta_j = \eta_j^* := (\log p_{Z_j})', \quad j = 1, \dots, k, \quad (192)$$

where p_{Z_j} is the density of Z_j . Assume that each p_{Z_j} is strictly positive and twice differentiable with continuous second derivative, that $(1+z^2)p_{Z_j}'' \in L^1(\mathbb{R})$, that the expectations in Eqs. (182) and (183) converge absolutely, and that the boundary conditions

$$z p_{Z_j}(z) \rightarrow 0, \quad z^2 p_{Z_j}'(z) \rightarrow 0, \quad |z| \rightarrow \infty \quad (193)$$

hold. Then Eq. (182) holds automatically,

$$\zeta_j = \mathcal{I}(Z_j) \sigma_j^2 \geq 1, \quad \mathcal{I}(Z_j) := \mathbb{E}[\eta_j^*(Z_j)^2], \quad (194)$$

with equality if and only if Z_j is Gaussian, and $\gamma_j < 1$. Hence the inequalities Eq. (186) hold if and only if at most one of the sources is Gaussian.

Proof. Write $p := p_{Z_j}$ and $\eta := \eta_j = \eta_j^*$, so that $\eta p = p'$ and $\eta' p = p'' - (p')^2/p$; it is Eq. (192) that makes the first of these identities available, and it is what turns each of the expectations below into an integral of p and its derivatives. The second boundary condition in Eq. (193)

forces $p'(z) \rightarrow 0$, and $\int p'' = \lim_{R \rightarrow \infty} (p'(R) - p'(-R)) = 0$; the integral exists because $p'' \in L^1$. Normalisation: integration by parts, with vanishing boundary term by the first condition in Eq. (193), gives $\mathbb{E}[\eta(Z_j)Z_j] = \int zp'(z) dz = -\int p = -1$. For β_j ,

$$\beta_j = \int \left(\frac{p''}{p} - \frac{(p')^2}{p^2} \right) p = \int p'' - \mathbb{E}[\eta(Z_j)^2] = -\mathcal{I}(Z_j). \quad (195)$$

The inequality is Cauchy–Schwarz applied to the normalisation:

$$1 = (\mathbb{E}[\eta(Z_j)Z_j])^2 \leq \mathbb{E}[\eta(Z_j)^2] \mathbb{E}[Z_j^2] = \mathcal{I}(Z_j)\sigma_j^2 = \zeta_j, \quad (196)$$

with equality if and only if $\eta(Z_j)$ and Z_j are almost surely proportional, i.e. $(\log p)'(z) = -z/\sigma_j^2$, i.e. p is the $\mathcal{N}(0, \sigma_j^2)$ density. For γ_j , integrating by parts twice – the first boundary term vanishing by the second condition in Eq. (193) and the second by the first condition – we get $\int z^2 p'' = -2 \int zp' = 2$, so

$$\gamma_j = \int z^2 \left(\frac{p''}{p} - \eta^2 \right) p = 2 - \mathbb{E}[\eta(Z_j)^2 Z_j^2] \leq 2 - (\mathbb{E}[\eta(Z_j)Z_j])^2 = 1, \quad (197)$$

again by Cauchy–Schwarz, with equality only if $\eta(Z_j)Z_j$ is almost surely constant, i.e. $p(z) \propto |z|^{-1}$, which is not integrable; so $\gamma_j < 1$ strictly. Finally, $\zeta_j \geq 1$ for all j with equality exactly at the Gaussian sources, so $\zeta_j \zeta_l > 1$ for all $j \neq l$ if and only if at most one ζ_j equals 1. $\square$

Remark 7.24 (On the hypotheses of Proposition 7.22 and Corollary 7.23). The decay condition $\eta_j(z)p(z) \rightarrow 0$ of Proposition 7.22 is what makes the boundary term in Stein’s identity vanish; it holds whenever η_j has at most polynomial growth, and in particular for all three nonlinearities of Lemma 7.18. In Corollary 7.23 the second derivative of p_{Z_j} is needed because $\eta_j^* = p''/p - (p'/p)^2$, so Theorem 7.20 presupposes it anyway, and its continuity is what makes η_j continuously differentiable; Theorem 7.20 asks for one derivative more. Note also what Corollary 7.23 does and does not deliver: it gives the *inequalities* Eq. (186), and these coincide with the hypothesis of Corollary 7.4; but to turn them into local stability one still needs the regularity Eq. (181), which a true score need not satisfy.

Corollary 7.23 is the conceptual answer to “why does gradient descent find the right answer”. With correctly specified scores, $\eta_j = \eta_j^*$, the inequalities Eq. (186) hold exactly when the identifiability theory declares the model identifiable; subject to the regularity caveat of Remark 7.24, a separating solution is then a locally stable equilibrium of the mean dynamics. The converse is false, and it is worth being explicit about it: *stable non-separating equilibria can exist even under a perfectly specified model*. For $k = 2$ with Z_1, Z_2 i.i.d. from the smooth, strictly positive, non-Gaussian density $0.4\mathcal{N}(-1.5, 0.25^2) + 0.2\mathcal{N}(0, 0.25^2) + 0.4\mathcal{N}(1.5, 0.25^2)$ and $\eta_j = \eta_j^*$ its exact score, the matrix $R^* = 0.80993 \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ is a stationary point of Eq. (176) whose linearisation has all four eigenvalues strictly negative, so it is locally attracting – and it is not a generalised permutation matrix, so $C = R^*Z$ has dependent components. Theorem 7.20 says which *separating* solutions attract; it does not say that nothing else does. The remaining question is what happens when $\eta_j \neq \eta_j^*$, which in practice is always.

Corollary 7.25 (The cubic nonlinearity: kurtosis is exactly the criterion). *Let $\eta_j(u) := -u^3$ and let Z_j have $\mathbb{E}[Z_j] = 0$, $\mathbb{E}[Z_j^4] < \infty$ and be normalised as in Lemma 7.18, so that $\mathbb{E}[Z_j^4] = 1$. Then $\gamma_j = -3$ and*

$$\zeta_j = 3\sigma_j^4 = \frac{3}{3 + \text{kurt}(Z_j)}. \quad (198)$$

Hence $\zeta_j > 1$ if and only if $\text{kurt}(Z_j) < 0$, i.e. if and only if Z_j is sub-Gaussian in the sense of Definition 3.21.

Proof. $\eta'_j(u) = -3u^2$, so $\beta_j = -3\sigma_j^2$ and $\zeta_j = 3\sigma_j^4$, while $\gamma_j = -3\mathbb{E}[Z_j^4] = -3$. By [Definition 3.21](#), $\text{kurt}(Z_j) = \mathbb{E}[Z_j^4]/\sigma_j^4 - 3 = \sigma_j^{-4} - 3$, so $\sigma_j^4 = (3 + \text{kurt}(Z_j))^{-1}$ and $\zeta_j = 3/(3 + \text{kurt}(Z_j))$, which exceeds 1 exactly when $\text{kurt}(Z_j) < 0$. $\square$

Remark 7.26 (The hyperbolic tangent, and what the kurtosis rule really is). For the two tanh-based model scores the criterion $\zeta_j > 1$ becomes an explicit inequality. With the normalisation [Eq. \(182\)](#):

- $\eta_j(u) = -\tanh(u)$, so that $\mathbb{E}[\tanh(Z_j)Z_j] = 1$ and $\beta_j = -\mathbb{E}[\text{sech}^2(Z_j)]$; the condition $\zeta_j > 1$ reads

$$\mathbb{E}[Z_j^2] \mathbb{E}[\text{sech}^2(Z_j)] > \mathbb{E}[\tanh(Z_j) Z_j]. \quad (199)$$

- $\eta_j(u) = \tanh(u) - u$, so that $\beta_j = -\mathbb{E}[\tanh^2(Z_j)]$; here the condition $\zeta_j > 1$ reads

$$\mathbb{E}[Z_j^2] \mathbb{E}[\tanh^2(Z_j)] > 1. \quad (200)$$

[Eq. \(199\)](#) is a *Stein discrepancy*: by the identity in the proof of [Proposition 7.22](#), its two sides are equal for every Gaussian Z_j and for every test function, so the inequality measures departure from Gaussianity in the direction picked out by tanh. It is exactly the switching statistic of [Lee et al. \(1999, eq. \(2.28\)\)](#).

It is tempting – and it is what the ICA literature usually says – to summarise this as “use $-\tanh$ for super-Gaussian sources and $\tanh - \text{id}$ for sub-Gaussian ones”. That rule is a *heuristic*, not a theorem: the sign of $\text{kurt}(Z_j)$ is exactly the criterion only for the cubic nonlinearity ([Corollary 7.25](#)), and for tanh the correct criterion is [Eq. \(199\)](#), which must be checked. It does hold for the standard source models; [Table 2](#) lists the values. In general it fails in both directions, and the failures are not exotic. For the symmetric three-point family $\mathbb{P}[Y = \pm a] = p/2$, $\mathbb{P}[Y = 0] = 1 - p$, normalised as in [Lemma 7.18](#), write $u := ca$ for the rescaled atom; the normalisation reads $pu \tanh u = 1$ and one computes

$$\zeta = \frac{u}{\tanh u} - 1, \quad \text{kurt} = u \tanh u - 3. \quad (201)$$

So $\zeta = 1$ happens at $u = 2 \tanh u$, i.e. at $\text{kurt} \approx -1.1664$, and every member of the family with $-1.1664 < \text{kurt} < 0$ is *sub*-Gaussian yet has $\zeta > 1$, so that $-\tanh$ separates it. In the other direction, mass placed far out in units of σ inflates kurt while saturating tanh: the four-atom law with $\mathbb{P}[Y = \pm M] = q/2$ and $\mathbb{P}[Y = \pm 1] = (1 - q)/2$ has $\text{kurt} \approx 451$ but $\zeta \approx 0.46$ for $q = 5 \cdot 10^{-6}$, $M = 100$, so two such *super*-Gaussian sources make the separating solution unstable for $-\tanh$. Note finally that the sign of the discrepancy in [Eq. \(199\)](#) is scale dependent – it is proportional to kurt only in the small-scale limit – so the criterion is meaningful only at the fixed-point scale of [Lemma 7.18](#); cf. [Remark 7.19](#).

Remark 7.27 (What is approximate, and why it does not spoil convergence). It is worth separating three things that are easy to confuse.

- The step [Eq. \(172\)](#) is *exact*: it is the gradient of the exact objective for the metric ([Proposition 7.13](#)). Nothing is approximated there, and in particular no Hessian is being approximated ([Remark 7.14](#)).
- The model scores η_j are an approximation to the true scores η_j^* , since the true source densities are unknown. Their entire effect on whether the algorithm converges to a separating solution is captured by [Theorem 7.20](#): a model score with $\eta_j \neq \eta_j^*$ is harmless as long as $\gamma_j < 1$, $\zeta_j > 0$ and $\zeta_j \zeta_l > 1$ – and for the three nonlinearities of [Lemma 7.18](#) the first two are automatic, since $\eta'_j < 0$ off a Lebesgue null set and Z_j is non-degenerate,

Law of Z_j	kurt(Z_j)	ζ_j for $\eta = -\tanh$	ζ_j for $\eta = \tanh - \text{id}$	ζ_j for $\eta(u) = -u^3$
Gaussian	0	1.000	1.000	1.000
Laplace	+3	1.414	0.726	0.500
Student t_5	+6	1.254	0.808	0.333
Student t_8	+1.5	1.126	0.895	0.667
Logistic	+1.2	1.134	0.889	0.714
Uniform	-1.2	0.762	1.273	1.667

Table 2: The stability quantity ζ_j of Eq. (183) for the three standard nonlinearities. Each law is taken in its standard normalisation – $\mathcal{N}(0, 1)$; Laplace and logistic with unit scale parameter; t_ν standard; uniform on $[-1, 1]$ – and is then rescaled by the unique factor $c > 0$ of Lemma 7.18, which is what makes ζ_j well defined (Remark 7.19); the tabulated values do not depend on the starting normalisation. Stability of a separating solution requires $\zeta_j \zeta_l > 1$ for all $j \neq l$, for which $\zeta_j > 1$ for all j is sufficient; the remaining condition $\gamma_j < 1$ is automatic here, since $\eta'_j \leq 0$ for all three nonlinearities and hence $\gamma_j \leq 0$. Every Gaussian entry equals 1 exactly, as Proposition 7.22 predicts, and the cubic column is $3/(3 + \text{kurt})$ exactly, as Corollary 7.25 predicts. The values were obtained by numerical quadrature.

so $\gamma_j < 0$ and $\zeta_j > 0$. This is why one only needs to know each source *qualitatively*: in practice only the sign of $\zeta_j - 1$, rather than the density. For the source families of Table 2 that sign is determined by the sign of the excess kurtosis, for each of the three nonlinearities separately, but the correspondence does *not* hold in general – see the warning and the counterexamples in Remark 7.26, and note that ζ_j is a statement about neither kurtosis nor tails.

- (iii) What $\eta_j \neq \eta_j^*$ does cost is *statistical efficiency*. The maximum likelihood estimator with the true scores attains the Cramér–Rao bound asymptotically; with misspecified scores the separating solutions remain stationary points, at the scale fixed by Lemma 7.18, and remain locally attracting whenever Theorem 7.20 applies, but the asymptotic variance of the resulting estimator is larger. This is the semiparametric picture of Amari and Cardoso (1997); see also Cardoso (1998b).

Two caveats should be stated plainly. First, Theorem 7.20 is a *local* statement. The objective Eq. (163) is not concave in W , and Eq. (176) has stationary points that are not separating, some of which can themselves be attracting. The theorem says when a separating solution attracts, not that the algorithm finds one from an arbitrary start. Second, the online form Algorithm 7.28 below replaces the expectation in Eq. (172) by a single-sample estimate. It is thus a Robbins–Monro stochastic approximation scheme (Robbins and Monro, 1951) for the mean dynamics Eq. (176) – provided the samples are i.i.d. across t , or at least mixing enough that the single-sample estimate is a martingale difference plus a vanishing bias; with step sizes satisfying $\sum_n \alpha_n = \infty$ and $\sum_n \alpha_n^2 < \infty$, and under the usual regularity and boundedness conditions, its trajectories track the associated ordinary differential equation

$$\dot{R} = \left(I_k + \mathbb{E}[\eta(RZ)(RZ)^\top] \right) R \quad (202)$$

and converge almost surely to one of its locally stable equilibria – which Theorem 7.20 identifies. A constant step size α , as used in practice, gives convergence only to a neighbourhood whose size is $O(\alpha)$.

7.5 The algorithm

Algorithm 7.28 (Equivariant online ICA for the complete noiseless model). *Input:* data $x(1), \dots, x(T)$ in $\mathbb{R}^k$, centred; a step size $\alpha > 0$; model scores $\eta_1, \dots, \eta_k$ chosen according to [Theorem 7.20](#), in practice as in [Remark 7.29](#); an invertible initial $W \in \mathbb{R}^{k \times k}$, e.g. $W = I_k$.

Repeat over the data, until convergence: for each sample $x(t)$,

- (1) $c := W x(t)$ (current estimate of the sources)
- (2) $y := [\eta_1(c_1), \dots, \eta_k(c_k)]^\top$ (nonlinearity applied componentwise)
- (3) $W \leftarrow W + \alpha(W + y c^\top W) = W + \alpha(I_k + y c^\top)W$.

Output: the unmixing matrix W , and the reconstructed sources obtained by running step (1) once more with the final W , that is $c(t) = W x(t)$; they are determined up to permutation and scale.

Remark 7.29 (Reading the algorithm). In practice one takes $\eta_j = -\tanh$ for super-Gaussian and $\eta_j = \tanh -\text{id}$ for sub-Gaussian sources, with the caveats of [Remark 7.26](#). Step (3) is [Eq. \(172\)](#) with the expectation replaced by the current sample, since $\mathbb{E}[\eta(C)C^\top]$ is estimated by $y c^\top$. Written out as $W + \alpha(W + \eta(Wx) x^\top W^\top W)$ it is exactly the update of [Amari et al. \(1996\)](#), [Cardoso and Laheld \(1996\)](#) and [MacKay \(1996\)](#) – see also [MacKay \(2003, Chapter 34\)](#) for the same derivation in an archival source; the infomax algorithm of [Bell and Sejnowski \(1995\)](#) is the same update in its plain-gradient form, and the two objectives coincide, as observed by [Cardoso \(1997\)](#). Each step costs $O(k^2)$ operations and no matrix inversion or decomposition: $y c^\top W$ is computed as $y (c^\top W)$, a vector–matrix product followed by an outer product, each $O(k^2)$. By [Proposition 7.15](#) the whole dynamics is a dynamics on the global system matrix $R = WA$, so the mixing matrix enters only through $R_0 = W_0 A$: once started, the algorithm behaves identically for every A giving the same R_0 , however badly conditioned A itself is. This is what distinguishes the update from plain gradient ascent on $\mathcal{L}$, whose trajectory depends on A throughout.

7.6 LiNGAM: a causal order removes the permutation

[Corollary 7.4](#) leaves the sources determined only up to permutation, sign and scale. In causal discovery one adds structural assumptions that remove exactly those remaining ambiguities. The basic such model is the linear non-Gaussian acyclic model, LiNGAM, of [Shimizu et al. \(2006\)](#). We write Θ for the matrix of structural coefficients, which [Shimizu et al. \(2006\)](#) call B ; the letter B is already taken in these notes, and R is the global system matrix of [Proposition 7.15](#).

Assumption 7.30 (LiNGAM). The observed random vector $X \in \mathbb{R}^k$ satisfies the structural equations

$$X = \Theta X + Z, \tag{203}$$

where $\Theta \in \mathbb{R}^{k \times k}$ and the disturbance vector $Z \in \mathbb{R}^k$ satisfy:

- (i) *Acyclicity*. There is a bijection $\pi: \{1, \dots, k\} \rightarrow \{1, \dots, k\}$, the *causal order*, such that $\Theta_{ab} = 0$ whenever $\pi(a) \leq \pi(b)$. Equivalently, Θ becomes strictly lower triangular after simultaneously permuting rows and columns by π ; the associated directed graph, with an edge $b \rightarrow a$ whenever $\Theta_{ab} \neq 0$, is acyclic.
- (ii) *Independent disturbances, causal sufficiency*. The components $Z_1, \dots, Z_k$ of the disturbance vector are mutually independent and almost surely non-constant.

(iii) *Non-Gaussianity*. At most one Z_j is Gaussian.

Lemma 7.31 (LiNGAM is a complete noiseless ICA model). *Under [Assumption 7.30](#) (i) the matrix $I_k - \Theta$ is invertible with $\det(I_k - \Theta) = 1$, and [Eq. \(203\)](#) is equivalent to*

$$X = AZ, \quad A := (I_k - \Theta)^{-1}, \quad W := A^{-1} = I_k - \Theta. \quad (204)$$

This is a complete noiseless ICA model in the sense of [Assumption 7.2](#) (i)–(iv), and the unmixing matrix W has all diagonal entries equal to 1.

Proof. Let P_π be the permutation matrix of the causal order, so that $N := P_\pi \Theta P_\pi^\top$ is strictly lower triangular and hence $N^k = 0$; then $\Theta^k = P_\pi^\top N^k P_\pi = 0$, so Θ is nilpotent, $I_k - \Theta$ is invertible with inverse $\sum_{j=0}^{k-1} \Theta^j$, and $\det(I_k - \Theta) = \det(I_k - N) = 1$ because $I_k - N$ is lower triangular with unit diagonal. Rearranging [Eq. \(203\)](#) gives $(I_k - \Theta)X = Z$, which is [Eq. \(204\)](#). The diagonal of $W = I_k - \Theta$ is 1 because $\Theta_{aa} = 0$ by (i) with $b = a$. $\square$

The normalisation [Assumption 7.2](#) (v) is deliberately *not* imposed here: LiNGAM fixes the scale of the sources through $\text{diag}(W) = I_k$ rather than through either of the conventions of [Remark 7.3](#).

Corollary 7.32 (Identifiability of LiNGAM). *Suppose X satisfies [Assumption 7.30](#) with two parameter sets, that is*

$$\Theta^{(1)}X + Z^{(1)} = X = \Theta^{(2)}X + Z^{(2)}, \quad (205)$$

where each $\Theta^{(i)}$ satisfies (i) and each $Z^{(i)}$ satisfies (ii) and (iii). Then

$$\Theta^{(1)} = \Theta^{(2)}, \quad Z^{(1)} = Z^{(2)} \quad \text{almost surely.} \quad (206)$$

Proof. By [Lemma 7.31](#) we have two representations $X = A^{(i)}Z^{(i)}$ with $A^{(i)} = (I_k - \Theta^{(i)})^{-1}$ invertible, so [Corollary 7.4](#) applies (with $\mu^{(i)} = 0$) and yields a permutation matrix P and an invertible diagonal Λ with $A^{(2)} = A^{(1)}P\Lambda$, together with $Z^{(1)} = P\Lambda Z^{(2)} + c$. Inverting, and writing $W^{(i)} := I_k - \Theta^{(i)}$,

$$W^{(2)} = \Lambda^{-1}P^{-1}W^{(1)}. \quad (207)$$

The matrix $M := \Lambda^{-1}P^{-1}$ is a generalised permutation matrix: there are a permutation σ and non-zero scalars $m_1, \dots, m_k$ with $M_{ac} = m_a \delta_{c, \sigma(a)}$, so that [Eq. \(207\)](#) reads

$$W_{ab}^{(2)} = m_a W_{\sigma(a), b}^{(1)}, \quad a, b = 1, \dots, k. \quad (208)$$

Both $W^{(i)}$ have unit diagonal by [Lemma 7.31](#), so taking $b = a$ in [Eq. \(208\)](#) gives

$$1 = W_{aa}^{(2)} = m_a W_{\sigma(a), a}^{(1)}, \quad \text{hence} \quad W_{\sigma(a), a}^{(1)} \neq 0 \quad \text{for all } a. \quad (209)$$

Now use acyclicity of $\Theta^{(1)}$: for $c \neq b$ we have $W_{cb}^{(1)} = -\Theta_{cb}^{(1)}$, which is non-zero only if $\pi_1(c) > \pi_1(b)$, where π_1 is the causal order of $\Theta^{(1)}$. So [Eq. \(209\)](#) forces, for every a ,

$$\text{either } \sigma(a) = a \quad \text{or} \quad \pi_1(\sigma(a)) > \pi_1(a). \quad (210)$$

Suppose $\sigma \neq \text{id}$ and let a lie on a non-trivial cycle of σ , of length $L \geq 2$. Then $\sigma^j(a) \neq \sigma^{j+1}(a)$ for every j , so

$$\pi_1(a) < \pi_1(\sigma(a)) < \dots < \pi_1(\sigma^L(a)) = \pi_1(a), \quad (211)$$

a contradiction. Hence $\sigma = \text{id}$, and then [Eq. \(209\)](#) gives $m_a = 1/W_{aa}^{(1)} = 1$, so $M = I_k$. Therefore $W^{(2)} = W^{(1)}$, i.e. $\Theta^{(2)} = \Theta^{(1)}$, and consequently $Z^{(2)} = W^{(2)}X = W^{(1)}X = Z^{(1)}$. $\square$

So the entire structural model – graph, coefficients and disturbance distributions – is determined by the law of X ; the causal order itself is identified up to the ordering of variables that are not ancestrally related, which is exactly the freedom a topological sort leaves.

Remark 7.33 (What each assumption does). It is worth seeing which ambiguity of [Corollary 7.4](#) is removed by which structural assumption. The scale ambiguity Λ is removed by the *form* of [Eq. \(203\)](#): the coefficient of X_a in its own equation is fixed to 1, which is what makes W have unit diagonal. The permutation ambiguity P is removed by *acyclicity*, through the cycle argument above: a non-trivial relabelling would have to increase the causal order all the way around a cycle. [Assumption 7.30](#) (ii) is causal sufficiency: mutual independence of the disturbances says exactly that there are no hidden common causes. Non-Gaussianity is what makes [Corollary 7.4](#) available in the first place; without it one recovers the graph at best up to its Markov equivalence class, and then only under a faithfulness assumption. Note also that [Shimizu et al. \(2006\)](#) assume *all* disturbances non-Gaussian, whereas [Corollary 7.32](#) needs only “at most one”, because that is all [Corollary 7.4](#) needs.

Algorithm 7.34 (LiNGAM by equivariant ICA, after [Shimizu et al., 2006](#)). *Input*: data $x(1), \dots, x(T)$ in $\mathbb{R}^k$.

- (1) Centre the data.
- (2) Run [Algorithm 7.28](#) to obtain an estimate $\widehat{W}$ of the unmixing matrix. By [Corollary 7.4](#) it estimates $\Lambda^{-1}P^\top W$ for the true $W = I_k - \Theta$ and some unknown generalised permutation.
- (3) *Undo the permutation.* Find the row permutation P_1 that makes all diagonal entries of $P_1\widehat{W}$ non-zero; in practice one minimises $\sum_j |(P_1\widehat{W})_{jj}|^{-1}$ over permutations, which is a linear assignment problem. Put $W' := P_1\widehat{W}$.
- (4) *Undo the scaling.* Divide each row of W' by its own diagonal entry, that is, put $D := \text{diag}(W'_{11}, \dots, W'_{kk})$ and $W'' := D^{-1}W'$, so that W'' has unit diagonal.
- (5) *Read off the coefficients.* $\widehat{\Theta} := I_k - W''$.
- (6) *Read off the causal order.* Find the permutation π making $\widehat{\Theta}$ as close to strictly lower triangular as possible, e.g. by repeatedly setting the smallest-magnitude entries to zero until a causal order exists. Optionally prune the remaining small coefficients.

Output: the coefficient matrix $\widehat{\Theta}$, the causal order π , and the estimated disturbances $\widehat{z}(t) := W''x(t)$.

Remark 7.35 (Why steps (3) and (6) are well posed). In the population limit, where $\widehat{W} = \Lambda^{-1}P^\top W$ exactly, the proof of [Corollary 7.32](#) is precisely the statement that step (3) has a solution and that it is unique: the cycle argument shows that $P_1 = P$ is the only row permutation producing a nowhere-zero diagonal, and step (4) then recovers W exactly. Likewise step (6) succeeds because the true $\widehat{\Theta}$ is permutable to strictly lower triangular form. Note also that step (2) invokes [Algorithm 7.28](#), which was derived under the standing assumption of [Section 7.2](#) that each source has a strictly positive differentiable density; [Assumption 7.30](#) does not require this, so the assumption has to be added if the ICA-based route is taken. With finite samples neither search is exact, which is why both become combinatorial problems with tolerances; see [Shimizu et al. \(2006, Sections 4–5\)](#) for the details and [Shimizu et al. \(2011\)](#) for *DirectLiNGAM*, a later algorithm that estimates the causal order directly by iterated regression and independence testing, and so avoids the ICA step – and with it both the local convergence caveats of [Remark 7.27](#) and the two combinatorial searches above.

7.7 Generalisations: a brief overview

The model of these notes – one linear mixture, instantaneous, with independent components – has been extended in many directions. This subsection is a short guide to the landscape, with references rather than derivations.

Relaxing linearity. The obvious generalisation replaces the mixing matrix by a diffeomorphism, $X = f(Z)$ with Z having independent components. This model is *never* identifiable: Hyvärinen and Pajunen (1999) show that any random vector X with a positive density admits a representation $X = f(Z)$ with mutually independent Z_j , obtained from the conditional-quantile construction of Darrois (1953), which returns uniform sources; composing that solution with the measure-preserving automorphisms of the unit cube then produces infinitely many further ones. Identifiability has to be bought back with extra structure. The dominant device is an observed auxiliary variable U – a time index, a segment label, an experimental condition – with respect to which the sources are conditionally independent and sufficiently variable,

$$p_{Z|U}(z | u) = \prod_{j=1}^k p_{Z_j|U}(z_j | u). \quad (212)$$

Under such assumptions the nonlinear model becomes identifiable up to permutation and componentwise transformations; see Hyvärinen and Morioka (2016) for time-contrastive learning, where U is a time segment label; Hyvärinen and Morioka (2017) for temporally dependent sources, where the role of U is played by the previous time point; and Khemakhem et al. (2020) for the identifiable variational autoencoder, which unifies these results. A different route constrains the geometry of f rather than the statistics of Z : *independent mechanism analysis* (Gresele et al., 2021) requires the columns of the Jacobian of f to be orthogonal, which by Gresele et al. (2021, Theorem 4.7) excludes the Darrois solutions of Hyvärinen and Pajunen (1999) without an auxiliary variable. It is not, however, known to deliver full identifiability on its own.

Relaxing independence. If the sources fall into groups that are independent *across* groups but dependent *within*, one obtains multidimensional ICA, or independent subspace analysis (Cardoso, 1998a): the mixing matrix is identified only up to a permutation of equal-dimensional blocks and an invertible transformation within each block – the analogue, for a general block, of the orthogonal ambiguity inside a Gaussian block in Remark 7.5.

Several datasets at once. Given M related datasets $X^{(m)} = A^{(m)}Z^{(m)}$, $m = 1, \dots, M$, one may, in addition to independence within each dataset, tie corresponding sources across datasets. Independent vector analysis (Kim et al., 2006) keeps the components independent within a dataset but couples the j -th components across datasets, $j = 1, \dots, k$, into a single source component vector; this aligns the permutations across datasets, leaving only one global permutation of the k component vectors. Multi-view ICA goes further and splits the sources into a part shared by all views and parts specific to each,

$$X^{(m)} = A^{(m)} \begin{bmatrix} S \\ Z^{(m)} \end{bmatrix} + E^{(m)}, \quad m = 1, \dots, M, \quad (213)$$

with a shared source vector S and view-specific sources $Z^{(m)}$. In Pandeva and Forré (2023a) each $A^{(m)}$ is square and invertible and the noise is isotropic Gaussian *in the latent space*,

so that $E^{(m)} = A^{(m)}\epsilon^{(m)}$ with $\epsilon^{(m)} \sim \mathcal{N}(0, \sigma^2 I_{k_m})$ and the observed noise covariance is $\sigma^2 A^{(m)} A^{(m)\top}$. See that paper for identifiability results and an estimation procedure, [Pandeva and Forré \(2023b\)](#) for its use in integrating omics modalities, and [Pandeva et al. \(2025\)](#) for a related multi-view latent-variable model that replaces source independence by a shared sparse precision matrix and targets co-expression network inference.

Noise and overcompleteness. The noisy and overcomplete models – $X = AZ + E$ with $A \in \mathbb{R}^{p \times k}$ and possibly $k > p$ – are exactly the setting of [Sections 5](#) and [6](#); note that identifiability of A there does *not* entail that the sources can be recovered pointwise. Additive Gaussian noise leaves them determined only up to a Gaussian summand, by [Theorem 5.5](#), even when A has a left inverse; and in the overcomplete case $k > p$ the matrix A has no left inverse at all.

Causal discovery. Beyond LiNGAM, dropping linearity while keeping additive noise gives the additive noise models of [Hoyer et al. \(2008\)](#), in which $X_a = f_a(X_{\text{pa}(a)}) + Z_a$ and identifiability again comes from a non-Gaussianity- or nonlinearity-induced asymmetry between cause and effect.

Further reading. [Hyvärinen et al. \(2001\)](#) remains the standard textbook; [Comon and Jutten \(2010\)](#) is a comprehensive edited handbook covering algorithms, identifiability and applications.

A Proofs of the Classical Results on Characteristic Functions

This appendix proves the four results of [Section 3](#) that were stated there without proof: the uniqueness and inversion theorem ([Theorem 3.4](#)), Lévy’s continuity theorem ([Theorem 3.7](#)), Marcinkiewicz’ theorem ([Theorem 3.17](#)) and Cramér’s decomposition theorem ([Theorem 3.20](#)). Together with [Section B](#), which proves [Theorem 4.2](#), this makes every proof in these notes refer only to other results of these notes, together with the following standard background, which we use without proof:

- (a) measure theory: monotone and dominated convergence (hence differentiation under the integral sign), Fatou’s lemma, Fubini’s theorem, and the fact that a Borel probability measure on $\mathbb{R}^d$ is determined by its integrals against continuous functions of compact support;
- (b) Prokhorov’s theorem: a tight sequence of Borel probability measures on $\mathbb{R}^d$ has a weakly convergent subsequence ([Klenke, 2020](#), Chapter 13), ([Kallenberg, 2021](#), Chapter 5);
- (c) complex analysis of one variable: Morera’s theorem, the identity theorem, the mean value property of harmonic functions, and the fact that a nowhere vanishing entire function is exp of an entire function;
- (d) elementary real analysis: Taylor’s theorem with Peano remainder, Jensen’s and Lyapunov’s inequalities, existence of a median, the orthogonality relations for the trigonometric system together with termwise integration of uniformly convergent series, and the sub-subsequence criterion for convergence of real sequences.

[Lemma A.3](#) below supplies the converse half of [Proposition 3.9](#). Its proof *does* use the forward half; but the forward half is proved where it is stated and does not use the converse, so there is no circularity. Two further standard facts are used outside [Section 3](#) and its appendix: Osgood’s lemma in the several-variable half of [Theorem 3.13](#) (which is never applied), and, in

Section 7, matrix calculus – Jacobi’s formula for $\partial \log|\det W|$ – together with the unstable-manifold theorem for maps in Theorem 7.20. Apart from these, the only *result* that a proof in these notes uses without our having proved it is Bochner’s Theorem 3.3, and it is used exactly once, in Remark 3.8, where it could be replaced by an appeal to Pólya’s criterion or by a direct Fourier computation. (Two more classical facts are quoted inside remarks – Pólya’s criterion itself and the Lukacs characterisation in Remark 3.16 – but no proof depends on them.)

A.1 Uniqueness and inversion

Lemma A.1 (Gaussian Fourier identity). *For every $\varepsilon > 0$ and every $x \in \mathbb{R}$,*

$$\int_{\mathbb{R}} e^{-itx} \exp\left(-\frac{1}{2}\varepsilon^2 t^2\right) dt = \frac{\sqrt{2\pi}}{\varepsilon} \exp\left(-\frac{x^2}{2\varepsilon^2}\right). \quad (214)$$

Consequently, for $x \in \mathbb{R}^d$,

$$(2\pi)^{-d} \int_{\mathbb{R}^d} e^{-it^\top x} \exp\left(-\frac{1}{2}\varepsilon^2 \|t\|^2\right) dt = \gamma_\varepsilon(x) := (2\pi\varepsilon^2)^{-d/2} \exp\left(-\frac{\|x\|^2}{2\varepsilon^2}\right), \quad (215)$$

the Lebesgue density of $\mathcal{N}(0, \varepsilon^2 I_d)$.

Proof. Write $I(x)$ for the left hand side of Eq. (214). Differentiation under the integral sign is legitimate, since $|t| \exp(-\frac{1}{2}\varepsilon^2 t^2)$ is integrable, and an integration by parts gives

$$I'(x) = \int_{\mathbb{R}} (-it) e^{-itx} e^{-\varepsilon^2 t^2/2} dt = \frac{i}{\varepsilon^2} \int_{\mathbb{R}} e^{-itx} \frac{d}{dt} \left(e^{-\varepsilon^2 t^2/2} \right) dt = -\frac{x}{\varepsilon^2} I(x), \quad (216)$$

the boundary terms vanishing. Hence $I(x) = I(0) \exp(-x^2/(2\varepsilon^2))$, and $I(0) = \int_{\mathbb{R}} e^{-\varepsilon^2 t^2/2} dt = \sqrt{2\pi}/\varepsilon$. This is Eq. (214); Eq. (215) follows by taking the product over the d coordinates. $\square$

Proof of Theorem 3.4. Let X have law μ and let $G \sim \mathcal{N}(0, I_d)$ be independent of X , both defined on one probability space. Fix $\varepsilon > 0$ and put $X_\varepsilon := X + \varepsilon G$. Conditioning on X shows that X_ε has the Lebesgue density $f_\varepsilon(x) = \int_{\mathbb{R}^d} \gamma_\varepsilon(x - y) \mu(dy)$. Inserting Eq. (215) and applying Fubini’s theorem – legitimate because the integrand has modulus $\exp(-\frac{1}{2}\varepsilon^2 \|t\|^2)$, which is integrable on $\mathbb{R}^d \times \mathbb{R}^d$ against the probability measure μ – we obtain

$$f_\varepsilon(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} e^{-it^\top(x-y)} e^{-\varepsilon^2 \|t\|^2/2} dt \mu(dy) \quad (217)$$

$$= (2\pi)^{-d} \int_{\mathbb{R}^d} e^{-it^\top x} e^{-\varepsilon^2 \|t\|^2/2} \varphi_\mu(t) dt. \quad (218)$$

So f_ε , and hence the law of X_ε , is determined by φ_μ alone.

Now let $g: \mathbb{R}^d \rightarrow \mathbb{R}$ be bounded and continuous. Since $X_\varepsilon \rightarrow X$ pointwise on the underlying probability space as $\varepsilon \downarrow 0$, dominated convergence gives

$$\int_{\mathbb{R}^d} g(x) f_\varepsilon(x) dx = \mathbb{E}[g(X_\varepsilon)] \xrightarrow{\varepsilon \downarrow 0} \mathbb{E}[g(X)] = \int_{\mathbb{R}^d} g d\mu. \quad (219)$$

The left hand side depends on μ only through φ_μ , so $\int g d\mu$ is determined by φ_μ for every bounded continuous g , and therefore μ is determined by φ_μ . This proves part (i).

For part (ii) assume in addition $\varphi_\mu \in L^1(\mathbb{R}^d)$ and define

$$f(x) := (2\pi)^{-d} \int_{\mathbb{R}^d} e^{-it^\top x} \varphi_\mu(t) dt. \quad (220)$$

Then f is bounded by $(2\pi)^{-d}\|\varphi_\mu\|_{L^1}$ and continuous, by dominated convergence. Comparing with Eq. (217),

$$\sup_{x \in \mathbb{R}^d} |f_\varepsilon(x) - f(x)| \leq (2\pi)^{-d} \int_{\mathbb{R}^d} |e^{-\varepsilon^2 \|t\|^2/2} - 1| |\varphi_\mu(t)| dt \xrightarrow{\varepsilon \downarrow 0} 0 \quad (221)$$

by dominated convergence, with dominating function $2|\varphi_\mu|$. Since every f_ε is a probability density and $f_\varepsilon \rightarrow f$ pointwise, the limit f is real valued and $f \geq 0$, so $\nu(dx) := f(x) dx$ defines a Borel measure on $\mathbb{R}^d$. Let now g be continuous with compact support. Uniform convergence on that compact support gives $\int g f_\varepsilon dx \rightarrow \int g f dx$, while Eq. (219) gives $\int g f_\varepsilon dx \rightarrow \int g d\mu$, so that

$$\int_{\mathbb{R}^d} g d\mu = \int_{\mathbb{R}^d} g d\nu, \quad g \in C_c(\mathbb{R}^d). \quad (222)$$

Choosing $g_k \in C_c(\mathbb{R}^d)$ with $0 \leq g_k \uparrow 1$ pointwise and applying monotone convergence on both sides of Eq. (222) gives $\nu(\mathbb{R}^d) = \mu(\mathbb{R}^d) = 1$, so ν is a Borel probability measure. By Eq. (222) and background fact (a), $\mu = \nu$; that is, μ has the density f . $\square$

A.2 Lévy's continuity theorem

Lemma A.2 (Tail bound from the characteristic function near the origin). *Let Y be a real-valued random variable and $\delta > 0$. Then*

$$\mathbb{P}\left[|Y| \geq \frac{2}{\delta}\right] \leq \frac{1}{\delta} \int_{-\delta}^{\delta} (1 - \operatorname{Re} \varphi_Y(t)) dt. \quad (223)$$

Proof. By Fubini's theorem, using $\operatorname{Re} \varphi_Y(t) = \mathbb{E}[\cos(tY)]$,

$$\int_{-\delta}^{\delta} (1 - \operatorname{Re} \varphi_Y(t)) dt = 2 \int_0^{\delta} \mathbb{E}[1 - \cos(tY)] dt = 2\delta \mathbb{E}\left[1 - \frac{\sin(\delta Y)}{\delta Y}\right], \quad (224)$$

with the convention $\sin(u)/u := 1$ at $u = 0$. The integrand $1 - \sin(u)/u$ is non-negative for all $u \in \mathbb{R}$, and for $|u| \geq 2$ it is at least $1 - 1/|u| \geq \frac{1}{2}$. Hence

$$2\delta \mathbb{E}\left[1 - \frac{\sin(\delta Y)}{\delta Y}\right] \geq 2\delta \cdot \frac{1}{2} \mathbb{P}[|\delta Y| \geq 2] = \delta \mathbb{P}\left[|Y| \geq \frac{2}{\delta}\right], \quad (225)$$

which is Eq. (223). $\square$

Proof of Theorem 3.7. (i) If $X_n \Rightarrow X$ then, for each fixed t , the functions $x \mapsto \cos(t^\top x)$ and $x \mapsto \sin(t^\top x)$ are bounded and continuous, so $\varphi_{X_n}(t) = \mathbb{E}[\cos(t^\top X_n)] + i \mathbb{E}[\sin(t^\top X_n)]$ converges to $\varphi_X(t)$.

(ii) Assume $\varphi_{X_n} \rightarrow \phi$ pointwise with ϕ continuous at 0. Since $\varphi_{X_n}(0) = 1$ for all n we have $\phi(0) = 1$.

Tightness. Let $\varepsilon > 0$. By continuity of ϕ at 0 and $\phi(0) = 1$ we may choose $\delta > 0$ so small that

$$\frac{1}{\delta} \int_{-\delta}^{\delta} (1 - \operatorname{Re} \phi(se_j)) ds < \frac{\varepsilon}{2d}, \quad j = 1, \dots, d. \quad (226)$$

By Eq. (10) the characteristic function of the j -th coordinate $e_j^\top X_n$ is $s \mapsto \varphi_{X_n}(se_j)$, and $|1 - \operatorname{Re} \varphi_{X_n}(se_j)| \leq 2$, so dominated convergence on the bounded interval $[-\delta, \delta]$ gives

$$\frac{1}{\delta} \int_{-\delta}^{\delta} (1 - \operatorname{Re} \varphi_{X_n}(se_j)) ds \xrightarrow{n \rightarrow \infty} \frac{1}{\delta} \int_{-\delta}^{\delta} (1 - \operatorname{Re} \phi(se_j)) ds < \frac{\varepsilon}{2d}. \quad (227)$$

By [Lemma A.2](#) there is therefore an N with $\mathbb{P}[|e_j^\top X_n| \geq 2/\delta] < \varepsilon/d$ for all $n \geq N$ and all $j = 1, \dots, d$, whence $\mathbb{P}[\|X_n\|_\infty \geq 2/\delta] < \varepsilon$ for $n \geq N$. Enlarging the compact set $[-2/\delta, 2/\delta]^d$ so as to accommodate the finitely many laws of $X_1, \dots, X_{N-1}$ as well, we conclude that $(\mathcal{L}(X_n))_{n \in \mathbb{N}}$ is tight.

Identification of the limit. Let $(X_{n_l})_l$ be an arbitrary subsequence. By Prokhorov's theorem it has a further subsequence converging weakly to the law of some random vector X . By part (i) the characteristic function of that limit is $\lim_l \varphi_{X_{n_l}} = \phi$; in particular ϕ is a characteristic function. By [Theorem 3.4](#) (i) every subsequential weak limit has the same law, namely the one with characteristic function ϕ . Now fix a bounded continuous $g: \mathbb{R}^d \rightarrow \mathbb{R}$. By the previous paragraph, every subsequence of the real sequence $(\mathbb{E}[g(X_n)])_n$ has a further subsequence converging to $\mathbb{E}[g(X)]$, and a real sequence with that property converges to $\mathbb{E}[g(X)]$. As g was an arbitrary bounded continuous function, $X_n \Rightarrow X$. $\square$

A.3 Entire characteristic functions

Both remaining proofs rest on the following observation: a characteristic function cannot be analytic “by accident”. As soon as it agrees with an entire function near the origin, the underlying law has Gaussian-type integrability.

Lemma A.3 (Moments from differentiability at the origin). *Let Y be a real-valued random variable, $n \in \mathbb{N}$, and suppose that φ_Y is $2n$ times differentiable at $t = 0$. Then $\mathbb{E}[Y^{2n}] < \infty$.*

Proof. We induct on n , the case $n = 0$ being trivial. Assume $\varphi_Y^{(2n+2)}(0)$ exists. Then $\varphi_Y^{(2n+1)}$ exists on a neighbourhood of the origin, so φ_Y is in particular $2n$ times differentiable at 0 and the induction hypothesis gives $\mathbb{E}[Y^{2n}] < \infty$. By [Proposition 3.9](#) the function φ_Y is $2n$ times continuously differentiable on $\mathbb{R}$ with $\varphi_Y^{(2n)}(t) = (-1)^n \mathbb{E}[Y^{2n} e^{itY}]$. Put $c := \mathbb{E}[Y^{2n}]$. If $c = 0$ then $Y = 0$ almost surely and there is nothing to prove, so assume $c > 0$ and let ϱ be the probability measure with $\varrho(dy) := c^{-1} y^{2n} \mathcal{L}(Y)(dy)$. Then $(-1)^n c^{-1} \varphi_Y^{(2n)}$ is the characteristic function φ_ϱ of ϱ , and by assumption it is twice differentiable at 0. Writing $h := \varphi_\varrho$ and using $h(t) + h(-t) - 2h(0) = -2 \int (1 - \cos(ty)) \varrho(dy)$ we get, by Fatou's lemma,

$$\int_{\mathbb{R}} y^2 \varrho(dy) \leq \liminf_{t \rightarrow 0} \int_{\mathbb{R}} \frac{2(1 - \cos(ty))}{t^2} \varrho(dy) = - \lim_{t \rightarrow 0} \frac{h(t) + h(-t) - 2h(0)}{t^2} = -h''(0) < \infty, \quad (228)$$

because $2(1 - \cos(ty))/t^2 \rightarrow y^2$ pointwise as $t \rightarrow 0$, and because the second symmetric difference quotient converges to $h''(0)$ whenever the latter exists, by Taylor's theorem with Peano remainder. Hence $\mathbb{E}[Y^{2n+2}] = c \int y^2 \varrho(dy) < \infty$. $\square$

Lemma A.4 (Entire characteristic functions have all exponential moments). *Let Y be a real-valued random variable and suppose that there are $\delta > 0$ and an entire function F with $\varphi_Y(t) = F(t)$ for all $t \in (-\delta, \delta)$. Then*

$$\mathbb{E}[\exp(a|Y|)] < \infty \quad \text{for every } a > 0, \quad (229)$$

the characteristic function φ_Y extends to the entire function $z \mapsto \mathbb{E}[e^{izY}]$, and this extension equals F on all of $\mathbb{C}$.

Proof. Since F is entire, φ_Y is infinitely differentiable on $(-\delta, \delta)$, so by [Lemma A.3](#) all even moments of Y are finite, hence by Lyapunov's inequality all moments are, and by

Proposition 3.9 $\varphi_Y^{(m)}(0) = i^m \mathbb{E}[Y^m]$ for every m . These are also the derivatives of F at 0, so the Taylor series of F at the origin is

$$F(z) = \sum_{m \geq 0} \frac{i^m \mathbb{E}[Y^m]}{m!} z^m, \quad (230)$$

and, F being entire, this series converges absolutely for every $z \in \mathbb{C}$. In particular, for every $a > 0$,

$$\sum_{k \geq 0} \frac{a^{2k}}{(2k)!} \mathbb{E}[Y^{2k}] < \infty. \quad (231)$$

Using $\exp(a|y|) \leq \exp(ay) + \exp(-ay)$ and monotone convergence,

$$\mathbb{E}[\exp(a|Y|)] \leq \mathbb{E}[e^{aY}] + \mathbb{E}[e^{-aY}] = 2 \sum_{k \geq 0} \frac{a^{2k}}{(2k)!} \mathbb{E}[Y^{2k}] < \infty. \quad (232)$$

By [Theorem 3.13](#) the map $z \mapsto \mathbb{E}[e^{izY}]$ is therefore entire and restricts to φ_Y on $\mathbb{R}$. It agrees with F on the interval $(-\delta, \delta)$, hence on all of $\mathbb{C}$ by the identity theorem. $\square$

A.4 Marcinkiewicz' theorem

Proof of Theorem 3.17. By assumption $\varphi_Y = e^g$ on a neighbourhood of the origin, with a polynomial g ; replacing g by $g - g(0)$ – which changes e^g not at all, and the degree only in the trivial case of a constant g , since $e^{g(0)} = \varphi_Y(0) = 1$ forces $g(0) \in 2\pi i \mathbb{Z}$ – we may and do assume $g(0) = 0$. The function e^g is entire, so [Lemma A.4](#) applies: Y has all exponential moments, and

$$\varphi_Y(z) = \mathbb{E}[e^{izY}] = \exp(g(z)), \quad z \in \mathbb{C}. \quad (233)$$

Write $g(z) = \sum_{m=0}^n c_m z^m$ with $c_n \neq 0$ and suppose, aiming at a contradiction, that $n \geq 3$.

Since Y has all exponential moments, [Proposition 3.15](#) applies on every horizontal strip and gives, for $z = x + iy$ with $x, y \in \mathbb{R}$,

$$\exp(\operatorname{Re} g(x + iy)) = |\varphi_Y(x + iy)| \leq \varphi_Y(iy) = \exp(\operatorname{Re} g(iy)), \quad (234)$$

where we used that $\varphi_Y(iy) = \mathbb{E}[e^{-yY}] > 0$ is real, so that $\varphi_Y(iy) = \exp(\operatorname{Re} g(iy))$. Taking logarithms,

$$u(x, y) \leq u(0, y), \quad u(x, y) := \operatorname{Re} g(x + iy), \quad x, y \in \mathbb{R}. \quad (235)$$

Write $c_n = |c_n|e^{i\psi}$ and, for $\theta \in [0, 2\pi)$ and $r > 0$, put $z = re^{i\theta}$. Since g is a polynomial of degree n ,

$$u(r \cos \theta, r \sin \theta) = \operatorname{Re}(c_n r^n e^{in\theta}) + O(r^{n-1}) = |c_n| \cos(\psi + n\theta) r^n + O(r^{n-1}), \quad (236)$$

uniformly in θ . Likewise, for $y \in \mathbb{R}$,

$$u(0, y) = \operatorname{Re}(c_n (iy)^n) + O(|y|^{n-1}) \leq |c_n| |y|^n + O(|y|^{n-1}). \quad (237)$$

Let θ^* be any maximiser of $\theta \mapsto \cos(\psi + n\theta)$, i.e. $\cos(\psi + n\theta^*) = 1$. Applying [Eq. \(235\)](#) with $x = r \cos \theta^*$ and $y = r \sin \theta^*$ and inserting [Eqs. \(236\)](#) and [\(237\)](#) gives

$$|c_n| r^n + O(r^{n-1}) \leq |c_n| |\sin \theta^*|^n r^n + O(r^{n-1}). \quad (238)$$

Dividing by r^n and letting $r \rightarrow \infty$ yields $|c_n| \leq |c_n| |\sin \theta^*|^n$, and since $c_n \neq 0$ and $|\sin \theta^*| \leq 1$ this forces $|\sin \theta^*| = 1$, i.e. $\theta^* \in \{\pi/2, 3\pi/2\}$.

But the equation $\cos(\psi + n\theta) = 1$ holds exactly when $\psi + n\theta \in 2\pi\mathbb{Z}$, so its solutions form an arithmetic progression of spacing $2\pi/n$, of which exactly n lie in $[0, 2\pi)$. For $n \geq 3$ there are thus at least three maximisers, while only two values of θ in $[0, 2\pi)$ satisfy $|\sin \theta| = 1$. This contradiction shows $n \leq 2$.

It remains to determine the coefficients. We have $g(t) = c_1 t + c_2 t^2$ with $g(0) = 0$. From $\varphi_Y(-t) = \overline{\varphi_Y(t)}$ (Proposition 3.2 (iii)) we get $\exp(g(-t)) = \exp(\overline{g(t)})$ for all real t , so the polynomial $t \mapsto g(-t) - \overline{g(t)}$ takes values in $2\pi i\mathbb{Z}$; being continuous and vanishing at $t = 0$, it vanishes identically. Hence $-c_1 t + c_2 t^2 = \overline{c_1} t + \overline{c_2} t^2$ for all real t , whence $c_2 = \overline{c_2}$ is real and $c_1 = -\overline{c_1}$ is purely imaginary, say $c_1 = i\mu$ with $\mu \in \mathbb{R}$. Then $|\varphi_Y(t)| = \exp(c_2 t^2) \leq 1$ for all $t \in \mathbb{R}$ forces $c_2 \leq 0$, so $c_2 = -\frac{1}{2}\sigma^2$ with $\sigma^2 := -2c_2 \in \mathbb{R}_{\geq 0}$. Therefore $\varphi_Y(t) = \exp(i\mu t - \frac{1}{2}\sigma^2 t^2)$, which by Notation 2.4 and Theorem 3.4 (i) means $Y \sim \mathcal{N}(\mu, \sigma^2)$. $\square$

A.5 Cramér's decomposition theorem

The growth estimate needed for Cramér's theorem is supplied by the following classical coefficient bound, which replaces an appeal to Hadamard's factorisation theorem.

Lemma A.5 (Borel–Carathéodory coefficient bound). *Let h be an entire function with $h(0) = 0$ and Taylor expansion $h(z) = \sum_{m \geq 1} a_m z^m$. Then for every $r > 0$ and every $m \geq 1$,*

$$|a_m| r^m \leq 4M(r), \quad M(r) := \max_{|z|=r} \operatorname{Re} h(z). \quad (239)$$

Proof. Note first that $M(r) \geq \operatorname{Re} h(0) = 0$ by the mean value property applied to the harmonic function $\operatorname{Re} h$. Write $a_m = \alpha_m + i\beta_m$ with $\alpha_m, \beta_m \in \mathbb{R}$. On the circle $z = re^{i\vartheta}$,

$$\operatorname{Re} h(re^{i\vartheta}) = \sum_{m \geq 1} r^m (\alpha_m \cos(m\vartheta) - \beta_m \sin(m\vartheta)), \quad (240)$$

the series converging uniformly in ϑ . Termwise integration gives the Fourier coefficients

$$\frac{1}{\pi} \int_0^{2\pi} \operatorname{Re} h(re^{i\vartheta}) \cos(m\vartheta) d\vartheta = r^m \alpha_m, \quad \frac{1}{\pi} \int_0^{2\pi} \operatorname{Re} h(re^{i\vartheta}) \sin(m\vartheta) d\vartheta = -r^m \beta_m, \quad (241)$$

while the mean value property gives $\int_0^{2\pi} \operatorname{Re} h(re^{i\vartheta}) d\vartheta = 2\pi \operatorname{Re} h(0) = 0$. Therefore, for either sign,

$$\pm r^m \alpha_m = \frac{1}{\pi} \int_0^{2\pi} \operatorname{Re} h(re^{i\vartheta}) (1 \pm \cos(m\vartheta)) d\vartheta \leq \frac{M(r)}{\pi} \int_0^{2\pi} (1 \pm \cos(m\vartheta)) d\vartheta = 2M(r), \quad (242)$$

where we used $1 \pm \cos(m\vartheta) \geq 0$ and $\operatorname{Re} h(re^{i\vartheta}) \leq M(r)$. Hence $|\alpha_m| r^m \leq 2M(r)$, and the same argument with $1 \pm \sin(m\vartheta)$ gives $|\beta_m| r^m \leq 2M(r)$. Thus $|a_m| r^m \leq (|\alpha_m| + |\beta_m|) r^m \leq 4M(r)$. $\square$

Proof of Theorem 3.20. Let $Y_1 \perp\!\!\!\perp Y_2$ with $S := Y_1 + Y_2 \sim \mathcal{N}(\mu, \sigma^2)$.

The degenerate case. If $\sigma^2 = 0$ then $\varphi_S = \varphi_{Y_1} \varphi_{Y_2}$ has modulus 1 everywhere, and since $|\varphi_{Y_j}| \leq 1$ this forces $|\varphi_{Y_j}| \equiv 1$. Then the symmetrised variable $Y_j - Y'_j$, with Y'_j an independent copy of Y_j , has characteristic function $|\varphi_{Y_j}|^2 \equiv 1$, hence is almost surely 0 by Theorem 3.4 (i); so Y_j is almost surely constant, i.e. degenerate Gaussian. Assume from now on $\sigma > 0$, and, replacing Y_1 by $Y_1 - \mu$, that $\mu = 0$.

Step 1: Gaussian tails. Let (Y'_1, Y'_2) be an independent copy of (Y_1, Y_2) and put $\tilde{Y}_j := Y_j - Y'_j$ and $\tilde{S} := \tilde{Y}_1 + \tilde{Y}_2 \sim \mathcal{N}(0, 2\sigma^2)$. The variables $\tilde{Y}_1, \tilde{Y}_2$ are independent and symmetric, so $\mathbb{P}[\tilde{Y}_2 \geq 0] \geq \frac{1}{2}$ and hence, for $x > 0$,

$$\mathbb{P}[\tilde{S} > x] \geq \mathbb{P}[\tilde{Y}_1 > x, \tilde{Y}_2 \geq 0] = \mathbb{P}[\tilde{Y}_1 > x] \mathbb{P}[\tilde{Y}_2 \geq 0] \geq \frac{1}{2} \mathbb{P}[\tilde{Y}_1 > x]. \quad (243)$$

A Chernoff bound for $\tilde{S} \sim \mathcal{N}(0, 2\sigma^2)$ gives $\mathbb{P}[\tilde{S} > x] \leq \exp(-x^2/(4\sigma^2))$, so by symmetry

$$\mathbb{P}[|\tilde{Y}_1| > x] \leq 4 \exp\left(-\frac{x^2}{4\sigma^2}\right), \quad x > 0. \quad (244)$$

Let θ be a median of Y_1 , so that $\mathbb{P}[Y'_1 \leq \theta] \geq \frac{1}{2}$ and $\mathbb{P}[Y'_1 \geq \theta] \geq \frac{1}{2}$. Then

$$\mathbb{P}[\tilde{Y}_1 > x] \geq \mathbb{P}[Y_1 > \theta + x, Y'_1 \leq \theta] \geq \frac{1}{2} \mathbb{P}[Y_1 > \theta + x], \quad (245)$$

and symmetrically for the left tail, so Eq. (244) yields $\mathbb{P}[|Y_1 - \theta| > x] \leq 16 \exp(-x^2/(4\sigma^2))$. Consequently $\mathbb{E}[\exp(a|Y_1|)] < \infty$ for every $a > 0$, and the same holds for Y_2 .

Step 2: entire extensions. By Theorem 3.13 both φ_{Y_1} and φ_{Y_2} extend to entire functions, given by $\varphi_{Y_j}(z) = \mathbb{E}[e^{izY_j}]$. The entire functions $z \mapsto \varphi_{Y_1}(z)\varphi_{Y_2}(z)$ and $z \mapsto \exp(-\frac{1}{2}\sigma^2 z^2)$ agree on $\mathbb{R}$, hence on $\mathbb{C}$ by the identity theorem:

$$\varphi_{Y_1}(z) \varphi_{Y_2}(z) = \exp\left(-\frac{1}{2}\sigma^2 z^2\right), \quad z \in \mathbb{C}. \quad (246)$$

In particular neither factor has a zero in $\mathbb{C}$.

Step 3: growth. Fix $j \in \{1, 2\}$ and let l be the other index. By Proposition 3.15, for $z = x + iy$, $|\varphi_{Y_j}(z)| \leq \varphi_{Y_j}(iy) = \mathbb{E}[e^{-yY_j}]$. By Eq. (246) at $z = iy$, $\varphi_{Y_j}(iy) \varphi_{Y_l}(iy) = \exp(\frac{1}{2}\sigma^2 y^2)$, and Jensen's inequality gives $\varphi_{Y_l}(iy) = \mathbb{E}[e^{-yY_l}] \geq \exp(-y \mathbb{E}[Y_l])$. Hence, with $C := |\mathbb{E}[Y_l]|$,

$$|\varphi_{Y_j}(z)| \leq \frac{\exp(\frac{1}{2}\sigma^2 y^2)}{\varphi_{Y_l}(iy)} \leq \exp\left(\frac{1}{2}\sigma^2 y^2 + C|y|\right) \leq \exp\left(\frac{1}{2}\sigma^2 |z|^2 + C|z|\right). \quad (247)$$

Step 4: conclusion. Since φ_{Y_j} is entire and nowhere zero on the simply connected domain $\mathbb{C}$, there is an entire h_j with $\varphi_{Y_j} = e^{h_j}$, and we may normalise $h_j(0) = 0$ because $\varphi_{Y_j}(0) = 1$. By Eq. (247),

$$\operatorname{Re} h_j(z) = \log |\varphi_{Y_j}(z)| \leq \frac{1}{2}\sigma^2 |z|^2 + C|z|, \quad (248)$$

so in the notation of Lemma A.5, $M(r) \leq \frac{1}{2}\sigma^2 r^2 + Cr$. For the Taylor coefficients a_m of h_j and every $m \geq 3$, Eq. (239) gives

$$|a_m| \leq \frac{4M(r)}{r^m} \leq \frac{2\sigma^2 r^2 + 4Cr}{r^m} \xrightarrow{r \rightarrow \infty} 0, \quad (249)$$

so $a_m = 0$ for all $m \geq 3$ and h_j is a polynomial of degree at most 2. Thus $\varphi_{Y_j} = \exp(h_j)$ with a polynomial h_j , and Marcinkiewicz' Theorem 3.17 shows that Y_j is Gaussian, possibly degenerate. $\square$

B A Proof of the Kagan–Linnik–Rao Theorem

This appendix proves Theorem 4.2, the one result that Sections 4 to 7 take as given. Its first part – the equality of images and ranks – was already proved in Proposition 4.4; what remains is the column dichotomy Theorem 4.2(1)–(2).

The strategy is the classical one that goes back to Marcinkiewicz (1939), Darmois (1953) and Skitovich (1954), and that is developed systematically in Kagan et al. (1973, Chapters 3 and 10) and Linnik and Ostrovskii (1977). It has three steps.

- (1) Take distinguished logarithms in the identity between the two characteristic functions. Because each representation is an independent sum of *one-dimensional* sources evaluated along the columns of the mixing matrix, the identity becomes a linear relation among *ridge functions* $t \mapsto \psi(a^\top t)$ on $\mathbb{R}^p$.
- (2) Kill all but one of these ridge functions with finite-difference operators: a ridge function in direction a is annihilated by the difference operator in any direction $h \perp a$, and the hypothesis that the columns are pairwise non-proportional is exactly what guarantees that each unwanted direction admits such an h that does *not* annihilate the term we want to keep.
- (3) What survives is a one-dimensional function all of whose iterated differences vanish. By Fréchet’s functional equation such a function is a polynomial, and Marcinkiewicz’ [Theorem 3.17](#) converts “polynomial cumulant generating function” into “Gaussian”.

Everything below uses only the elementary properties of characteristic functions collected in [Proposition 3.2](#) – in particular [Eqs. \(8\) and \(9\)](#) – the distinguished logarithm of [Lemma 3.10](#), Marcinkiewicz’ [Theorem 3.17](#), [Proposition 4.4](#) for the first part, and elementary calculus.

B.1 Finite differences

Definition B.1 (Difference operator). For a function F defined on a subset of $\mathbb{R}^p$ and a vector $h \in \mathbb{R}^p$ we write

$$(\Delta_h F)(t) := F(t + h) - F(t), \quad (250)$$

whenever both t and $t + h$ lie in the domain of F . Difference operators in different directions commute, and iterating gives the inclusion–exclusion formula

$$(\Delta_{h_1} \cdots \Delta_{h_m} F)(t) = \sum_{S \subseteq \{1, \dots, m\}} (-1)^{m-|S|} F\left(t + \sum_{r \in S} h_r\right), \quad (251)$$

which requires F to be defined at the 2^m points $t + \sum_{r \in S} h_r$.

A function of the form $t \mapsto \psi(a^\top t)$, constant along the hyperplane $a^\perp$, is called a *ridge function* in the direction a . The next two lemmas are the whole geometric input.

Lemma B.2 (Ridge functions are annihilated by orthogonal differences). *Let $a \in \mathbb{R}^p$, let ψ be a function on a subset of $\mathbb{R}$ and put $F(t) := \psi(a^\top t)$. If $h \in \mathbb{R}^p$ satisfies $a^\top h = 0$, then $\Delta_h F = 0$ on the domain of F .*

Proof. $F(t + h) = \psi(a^\top t + a^\top h) = \psi(a^\top t) = F(t)$. □

Lemma B.3 (Separating two non-proportional directions). *Let $b, d \in \mathbb{R}^p \setminus \{0\}$ be non-proportional. Then there exists $h \in \mathbb{R}^p$ with*

$$d^\top h = 0, \quad b^\top h \neq 0. \quad (252)$$

Moreover the set of such h is open in the hyperplane $d^\perp$, so h may be chosen with $\|h\|$ arbitrarily small, and sh is admissible for every $s \neq 0$.

Proof. Suppose no such h exists, i.e. $d^\perp \subseteq b^\perp$. Taking orthogonal complements reverses the inclusion, so $\text{span}\{b\} = (b^\perp)^\perp \subseteq (d^\perp)^\perp = \text{span}\{d\}$. As $b \neq 0$ this makes b a non-zero multiple of d , contradicting non-proportionality. The set $\{h \in d^\perp : b^\top h \neq 0\}$ is the complement in $d^\perp$ of the closed subspace $b^\perp \cap d^\perp$, hence open in $d^\perp$; and it is invariant under $h \mapsto sh$ for $s \neq 0$, so it contains vectors of arbitrarily small norm. □

The analytic input is the local form of Fréchet's functional equation. We prove it from scratch; the argument is a smoothing bootstrap followed by one differentiation.

Lemma B.4 (Fréchet's functional equation, local form). *Let $m \in \mathbb{N}$ and let $\rho > \sigma > 0$ and $\varepsilon > 0$ satisfy $\sigma + m\varepsilon \leq \rho$. Let $F: (-\rho, \rho) \rightarrow \mathbb{C}$ be continuous and assume*

$$(\Delta_{c_1} \cdots \Delta_{c_m} F)(u) = 0 \quad \text{whenever } |c_1|, \dots, |c_m| < \varepsilon \text{ and } |u| < \sigma. \quad (253)$$

Then F coincides on $(-\frac{\sigma}{4}, \frac{\sigma}{4})$ with a polynomial of degree at most $m - 1$.

Proof. Note first that the condition $\sigma + m\varepsilon \leq \rho$ makes Eq. (253) meaningful: by Eq. (251) the left hand side only involves the values of F at the points $u + \sum_{r \in S} c_r$, all of which lie in $(-\rho, \rho)$.

Put

$$\kappa := \frac{1}{2} \min\left(\varepsilon, \frac{\sigma}{2m^2}\right) > 0, \quad (254)$$

and define the symmetric averaging operator

$$(M_\kappa G)(u) := \frac{1}{2\kappa} \int_{-\kappa}^{\kappa} G(u + s) ds = \frac{1}{2\kappa} \int_{u-\kappa}^{u+\kappa} G(v) dv, \quad (255)$$

defined for $|u| < A - \kappa$ whenever G is continuous on $(-A, A)$.

Step 1: M_κ raises smoothness. If G is continuous on $(-A, A)$ then, by the fundamental theorem of calculus, $M_\kappa G$ is continuously differentiable on $(-A + \kappa, A - \kappa)$ with

$$(M_\kappa G)'(u) = \frac{1}{2\kappa} (G(u + \kappa) - G(u - \kappa)). \quad (256)$$

Consequently, if $G \in C^r$ on $(-A, A)$ then $M_\kappa G \in C^{r+1}$ on $(-A + \kappa, A - \kappa)$.

Step 2: the hypothesis says $(I - M_\kappa)^m F = 0$. We have

$$F(u) - (M_\kappa F)(u) = -\frac{1}{2\kappa} \int_{-\kappa}^{\kappa} (\Delta_s F)(u) ds, \quad (257)$$

and translations – hence all the operators Δ_s and M_κ – commute with one another, so Fubini's theorem gives

$$[(I - M_\kappa)^m F](u) = \frac{(-1)^m}{(2\kappa)^m} \int_{-\kappa}^{\kappa} \cdots \int_{-\kappa}^{\kappa} (\Delta_{s_1} \cdots \Delta_{s_m} F)(u) ds_1 \cdots ds_m = 0 \quad (258)$$

for $|u| < \sigma$: every s_r occurring in the integral satisfies $|s_r| \leq \kappa < \varepsilon$, so the integrand vanishes identically by Eq. (253).

Step 3: bootstrap. Expanding the binomial in Eq. (258) and solving for the $n = 0$ term,

$$F = \sum_{n=1}^m (-1)^{n+1} \binom{m}{n} M_\kappa^n F \quad \text{on } (-\sigma, \sigma). \quad (259)$$

Every term on the right applies M_κ at least once. Hence, by Step 1, if $F \in C^r$ on $(-A, A)$ for some $A \leq \rho$, then the right hand side of Eq. (259) is C^{r+1} on $(-A + m\kappa, A - m\kappa)$, and therefore $F \in C^{r+1}$ on $(-A', A')$ with $A' := \min(\sigma, A - m\kappa)$. Starting from $F \in C^0$ on $(-\rho, \rho)$ and iterating m times, and using $\rho - m\kappa \geq \rho - m\varepsilon \geq \sigma$ at the first step, we obtain $F \in C^m$ on $(-A_m, A_m)$ with $A_m \geq \sigma - (m - 1)m\kappa \geq \sigma - m^2\kappa \geq \frac{3}{4}\sigma$, where the last inequality uses $\kappa \leq \sigma/(4m^2)$.

Step 4: differentiate. Fix u with $|u| < \sigma/4$ and let $c_1, \dots, c_m$ range over $(-\delta, \delta)$, where $\delta := \frac{1}{2} \min(\varepsilon, \sigma/(2m))$, so that $|u| + m\delta < \frac{3}{4}\sigma \leq A_m$ and $\delta < \varepsilon$. By [Eq. \(251\)](#) and [Eq. \(253\)](#),

$$\sum_{S \subseteq \{1, \dots, m\}} (-1)^{m-|S|} F\left(u + \sum_{r \in S} c_r\right) = 0, \quad (260)$$

identically in $(c_1, \dots, c_m) \in (-\delta, \delta)^m$. All terms are C^m functions of $(c_1, \dots, c_m)$ by Step 3, so we may apply $\partial^m/\partial c_1 \cdots \partial c_m$. A summand indexed by $S \neq \{1, \dots, m\}$ does not depend on c_{r_0} for any $r_0 \notin S$ and is therefore annihilated; the summand indexed by $S = \{1, \dots, m\}$ contributes $F^{(m)}(u + c_1 + \cdots + c_m)$. Hence $F^{(m)}(u + c_1 + \cdots + c_m) = 0$, and setting $c_1 = \cdots = c_m = 0$ gives $F^{(m)}(u) = 0$. As u was arbitrary in the interval $(-\frac{\sigma}{4}, \frac{\sigma}{4})$, which is connected, F is there a polynomial of degree at most $m - 1$. $\square$

Remark B.5 (What the hypotheses of [Lemma B.4](#) do and do not say). Two features of [Eq. \(253\)](#) are essential. First, the steps $c_1, \dots, c_m$ are allowed to *vary* over a neighbourhood of the origin, which is what makes the averaging in [Eq. \(258\)](#) legitimate; for a single fixed step vector the conclusion is false, as the 1-periodic function $F(u) = \sin(2\pi u)$ with $m = 1$, $c_1 = 1$ shows. Second, only continuity of F is assumed; no differentiability is available a priori, and Step 3 is precisely what manufactures it. The global version of the statement – a continuous F on $\mathbb{R}$ with $\Delta_{c_1} \cdots \Delta_{c_m} F \equiv 0$ for all $c_1, \dots, c_m \in \mathbb{R}$ is a polynomial of degree $< m$ – is due to [Fréchet \(1909\)](#), and the equation carries his name. Restricted-domain versions, in which the increments and the base point are confined to a subset, have been studied since; [Ger \(1994\)](#) shows that a solution on a sufficiently regular domain always extends to a polynomial function on the whole space. We have given a direct proof of the one-dimensional continuous case, which is all that is needed below and is short enough not to warrant a detour.

B.2 Proof of the column dichotomy

Proof of [Theorem 4.2 \(1\)–\(2\)](#). Write $k_i := k^{(i)}$, let $a_1, \dots, a_{k_1}$ be the columns of $A^{(1)}$ and $b_1, \dots, b_{k_2}$ the columns of $A^{(2)}$, and put

$$\phi_j := \varphi_{Z_j^{(1)}}, \quad \gamma_l := \varphi_{Z_l^{(2)}}, \quad c := \mu^{(2)} - \mu^{(1)}. \quad (261)$$

By [Eqs. \(8\)](#) and [\(9\)](#), the two representations [Eq. \(54\)](#) give, for every $t \in \mathbb{R}^p$,

$$\exp(it^\top \mu^{(1)}) \prod_{j=1}^{k_1} \phi_j(a_j^\top t) = \varphi_X(t) = \exp(it^\top \mu^{(2)}) \prod_{l=1}^{k_2} \gamma_l(b_l^\top t). \quad (262)$$

Step 1: the logarithmic form. By [Proposition 3.2 \(i\)–\(ii\)](#) each ϕ_j and each γ_l is continuous and equals 1 at the origin, so there is a $\delta_0 > 0$ such that all of them are non-vanishing on $(-\delta_0, \delta_0)$, and [Lemma 3.10](#) supplies the cumulant generating functions $\psi_j := \psi_{Z_j^{(1)}}$ and $\chi_l := \psi_{Z_l^{(2)}}$, continuous on $(-\delta_0, \delta_0)$ with $\psi_j(0) = \chi_l(0) = 0$ and $\phi_j = \exp(\psi_j)$, $\gamma_l = \exp(\chi_l)$ there. Choose $R > 0$ so small that $|a_j^\top t| < \delta_0$ and $|b_l^\top t| < \delta_0$ for all j, l whenever $\|t\| < R$, and define on the ball $B_R(0) \subseteq \mathbb{R}^p$

$$\Phi(t) := \sum_{j=1}^{k_1} \psi_j(a_j^\top t) - \sum_{l=1}^{k_2} \chi_l(b_l^\top t) - it^\top c. \quad (263)$$

Then Φ is continuous, $\Phi(0) = 0$, and by [Eq. \(262\)](#) $\exp(\Phi) \equiv 1$ on $B_R(0)$. Hence $\Phi/(2\pi i)$ is a continuous, integer-valued function on the connected set $B_R(0)$ vanishing at the origin, so it

vanishes identically:

$$\sum_{j=1}^{k_1} \psi_j(a_j^\top t) - \sum_{l=1}^{k_2} \chi_l(b_l^\top t) = i t^\top c, \quad t \in B_R(0). \quad (264)$$

This is a linear relation among ridge functions, and it is the only consequence of the hypotheses that we shall use.

Step 2: choosing the directions. We treat both parts at once. Fix the index $l_0 \in \{1, \dots, k_2\}$ of the column under consideration and let

$$e := \begin{cases} b_{l_0}, & \text{in case (1),} \\ a_{j_0}, & \text{in case (2), where } b_{l_0} = \lambda a_{j_0}, \end{cases} \quad (265)$$

and let $\mathcal{D}$ be the multiset of the remaining column directions,

$$\mathcal{D} := \begin{cases} \{a_1, \dots, a_{k_1}\} \uplus \{b_l : l \neq l_0\}, & \text{in case (1),} \\ \{a_j : j \neq j_0\} \uplus \{b_l : l \neq l_0\}, & \text{in case (2),} \end{cases} \quad (266)$$

where $\uplus$ is the disjoint union, so that $|\mathcal{D}| = k_1 + k_2 - 1$ in case (1) and $k_1 + k_2 - 2$ in case (2) even if the same vector happens to be a column of both matrices. We claim that no $d \in \mathcal{D}$ is proportional to e . In case (1) this holds for $d = a_j$ by the hypothesis of part (1) and for $d = b_l$, $l \neq l_0$, because the columns of $A^{(2)}$ are pairwise non-proportional. In case (2) it holds for $d = a_j$, $j \neq j_0$, because the columns of $A^{(1)}$ are pairwise non-proportional, and for $d = b_l$, $l \neq l_0$, because $b_l \parallel a_{j_0}$ would give $b_l \parallel \lambda a_{j_0} = b_{l_0}$, again contradicting pairwise non-proportionality. All columns are non-zero by hypothesis (i), so $e \neq 0$ and $d \neq 0$.

Let $m_0 := |\mathcal{D}|$ and set $m := \max(m_0, 2)$. For $r = 1, \dots, m_0$ apply [Lemma B.3](#) to the pair (e, d_r) : it yields $h_r \in \mathbb{R}^p$ with $d_r^\top h_r = 0$ and $\alpha_r := e^\top h_r \neq 0$, and h_r may be taken with $\|h_r\| < R/(2m)$. For $r = m_0 + 1, \dots, m$ (there are at most two such r , and only when $m_0 < 2$) simply put $h_r := \tau e / \|e\|^2$ with $\tau > 0$ small, so that $\alpha_r = \tau \neq 0$ and $\|h_r\| < R/(2m)$ as well. Note that each h_r may be replaced by $s_r h_r$ for any $s_r \in (-1, 1) \setminus \{0\}$ without disturbing any of these properties, and that this replaces α_r by $s_r \alpha_r$.

Step 3: annihilating all but one term. Apply the commuting operators $\Delta_{h_1}, \dots, \Delta_{h_m}$ to [Eq. \(264\)](#). This is legitimate on $B_{R/2}(0)$, because $\|t\| < R/2$ and $\|h_r\| < R/(2m)$ force $\|t + \sum_{r \in S} h_r\| < R$ for every $S \subseteq \{1, \dots, m\}$. Each ridge function indexed by a direction $d_r \in \mathcal{D}$ is annihilated by Δ_{h_r} ([Lemma B.2](#)), hence by the whole composition; and the affine term $i t^\top c$ is annihilated by any two of the operators, of which there are $m \geq 2$. What survives is

$$\Delta_{h_1} \cdots \Delta_{h_m} [\Xi(e^\top t)] = 0, \quad \|t\| < R/2, \quad \text{where} \quad \Xi := \begin{cases} -\chi_{l_0}, & \text{in case (1),} \\ \psi_{j_0} - \chi_{l_0}(\lambda \cdot), & \text{in case (2).} \end{cases} \quad (267)$$

In case (2) we used that $b_{l_0}^\top t = \lambda a_{j_0}^\top t = \lambda e^\top t$, so that the two surviving ridge functions share the direction e and combine into a single one.

Step 4: Fréchet. Set $\rho := \|e\|R$. Every u with $|u| < \rho$ is of the form $u = e^\top t$ for some $t \in B_R(0)$, and for such t the choice of R puts every $a_j^\top t$ and every $b_l^\top t$ inside $(-\delta_0, \delta_0)$. In case (1), $\Xi = -\chi_{l_0}$ is evaluated at $u = b_{l_0}^\top t$; in case (2) it is built from ψ_{j_0} evaluated at $u = a_{j_0}^\top t$ and from χ_{l_0} evaluated at $\lambda u = b_{l_0}^\top t$. In either case all arguments lie in $(-\delta_0, \delta_0)$, so Ξ is defined and continuous on $(-\rho, \rho)$. Because $e \neq 0$, the linear functional $t \mapsto e^\top t$ maps $B_{R/2}(0)$

onto $(-\sigma, \sigma)$ with $\sigma := \|e\|R/2 = \rho/2$, and by Eq. (251) the left hand side of Eq. (267) equals $(\Delta_{\alpha_1} \cdots \Delta_{\alpha_m} \Xi)(e^\top t)$. Hence

$$(\Delta_{\alpha_1} \cdots \Delta_{\alpha_m} \Xi)(u) = 0, \quad |u| < \sigma. \quad (268)$$

Replacing each h_r by $s_r h_r$ with $s_r \in (-1, 1)$, as permitted in Step 2, replaces α_r by $s_r \alpha_r$ without affecting anything else, and as s_r ranges over $(-1, 1)$ the number $s_r \alpha_r$ ranges over the whole interval $(-|\alpha_r|, |\alpha_r|)$. So Eq. (268) holds with α_r replaced by any c_r with $0 < |c_r| < \varepsilon := \min_r |\alpha_r|$; and for $c_r = 0$ it holds trivially, because $\Delta_0 = 0$. This is exactly the hypothesis Eq. (253). Finally $|\alpha_r| = |e^\top h_r| \leq \|e\| \|h_r\| < \|e\| R/(2m) = \rho/(2m)$, so $\sigma + m\varepsilon < \rho/2 + \rho/2 = \rho$ and Lemma B.4 applies. It makes Ξ a polynomial of degree at most $m - 1$ on $(-\frac{\sigma}{4}, \frac{\sigma}{4})$, a neighbourhood of the origin in $\mathbb{R}$.

Step 5: conclusion. In case (1) we have obtained that $\chi_{l_0} = -\Xi$ is a polynomial near the origin, so

$$\varphi_{Z_{l_0}^{(2)}}(t) = \exp(\chi_{l_0}(t)) \quad (269)$$

near the origin with a polynomial exponent. By Marcinkiewicz' Theorem 3.17, $Z_{l_0}^{(2)}$ is a (possibly degenerate) Gaussian random variable, which is assertion (1).

In case (2), $\Xi(t) = \psi_{j_0}(t) - \chi_{l_0}(\lambda t)$ is a polynomial near the origin; writing $g := -\Xi$ and exponentiating,

$$\varphi_{Z_{l_0}^{(2)}}(\lambda t) = \exp(\chi_{l_0}(\lambda t)) = \exp(\psi_{j_0}(t)) \exp(g(t)) = \varphi_{Z_{j_0}^{(1)}}(t) \cdot \exp(g(t)) \quad (270)$$

near the origin, which is Eq. (56). For the final assertion of (2), note that this is exactly the situation of Remark 3.18 (i) *with the factor λ present*, which is why that remark was stated with the factor: it gives directly that $Z_{l_0}^{(2)}$ is Gaussian if and only if $Z_{j_0}^{(1)}$ is. (The factor cannot be dropped: by Eq. (8) the left hand side is $\varphi_{\lambda Z_{l_0}^{(2)}}(t)$, and it is only because $\lambda \neq 0$ that $\lambda Z_{l_0}^{(2)}$ being Gaussian is equivalent to $Z_{l_0}^{(2)}$ being Gaussian; cf. Footnote 1.) $\square$

Remark B.6 (Reading the proof). Three points are worth isolating.

Where each hypothesis enters. That the columns are non-zero is used twice: in Step 2, to make Lemma B.3 applicable to the pairs (e, d_r) , and in Step 4, where it makes $e^\top t$ surject onto an interval, so that a statement about $t \in \mathbb{R}^p$ becomes a statement about $u \in \mathbb{R}$. Pairwise non-proportionality is used only in Step 2, to make Lemma B.3 applicable, and only for those columns that have to be separated from e : in case (2) for the columns of both matrices, in case (1) for the columns of $A^{(2)}$ only – there the separation of b_{l_0} from the columns of $A^{(1)}$ is the hypothesis of part (1) itself, and the columns of $A^{(1)}$ may perfectly well be proportional to one another. Mutual independence of the components, hypothesis (iii)(a), is used once, to factorise the characteristic functions in Eq. (262). Non-constancy of the sources, hypothesis (iii)(b), is not needed for (1)–(2) at all; it enters only through Proposition 4.4, i.e. in the first part of Theorem 4.2.

Why the degree bound is irrelevant. Step 4 produces a polynomial of degree at most $m - 1$, where $m = \max(k_1 + k_2 - 1, 2)$ in case (1) and $m = \max(k_1 + k_2 - 2, 2)$ in case (2); this bound grows with the number of sources and is of no use by itself. It is Marcinkiewicz' theorem that collapses it to 2 – which is precisely why Theorem 3.17 was called the single most important analytic input of these notes in Section 3.4.

Why the argument is local. Everything happens on a ball around the origin, because Lemma 3.10 produces the cumulant generating functions only there. This is not a defect: by Remark 3.8 a local identity between characteristic functions carries no information about

the laws in general, and the reason it does here is that the local conclusion is “exp of a polynomial”, a form rigid enough for [Theorem 3.17](#) to upgrade it to a global statement about the distribution.

Remark B.7 (Relation to the Darmois–Skitovich theorem). The classical theorem of [Darmois \(1953\)](#) and [Skitovich \(1954\)](#) says this: if $Y_1, \dots, Y_k$ are independent and the two linear forms $L_1 = \sum_j \alpha_j Y_j$ and $L_2 = \sum_j \beta_j Y_j$ are independent of *each other*, then every Y_j with $\alpha_j \beta_j \neq 0$ is Gaussian. It is the case $p = 2$ of [Theorem 4.2](#). Indeed, put $X := [L_1, L_2]^\top \in \mathbb{R}^2$ and compare the two representations

$$I_2 \begin{bmatrix} L_1 \\ L_2 \end{bmatrix} = X = \begin{bmatrix} \alpha_1 & \cdots & \alpha_k \\ \beta_1 & \cdots & \beta_k \end{bmatrix} Y, \quad (271)$$

so that $k^{(1)} = 2$ with $A^{(1)} = I_2$, and $k^{(2)} = k$ with $a_j^{(2)} = [\alpha_j, \beta_j]^\top$. A *non-zero* column $a_j^{(2)}$ is proportional to a column of I_2 precisely when $\alpha_j \beta_j = 0$ (zero columns having been removed by [Remark 4.1](#)), so for every j with $\alpha_j \beta_j \neq 0$ part (1) declares Y_j Gaussian. Two caveats: the columns $a_j^{(2)}$ need not be pairwise non-proportional, so one first passes to a normalised representation as in [Remark 4.1](#), which *merges* the sources belonging to proportional columns; and one then recovers the individual Y_j from the merged sum by finitely many applications of Cramér’s [Theorem 3.20](#), the case of an a.s. constant merged source being covered by the convention that constants are degenerate Gaussians ([Notation 2.4](#)).

The two-representation formulation of [Theorem 4.2](#) is what makes the result directly usable for identifiability, since it compares two *models* rather than two forms, and it is proved by the same finite-difference argument. See [Kagan et al. \(1973, Chapter 3\)](#) for the classical statement and [Ghurye and Olkin \(1962\)](#) for a multivariate version.

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